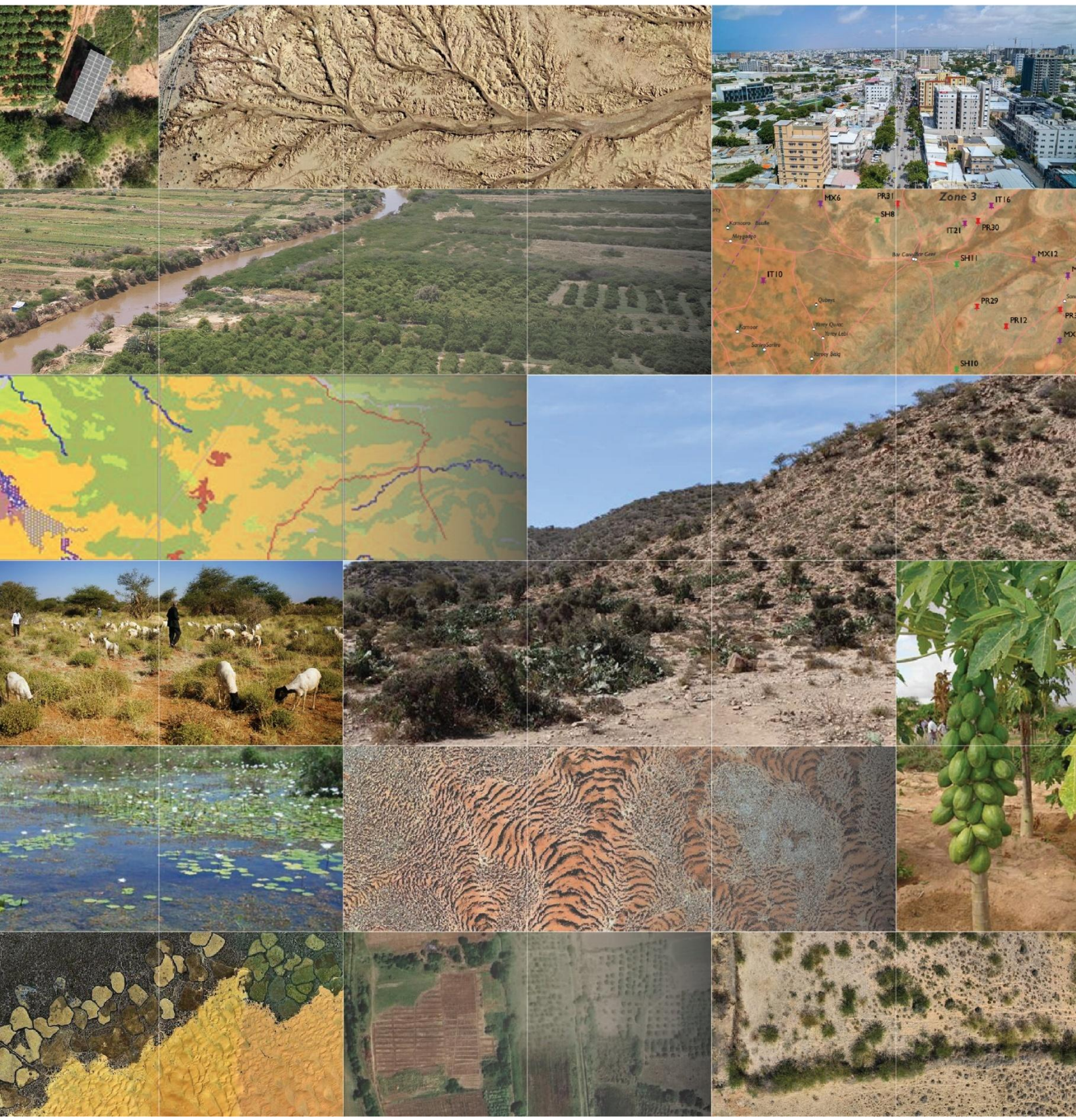




Food and Agriculture  
Organization of the  
United Nations

# LAND COVER REFERENCE SYSTEM FOR SOMALIA







# Land cover reference system for Somalia

by

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# Contributors

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# Foreword

Reliable and up-to-date land cover information is essential for sustainable resource management, environmental conservation and climate resilience. In Somalia, where natural resources are increasingly strained by climate variability, land degradation and competing land uses, a harmonized classification system is critical for supporting evidence-based decisions and planning.

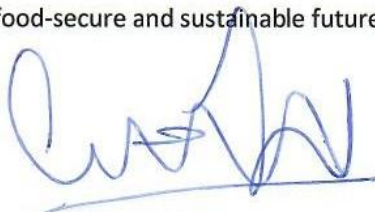
The LCRS for Somalia provides a scientifically rigorous and globally aligned framework for classifying and interpreting land cover across the country. Food and Agriculture Organization of the United Nations (FAO)'s LCML was used to ensure methodological consistency while being adaptable to Somalia's diverse environmental and socioeconomic conditions.

Aligned with FAO's global vision of a world free from hunger and malnutrition, this reference system contributes to the FAO's "four betters" – better production, better nutrition, a better environment, and a better life. By improving geospatial data accuracy, it will enhance land use planning, agricultural development and food security monitoring. Additionally, the system will play a key role in mitigating land degradation, strengthening climate resilience and supporting early warning mechanisms to reduce disaster risks and enhance adaptation strategies.

At the national level, the system will provide a reliable foundation for monitoring land use changes, informing rangeland and agricultural management and enabling integrated resource planning. It will support policymakers and practitioners in making data-driven decisions that enhance resilience, particularly in regions vulnerable to droughts and flooding.

The successful development of this framework reflects the dedication and expertise of the FAO teams and the government ministries of Somalia. Their commitment to maintaining high standards in land cover classification has been instrumental in ensuring that Somalia benefits from a robust and internationally recognized framework. Special appreciation goes to all those who have contributed their knowledge and technical expertise to this effort.

FAO remains committed to advancing geospatial information systems that support evidence-based policymaking, sustainable natural resource management and climate resilience in Somalia. This national LCRS will serve as a key tool for standardized land cover mapping to inform a more food-secure and sustainable future.



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The development of this land cover reference system (LCCS) for Somalia was led by the Somalia Land and Water Information Management (SWALIM) project unit of the Food and Agriculture Organization of the United Nations (FAO), with valuable support from FAO's Geospatial Unit, Land and Water Division (NSL). This effort aligns with FAO's Land Cover Meta Language (LCML), ensuring a robust and internationally recognized framework for land cover classification.

We acknowledge the dedicated efforts of the FAO SWALIM team, whose expertise in remote sensing, Geographic Information System (GIS) technology and LCCS played a pivotal role in ensuring the accuracy and reliability of this system. Their commitment to data collection, processing and analysis has been instrumental in advancing land cover monitoring in Somalia.

We also extend our gratitude to FAO's NSL, whose technical support significantly strengthened the methodology and analytical rigour of this document. Their guidance and expertise have been invaluable in refining the classification approach.

A special appreciation goes to Antonio Di Gregorio, whose foundational work on the LCML provided the methodological framework that guided this classification process. His contributions continue to shape global land cover classification standards.

Finally, we would like to recognize all colleagues in FAO, the Ministry of Agriculture (MoA) and SWALIM-established government Information Management Centres (IMCs) who contributed their time and expertise to reviewing and finalizing this document. Their collective efforts have resulted in a valuable resource for decision-makers, researchers and practitioners working on land cover monitoring and sustainable land management in Somalia.

# Abbreviations

|        |                                                                                                |
|--------|------------------------------------------------------------------------------------------------|
| FAO    | Food and Agriculture Organization of the United Nations                                        |
| CBD    | Convention on Biological Diversity                                                             |
| GIS    | Geographic Information System                                                                  |
| IMC    | Information Management Centre                                                                  |
| ISO    | International Organization for Standardization                                                 |
| LC     | land cover                                                                                     |
| LCCS   | Land Cover Classification System                                                               |
| LCHS   | land characterization system software                                                          |
| LCML   | Land Cover Meta Language                                                                       |
| LU     | land use                                                                                       |
| LCRS   | land cover reference system                                                                    |
| NCVC   | National Center for Vegetation Cover Development and Combating Desertification<br>Saudi Arabia |
| RLCM   | rapid land cover mapper                                                                        |
| SDG    | Sustainable Development Goal                                                                   |
| SWALIM | Somalia Water and Land Information Management project                                          |
| TC211  | technical committee 211                                                                        |
| UML    | unified modelling language                                                                     |
| UNCCD  | United Nations Convention to Combat Desertification                                            |
| UNFCCC | United Nations Framework Convention on Climate Change                                          |
| UNFF   | United Nations Forum on Forests                                                                |
| UNGGIM | United Nations Global Geospatial Information Management                                        |

# Executive summary

Accurate land cover assessment and classification have become a priority for various disciplines across a variety of spatial and temporal scales. Land cover is one of the 14 fundamental data theme layers recognized by the United Nations Global Geospatial Information Management (UNGGIM) due to its critical role in monitoring natural resources, supporting the Sustainable Development Goals (SDGs) and in addressing the requirements of the major multilateral environmental agreements, including the United Nations Framework Convention on Climate Change (UNFCCC), the Convention on Biological Diversity (CBD), the United Nations Convention to Combat Desertification (UNCCD), and the United Nations Forum on Forests (UNFF).

Despite its significance, the effective use and integration of land cover data are hindered by inconsistencies in classification methodologies, variations in spatial and temporal resolutions and vague nomenclature systems. Somalia, with its diverse plains, plateaus and highlands, poses a challenge for land cover classifications due to the varied methodologies employed by different ministries and stakeholders. The absence of a standardized framework complicates data harmonization, limiting interoperability and comparability across regions and affecting the accuracy of land cover estimation.

FAO has been supporting and assisting nations in implementing land cover standards based on the International Organization for Standardization (ISO) 19144-2 LCML (FAO and ISO, 2025).<sup>1</sup> This approach provides a robust method for classifying land cover across diverse landscapes using an object-oriented model that establishes clear relationships between land cover types at various scales, ensuring consistency and interoperability.

The purpose of this publication is to provide a LCRS for Somalia, enabling specialists and stakeholders to develop customized legends that meet their specific needs while ensuring data harmonization, standardization and seamless future integration. This manual aligns with international best practices, facilitating integration with global land cover initiatives. By providing a standardized reference system, this publication supports enhanced environmental monitoring, policy development and sustainable resource management at local, national and regional levels.

---

<sup>1</sup> **FAO (Food and Agriculture Organization of the United Nations) & ISO (International Organization of Standardization).** 2025. *Geographic information – Classification systems: Land Cover Meta Language (LCML)*. Part 2: ISO 19144-2:2023 (Edition 2). Rome, FAO. <https://openknowledge.fao.org/items/2859a0af-5fb5-4503-bff2-4319a81ebef1>

# 1. Introduction

Somalia's land resources are vital for agriculture, pastoral livelihoods and ecological sustainability. However, land degradation, deforestation, desertification, and climate variability continue to threaten food security, water resources and biodiversity (Omuto et al., 2009). Rapid land use changes, recurring droughts and floods further highlight the need for accurate, harmonized land cover data to guide sustainable management, disaster risk reduction and climate resilience (Nicholson, 2017).

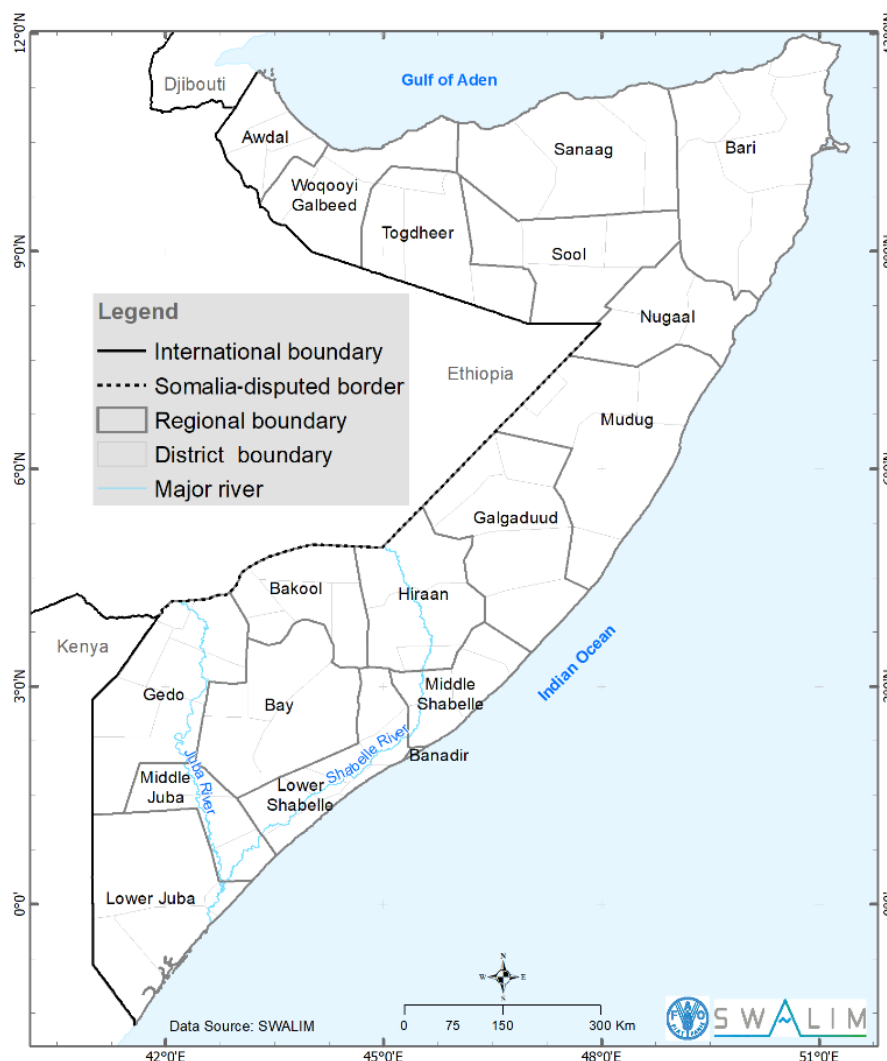
A standardized land cover reference system (LCRS) provides reliable and comparable data to support effective planning and decision-making. The internationally recognized Land Cover Meta Language (LCML) (FAO and ISO, 2025) offers a flexible and consistent way to classify Somalia's diverse landscapes, from arid rangelands and agropastoral areas to river basins and coastal ecosystems (Di Gregorio, 2023). This approach enhances data interoperability and ensures integration with international assessments and reporting frameworks to align with the United Nations (UN)'s Sustainable Development Goals (SDGs) (Calvo Buendia et al, 2019).

This national LCRS is Somalia's strategic framework for harmonized land cover monitoring, developed in line with international standards. It provides clear guidance for producing accurate land cover maps and supports key initiatives such as the Somalia Land and Water Information Management (SWALIM) project, the National Voluntary Land Degradation Neutrality Targets (Federal Republic of Somalia, 2020), the Integrated Land and Water Resources Management (ILWRM) project (FAO, 2025), and the Regreening Africa project (Resilient Landscapes, 2023). Strengthening Somalia's capacity to monitor land cover changes through the national LCRS will promote sustainable land management, improve natural resource governance and build climate resilience.

## 2. Land cover in Somalia – an overview

Somalia, located in the Horn of Africa, spans approximately 637 000 km<sup>2</sup> and possesses the longest coastline in mainland Africa, bordering the Indian Ocean and the Gulf of Aden (Figure 1). Somalia's physiography is characterized by diverse landscapes comprising arid and semi-arid rangelands, coastal plains, plateaus, escarpments, mountain ranges, and fertile riverine zones associated with the Shabelle and Juba rivers. These landscapes are shaped by underlying geological structures and climatic regimes, which collectively influence vegetation distribution, water availability and land use patterns (Venema, 2007).

Figure 1. Administrative boundaries of Somalia



Note: Refer to the disclaimer on page ii for the names and boundaries used in this map.

Sources: **FAO SWALIM (Somalia Water and Land Information Management)**. 2025. *Somalia Administrative Map*. Mogadishu.

**HDX (Humanitarian Data Exchange)**. 2024. *Somalia – Subnational Administrative Boundaries*. [Accessed on 5 August 2025]. <https://data.humdata.org/dataset/geoboundaries-admin-boundaries-for-somalia>. Licence: CC-BY-4.0.

**ESRI (Environmental Systems Research Institute)**. 2025. *ArcGIS Pro – Esri's enterprise geospatial platform*. [Accessed on 5 August 2025]. <https://www.esri.com/en-us/arcgis/>. Licence: CC-BY-4.0.



The climate of Somalia is predominantly arid to semi-arid, characterized by high temperatures, low and erratic precipitation, and significant spatial and temporal variability. Annual rainfall ranges from less than 20 mm in the northeastern coastal plains to between 200 and 500 mm in the southern regions and northern highlands. These climatic constraints critically determine the extent and composition of vegetation cover, patterns of resource use and the viability of different land management systems.

Somalia experiences a bimodal rainfall pattern interspersed with two dry seasons. The Gu season (mid-March to May) constitutes the primary rainy period and plays a crucial role in agricultural production and pasture regeneration. The secondary rainy season, Deyr (mid-September to November), provides supplementary moisture, albeit with reduced intensity and duration. Studies have however reported that the Deyr season has been getting wetter since the beginning of the twenty-first century (Muchiri, 2007). Conversely, the country is subjected to two dry spells: Jilaal (December to mid-March), the most prolonged and severe, often leads to acute water scarcity and pasture degradation, while Hagaa (July to September) is shorter but remains climatologically significant, particularly in the southern regions. In the north, particularly Somaliland, Karan rains are observed during the June to September period, which, when coupled with the Gu and Deyr rains, presents characteristics of a unimodal rainfall distribution extending from April to November.

These seasonal and interannual fluctuations exert a profound influence on land cover dynamics. Vegetation density, land use practices and surface hydrology respond to rainfall variability, drought episodes and flood events. Additionally, the mobility and adaptive strategies of pastoralist communities contribute to temporal and spatial variations in landscape configuration. Climatic stressors, particularly prolonged drought and erratic rainfall, exacerbate land degradation processes, including desertification and declining vegetative productivity (Monaci, Downie and Oduori, 2007).

Somalia's ecosystems can be broadly classified into five major ecological zones:

1. **Northern coastal plains (Guban):** A zone with arid to semi-arid conditions with sparse vegetation cover.
2. **Northern highlands:** A zone with areas of elevated precipitation supporting deciduous woodlands and evergreen bushland.
3. **Nugaal Valley:** A zone with predominantly semi-desert grasslands.
4. **Central rangelands:** A zone with open savanna interspersed with scattered bush, primarily used for extensive pastoralism.
5. **Southern plains:** A zone with relatively more productive landscapes including riverine forests and cultivated areas along the Shabelle and Juba rivers, supporting both rainfed and irrigated agriculture.

The evolution of land cover mapping in Somalia has undergone several phases. Initial national scale land use assessments were conducted during the 1980s. (Venema, 2007). However, following the collapse of central governance in 1991, systematic monitoring efforts were severely disrupted.

Between 1999 and 2002, the Food and Agriculture Organization of the United Nations (FAO) Africover project developed a digital, georeferenced land cover database for several African countries, including Somalia, at the scale of 1:250 000 (Venema, 2007). The initiative aimed to produce detailed and

standardized land cover datasets to support a wide range of applications, including environmental monitoring, natural resource management, agricultural planning, and sustainable development.

In Somalia, the project primarily utilized Landsat satellite imagery (USGS, 2007), which was visually interpreted to identify land cover features. This analysis was rigorously supported by extensive field verification to enhance accuracy (Venema, 2007). Notably, the Africover project represented the first application of FAO's Land Cover Classification System (LCCS) in the country.

Subsequently, the FAO Somalia Water and Land Information Management (SWALIM) project has played a key role in generating land cover information using medium- and high-resolution satellite imagery, supported by systematic ground validation. In 2007, FAO SWALIM undertook comprehensive wall-to-wall land cover mapping of the riverine agricultural areas along the Juba and Shabelle rivers, focusing on delineating zones suitable for irrigation. Concurrently, mapping activities were conducted in western Somaliland to support the development of 13 land suitability maps. Both exercises employed the FAO LCCS and relied on visual interpretation of multisource satellite imagery, including Landsat, validated through field surveys. These efforts laid the groundwork for evidence-based agricultural planning in two of Somalia's key agroecological regions (Monaci, Downie and Oduori, 2007) (Appendix 2).

Building on these earlier initiatives, SWALIM conducted a comprehensive statistical analysis of Somalia's land cover in 2014 (Leonardi, 2014) using the Rapid Land Cover Mapper (RLCM) dot-grid methodology developed by the United States Geological Survey (USGS) (Cotillon and Mathis, 2017). The approach relied on visual interpretation of land cover features at regularly spaced 500 m sampling points, significantly reducing workload while maintaining high statistical accuracy. A derived land cover map, based on these samples, was also produced, marking the first national scale land cover mapping since the FAO Africover project. The primary imagery used was ASTER, complemented by Landsat ETM data for areas with cloud cover or image gaps. Very high-resolution (VHR) imagery further supported the identification of agricultural zones and natural vegetation. The classification followed the FAO LCML standard (Leonardi, 2014) (Appendix 2).

In 2015, SWALIM carried out wall-to-wall land cover mapping at a scale of 1:50 000 for the Owdweyne and Burco districts (Ullah, 2016). This effort marked an advancement as it introduced a comprehensive land cover legend developed using the LCML. The mapping process involved the visual interpretation of polygons derived from the segmentation of Landsat-8 satellite imagery. These polygons were generated based on their spectral characteristic (Appendix 2).

Recent developments in land cover mapping in Somalia include the land cover dataset for the Jowhar Off-stream Storage Programme (JOSP), that at the time of publication of this document, is still under production (FAO SWALIM, forthcoming). The mapping activity is based on VHR satellite imagery and a detailed legend structured according to LCML, to enhance classification accuracy.

To address the critical need for standardization and consistency across these diverse mapping initiatives, FAO SWALIM developed the national LCRS, which is aligned with the FAO Land Cover Classification System (LCCS) (Di Gregorio and Leonardi, 2016) and ISO's Land Cover Meta Language (LCML) standard (FAO and ISO, 2025). This comprehensive framework ensures data interoperability and enables systematic tracking of land cover changes, land degradation trends and urban expansion patterns, thereby supporting both national resource management and international environmental monitoring initiatives (Rembold et al., 2015).

Continued improvements in land cover mapping are critical for supporting sustainable land management and climate resilience in Somalia. Despite progress, challenges persist, such as limited field access, rapid land use change and the need for frequent updates. Field validation remains constrained in areas affected by insecurity, limiting the spatial completeness of datasets. Future efforts should prioritize high resolution, regularly updated datasets, integration of emerging remote sensing technologies and alignment with international classification standards to ensure consistency, accuracy and operational relevance for planning and decision-making.

Looking ahead, Somalia's land cover monitoring efforts will benefit from continued investment in remote sensing capabilities, improved ground-truthing methodologies and strengthened interagency collaboration. These advancements will support sustainable land use planning, natural resource management and climate resilience strategies, while contributing to global environmental reporting frameworks such as the UN's SDGs and the United Nations Convention to Combat Desertification (UNCCD).

### 3. National land cover reference system

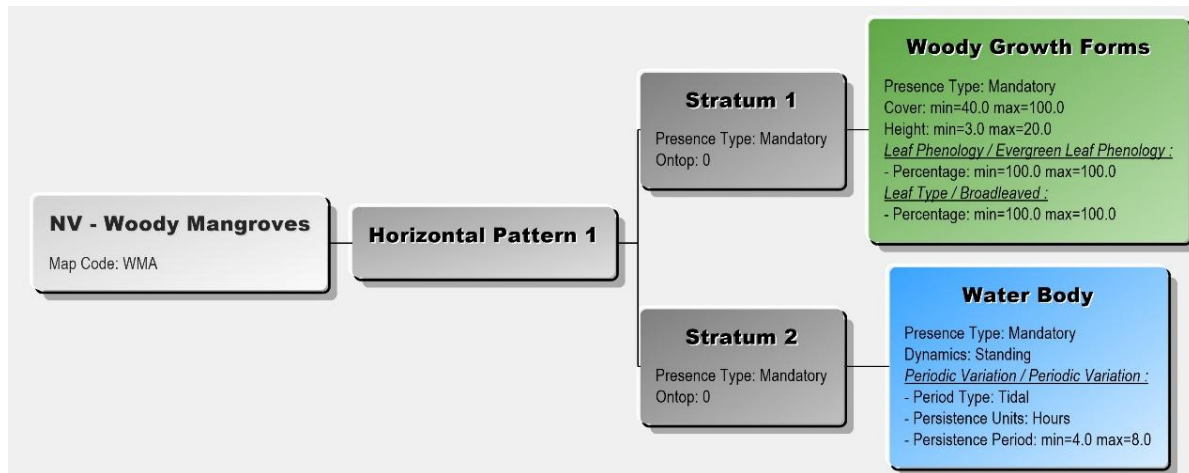
The importance and complexity of the creation of a national land cover reference system (LCRS) are very often underestimated. In many cases different subnational, national or regional legends in use are derived from the work of a single national institution that has not considered the development of a common reference system as a real national priority. Direct consequences of this situation are described as follows:

- **Difficult exchange of land cover and forest information:** The lack of standardized systems leads to communication gaps and data incompatibility between institutions.
- **Reduced efficiency of national databases:** Inconsistent classifications cause disorganized, duplicated and conflicting data, making retrieval inefficient.
- **Challenges in updating land cover information:** Incompatible datasets (past datasets may not align with new ones) hinder tracking environmental changes and long-term sustainability efforts.
- **Absence of a reference system for harmonization:** Without a standardized framework, data remains fragmented and difficult to integrate at national or regional levels.
- **Loss of financial and human resources:** Duplicated efforts arise when multiple institutions conduct similar land cover assessments without coordination, leading to wasted financial and human resources.

International standards play a crucial role in harmonizing national and regional classification systems (FAO and ISO, 2025; Mushtaq et al., 2024), but legend creation remains a local process that must align with specific geographic contexts. A national LCRS is more than just a legend, as it serves as a structured framework representing a country's land features through an object-oriented model that defines clear relationships between them (Gregorio et al., 2022; Jalal et al., 2019).. This system provides a graphical metamodel to enhance understanding, ensuring consistency and reducing ambiguities in land classification (Mosca et al., 2020). Regional and national agencies should follow a consistent national LCRS as their functional guideline to create specialized legends that are better equipped to reflect certain applications.

The national LCRS goes beyond a simple legend as it serves as a comprehensive manual that logically represents a country or region's land features. Using an object-oriented model, it defines clear relationships and inheritance among features, ensuring consistency and clarity. This system provides a structured, unambiguous classification of land features through a graphical metamodel, making it easier to understand and apply. By reducing ambiguities, it enables mapping agencies to develop specialized legends while maintaining harmonization and comparability across different mapping activities.

Figure 2. Unified modelling language (UML) representation of the graphical format of the *Woody mangroves* class



Source: Author's own elaboration.

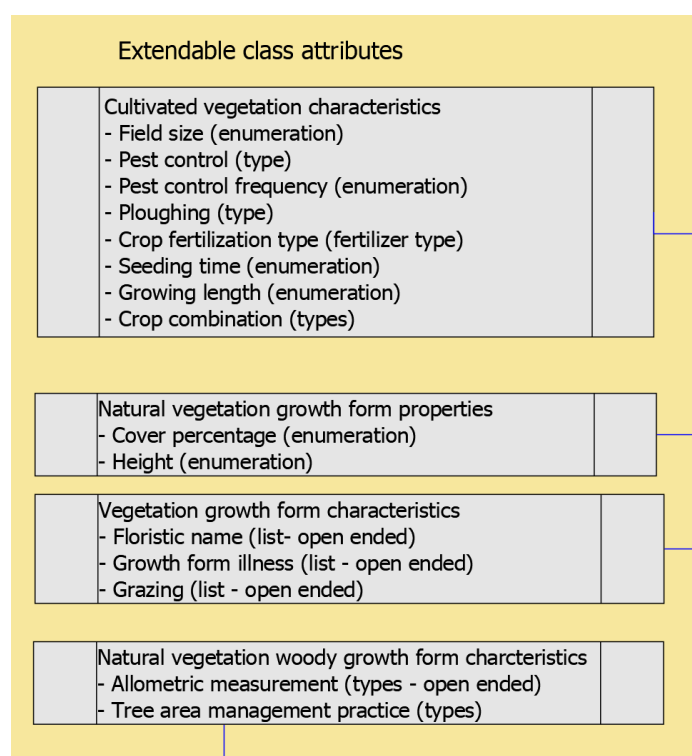
The system consists of metaclasses, which are not original classes but can be extended as needed. For each land cover metaclass in the system, there is a model available that defines it (Figure 2), both in a graphical format and in a specific technical format (XSD). Each model is stored in a specific national registry and also in the FAO land cover legend registry (Mushtaq et al., 2022) using FAO's LCCS version 3 (LCCSv3) and land characterization system software (LCHS) (FAO, 2016; Mosca et al., 2024). Therefore, when any national or international institution wishes to use those classes, they just need to refer through the map code to the registry, both for the class description and for any other advanced modelling and GIS operations that require the use of XSD markup language. In this framework, it is important to stress that the national LCRS must be considered a sort of manual to be consulted by any national (or international) institution or organization when involved in the design of a land cover mapping activity.

The system has different levels to define land cover metaclasses and the user has a complete overview when looking at the full graphic schema (see Section 6). The system is quite dynamic both in terms of choosing metaclasses from many levels as well as in terms of enriching and organizing them. Two last operational advice for users are as follows:

1. When creating their own system, the user can freely choose any number of different metaclasses from various levels of the reference schema. There are no limits on the levels, so the user can adjust the legend to be either broader or more detailed, depending on what they need. The classes used in the system form the basic structure of real-world land features. These metaclasses are like starting points that can be expanded if more information is needed for specific tasks. This can be done by utilizing the LCML shared dictionary of attributes, which are shown in the graphical model in special boxes (grey boxes). These boxes show how the attributes link to the land feature categories in the system.

Figure 3 shows examples of attributes in grey boxes that can be used to enhance the thematic content of the system's metaclasses, such as the general Natural and semi-natural vegetation class or the more specific Trees closed class. This list of attributes serves as a guide to standardize the enrichment of these classes and is flexible, allowing for further additions.

Figure 3. Examples of the land cover metalanguage attributes



Source: Author's own elaboration

2. When a specific metaclass inserted in the user-defined legend needs an increased number of details, the user should load it (the class) into the LCCSv3/LCHS software and upgrade or enrich the class with the extra attributes of the LCML standard. For consistency, it is recommended to maintain the original map code followed by a user-defined extended code (Appendix 4).

### 3.1. Flowchart of land cover reference system for Somalia

The overall flowchart structure (Section 6) is composed by three different elements, as follows:

1. Different types of metaclasses are distributed in different levels according to their level of generalization of the land features they represent (boxes with different colours).
2. Rules that define how each node or category can be further divided into more detailed land features (shown in red text annotations in the LCRS schema).
3. Expandable attributes of the metaclasses that allow adding more details about specific aspects of the land features are shown in grey boxes in Figure 3.

The graphical schema is presented in the form of a flowchart for an easier understanding of most users (see Section 6). However, some rules with a specific graphical symbol have been derived from the unified modelling language (UML):

- Inheritance (arrow)



- Association (diamond)

The rules (the text in red) that characterize each node are controlled by a specific object-based rule that is fully documented and explained in the next sections. The description consists as a specific synthetic text that clearly defines the conditions in terms of presence or absence or relevance of the basic LCML objects and attributes. An overview of the most important aspects of the system is described in Box 1.

#### Box 1. The national land cover reference system (LCRS) for Somalia: rules overview

- The model has been created using a flowchart. However, some rules of the Universal Modelling Language (UML) have been used, such as in Inheritance → and Association 
- The schema or flowchart is not a legend per se but more a graphic reference of how the land cover features are organized and logically ordered in this specific geographic context.
- The definition and ordering of the land cover feature of the schema strictly follow the object-oriented parametric approach as specified in ISO 19144-2 (FAO and NDISO, 2025), called a Land Cover Meta Language (LCML).
- The categories represented in the boxes are metaclasses. They represent the basic structure of specific land cover features that can be expandable using additional attributes using the shared dictionary of terms and attributes of LCML. Green boxes are metaclasses and grey boxes are expandable attributes.
- Each node of the schema and subsequent lower-level subdivision is regulated by a specific rule that is fully listed and documented.
- Each metaclass displayed in the lower level of the schema is defined according to the ISO 19144-2 LCML, using LCCSv3 and land characterization system software (LCHS) (FAO, 2016; Mosca *et al.*, 2020).
- To create a user-defined legend, the user needs to define at which level of each feature (metaclass) they want to stay, as for instance, no restriction of level exists in a level-based traditional hierarchical legend. For each specific feature to be a part of a legend, the user can choose any level or detail, such as a very generic one for feature A and a very detailed one for feature B. As specified in point 2, those classes can be expanded.

**Sources:** **FAO (Food and Agriculture Organization of the United Nations) & ISO (International Organization of Standardization).** 2025. ISO 19144 2:2023. Geographic information – Classification systems – Part 2: Land Cover Meta Language (LCML) In: *ISO*. Geneva, Switzerland. <https://openknowledge.fao.org/items/2859a0af-5fb5-4503-bff2-4319a81ebef1>

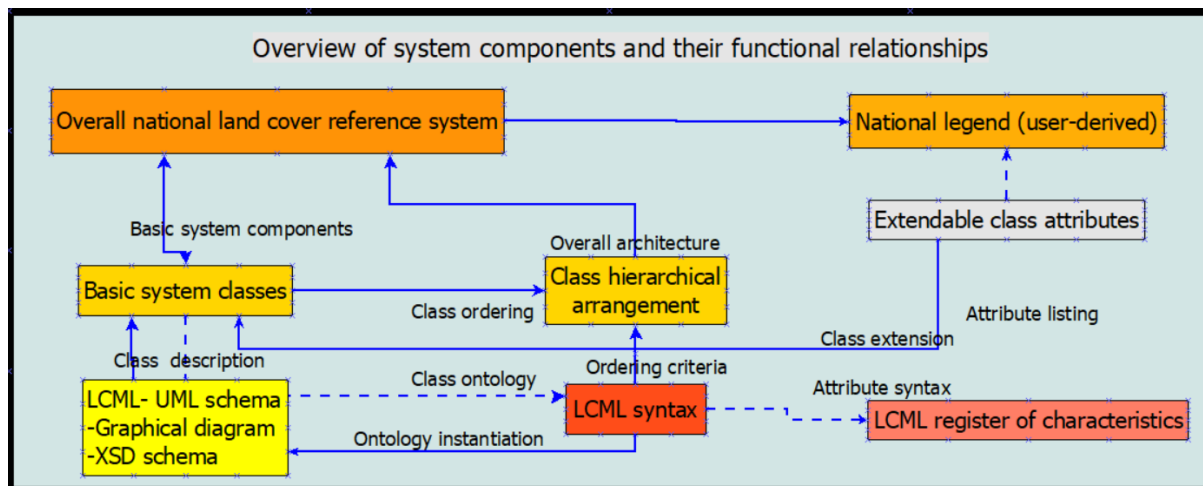
**FAO.** 2016. *Land Cover Classification System User Manual*. Rome. <https://openknowledge.fao.org/items/ea8e72e6-4610-4c7a-8571-8fcbdd24e346>

**Mosca, N., Di Gregorio, A., Henry, M., Jalal, R. & Blonda, P.** 2020. Object Based Similarity Assessment Using Land Cover Meta Language (LCML): Concept, Challenges, and Implementation. *IEEE Journal of Selected Topics in Applied Earth Observations and Remote Sensing*, 13: 3790-3805. <https://doi.org/10.1109/JSTARS.2020.3003888>

A full overview of the system components and their functional relationship is detailed in Figure 4 and is explained as follows:

- The LCML syntax gives the conceptual theoretical base for building up the system classes.
- The classes are arranged in different levels to clearly instantiate the class inheritance forming the proposed overall system architecture.
- Each system class is formally described with the most advanced mark-up languages.
- The existing LCML register of characteristics supports a list of possible extra attributes that can further expand the class information.
- Finally, the whole system will support any type of national or subnational legends that can be tailored to specific project needs, while at the same time being fully coherent with the main national LCRS.

Figure 4. Overview of system components and their functional relationships



Note: LCML = Land Cover Meta Language and UML = unified modelling language.

Source: Author's own elaboration.

## 4. Rules for the arrangement of metaclasses

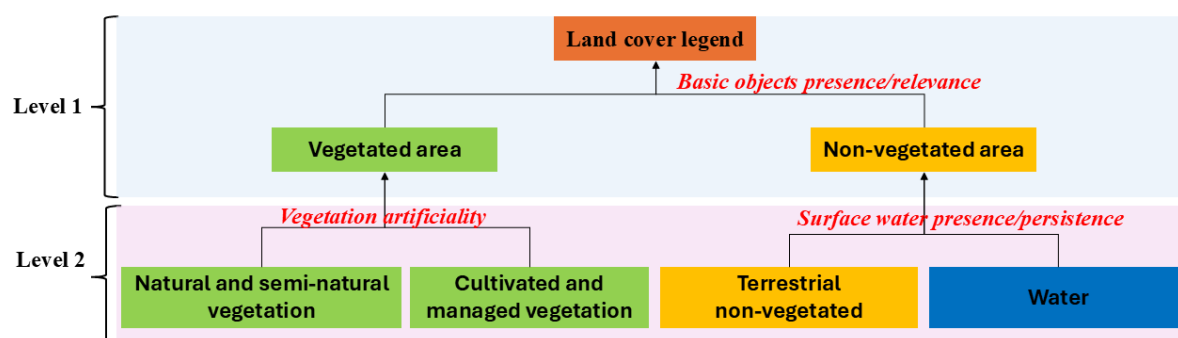
Somalia's national LCRS is structured into distinct levels based on the required level of detail. These levels follow the object-oriented rules and syntax of LCML. The classification relies on a physiognomic-structural approach, considering both the external appearance (such as trees, shrubs and buildings) and spatial arrangement (such as cover percentage and vegetation layering). Additional qualitative criteria, such as artificiality, soil conditions or species names, further refine the classification.

### 4.1. Rules for differentiating the first and second levels of schema metaclasses

The main separation of the basic land features at the **first level** is achieved as follows:

- **Basic objects presence/absence and/or relevance** is split into two main metaclasses:
  - **Vegetated (biotic) surfaces** defines areas that have a vegetative cover of at least four percent for at least two months of the year. This cover may consist of Woody growth forms including Trees and Shrubs, Herbaceous growth forms including Forbs and Graminoids or a combination.
  - **Non-vegetated (abiotic) surfaces** defines any surface with less than 4 percent of Vegetated growth forms.

Figure 5. First and second levels of the schema and the relative rule governing the main subdivisions



Source: Author's own elaboration.

The main separation of the basic land features at the **second level** are done according to **Vegetation artificiality** for the vegetated section and **Surface water presence/persistence** for the non-vegetated section.

**Vegetation artificiality** defines the status of the vegetation cover (whether it is subject to a natural process or is characterized by a type of human management). It is divided in two major groups:

1. **The Natural and semi-natural vegetation** class is explained as follows:
  - a. **Natural vegetation** defines an area where the vegetative cover is in balance with the abiotic and biotic forces of its biotope.
  - b. **Semi-natural vegetation** defines an area where the vegetation is not planted by humans but influenced by human actions. These may result from grazing activities, possibly overgrazing the natural phytocoenosis, or else from practices such as selective logging in a natural forest whereby the floristic composition has been changed. Previously cultivated areas that have been abandoned and where vegetation is regenerating are also included. Another example is the secondary vegetation that grows during the fallow time of shifting cultivation and where the human disturbance may be deliberate or inadvertent. Hence, the term Semi-natural vegetation refers to vegetation that has been affected by humans but has recovered to the point where the species composition and environmental and biological processes are either identical to, or in the process of obtaining, their undisturbed state.
2. **The Cultivated and managed vegetation** class defines an area where the natural vegetation has been removed or modified and replaced by other types of vegetative cover of anthropogenic origin. This vegetation is artificial and requires human activities to maintain it in the long term. Between human activities, or before starting crop cultivation, the surface can be temporarily without vegetative cover. Its seasonal phenological appearance can be regularly modified by humans (such as through tillage, harvest and irrigation). All vegetation that is planted or cultivated with an intent to harvest is included in this class (such as wheat fields, orchards, rubber, and teak plantations).

**Surface water presence/persistence** defines the status of the non-vegetated areas according to the presence, relevance or persistence of the water. It is divided into two areas:

1. **Terrestrial non-vegetated areas** defines areas with no regular presence of surface water.
2. **Water and associated areas** defines the systematic presence of water, even when the persistence period is very limited or ephemeral, as the land surface is still influenced by the water.

## 4.2. Natural and semi-natural vegetation (rules for differentiating the third and fourth levels of schema metaclasses)

In the **third level** (Figure 6) of the schema, when considering the **Natural and semi-natural vegetation** metaclass, the Vegetation edaphic conditions discriminative rule is applied.

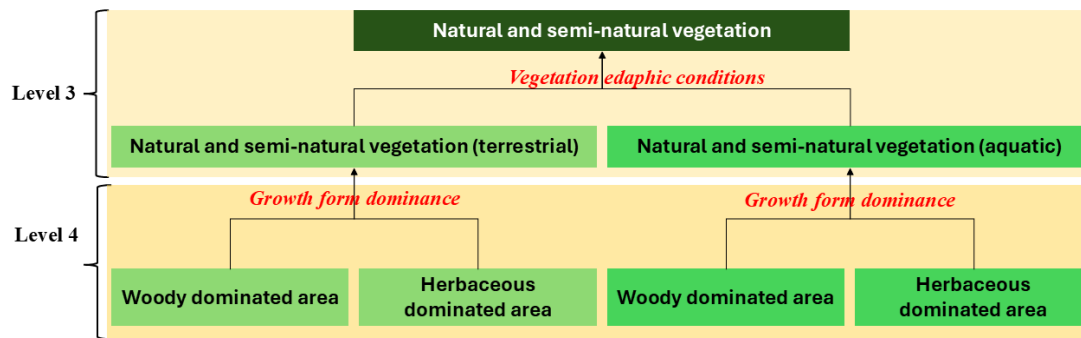
**Vegetation edaphic conditions** defines if and how a specific vegetated area is influenced by the presence of permanent, seasonal or sporadic water and contains two major groups:

1. **Natural and semi-natural vegetation (terrestrial)** defines areas where the vegetation growth relies on rainfall and its presence or absence is not influenced by the occurrence of permanent or seasonal surface water.
2. **Natural and semi-natural vegetation (aquatic or regularly flooded)** defines areas where the vegetative cover is influenced by the presence of surface water. The surface water presence can be permanent, seasonal or ephemeral, with a water table usually at or near the surface. Those areas additionally show a distinctive surface aspect influenced by water.

In the **fourth level** (Figure 6) of the schema for the two previously mentioned groups, the Growth form dominance rule is applied:

- **Growth form dominance** is the main domain that characterizes a specific vegetated area. Following the major structural vegetation classifications worldwide, the main criterion is characterized by the uppermost canopy layer logic. This means that the dominant layer goes from Woody growth forms (Trees or Shrubs or woody canopy) to Herbaceous growth form (Forbs or Graminoids). This general rule is subject **to a subcondition of cover**, as it is only valid if the dominant growth form has a cover from 10 to 100 percent (open to closed). If the growth form is less than 10 percent (sparse), the dominance goes to another growth form that has an open or closed cover. Based on this specific rule the fourth level of the **Natural and semi-natural vegetation (terrestrial)** is divided into the following:
  - **Woody dominated area** defines an area that has the presence of Woody growth forms with a cover from 10 to 100 percent. For areas where the cover percentage of the Woody element is less than 10 percent, the area falls into this class if the cover of the herbaceous element is not particularly important (less than 10 percent and persistent for at least two or more months a year). **Woody closed** refers to areas with woody vegetation covering 70 to 100 percent of the surface, often forming dense canopies with limited ground visibility. **Woody open** refers to areas with woody vegetation covering 10 to 70 percent of the surface, allowing significant light penetration and typically supporting a more diverse understory vegetation.
  - **A Herbaceous-dominated area** defines an area that has the presence of Herbaceous growth forms for at least more than two months a year, with a cover from 10 to 100 percent and where the presence of the Woody growth forms (Trees and/or Shrubs) is less than 10 percent. For areas where the Herbaceous element cover is less than 10 percent the area falls into this class if the cover of the Woody growth forms (Trees or Shrubs) is absent or sporadic.

Figure 6. Third and fourth level for *Natural and semi-natural vegetation*



Note: The rules governing the subdivision of *Natural and semi-natural vegetation* are in red.

Source: Author's own elaboration.

The fourth level of the **Natural and semi-natural vegetation (aquatic)** or **Regularly flooded vegetation** flows the same pattern of the terrestrial vegetation:

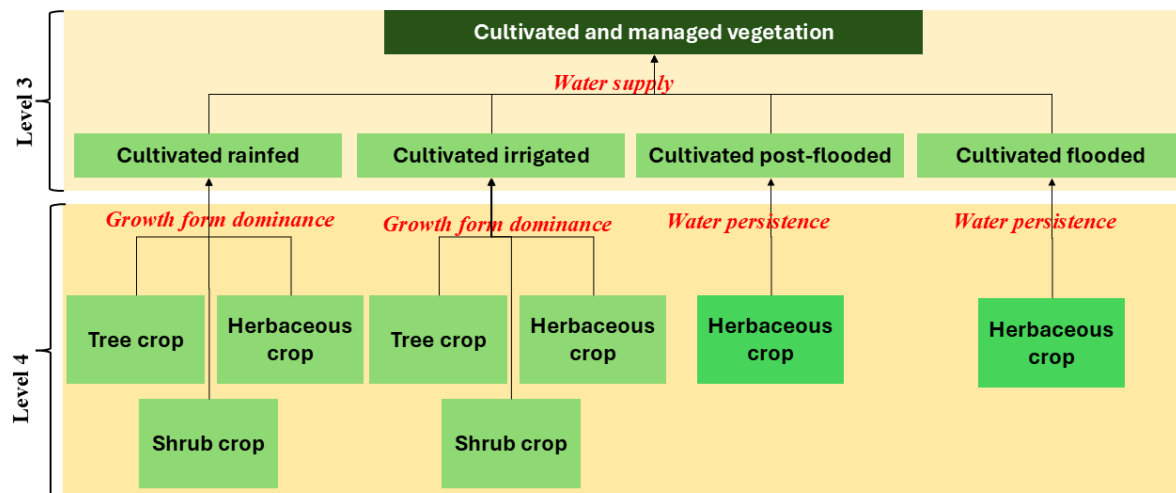
- **Woody dominated area** defines an area that has the presence of Woody growth forms with a cover from 10 to 100 percent. For areas where the cover percentage of the Woody element is less than 10 percent, the area is still included in this class if the cover of the Herbaceous element is not particularly important (less than 10 percent and persistent for at least two or more months a year).
- **Herbaceous-dominated area** defines an area that has the presence of Herbaceous growth forms for at least more than two months a year, with a cover from 10 to 100 percent and where the presence of Woody growth forms (Trees and/or Shrubs) is not significant (less than 10 percent). For areas where the cover percentage of the Herbaceous element is less than 10 percent, the area is still included in this class if the cover of the Woody growth forms (Trees and/or Shrubs) is absent or sporadic (from 1 to 3 percent).

### 4.3. Cultivated and managed vegetation (rules for differentiating the third and fourth levels of schema metaclasses)

In the **third level**, **Cultivated and managed vegetation** is governed by the water supply discriminative rule (Figure 7). **Water supply** refers to the main watering regime of a specific cropping area. It defines four major metaclasses:

1. **Rainfed** defines a crop where the water supply relies mainly on rainfall.
2. **Irrigated** defines a crop where the water regime relies mainly on providing an artificial, regular supply of water.
3. **Flooded** defines a crop that remains intentionally inundated for most of its growing stage.
4. **Post-flooded** defines a crop that is cultivated after floodwaters have receded.

Figure 7. Third and fourth level for *Cultivated and managed vegetation*



Note: The rules governing the subdivision of *Natural and semi-natural vegetation* are in red.

Source: Author's own elaboration.

In the **fourth level**, the Growth form dominance and Water persistence discriminative rules are applied (Figure 7).

Growth forms dominance is defined as the main vegetation domain that characterizes a specific cultivated area. Following the major structural vegetation classifications (as for **Natural and semi-natural vegetation**), the main criterion is characterized by uppermost canopy layer logic. It further defines the following major metaclasses:

- For rainfed areas:
  - **Tree crop** defines all cultivated trees, such as orchards and other types of plantations.
  - **Shrub crop** defines all cultivated woody perennial plants with persistent, woody stems that lack a defined main stem (mostly orchards and other types of plantations).
  - **Herbaceous crop** defines all crops with non-woody stems (Graminoids and Forbs).
- For irrigated areas:
  - **Tree crop** defines all cultivated trees such as Orchards and other types of plantations.
  - **Shrub crop** defines all cultivated woody perennial plants with persistent, woody stems that lack a defined main stem, mostly including Orchards and other types of plantations.
  - **Herbaceous crop** defines all crops with non-woody stems (Graminoids and Forbs).

**Water persistence** is the presence and duration of water supply for the growing of a specific crop and defines the following major metaclasses:

- For flooded areas:
  - **Herbaceous crop** defines all crops with non-woody stems (Graminoids and Forbs).
- For the post-flooded areas:
  - **Herbaceous crop** defines all crops with non-woody stems (Graminoids and Forbs).



## 4.4. Terrestrial non-vegetated areas (rules for differentiating the third and fourth levels of schema metaclasses)

In the **third level**, Terrestrial non-vegetated areas is governed by the Object artificiality types discriminative rule (see Figure 8).

**Object artificiality types** defines the nature of the bare objects (whether they are natural [Bare rocks Bare soil or Sand] or if they are artificially made [such as Buildings or Roads]). This specific rule defines two major metaclasses (Figure 8):

1. **Natural surfaces** defines all natural areas that do not have a vegetation cover, or areas where the vegetation cover is virtually absent (less than 4 percent).
2. **Artificial surfaces** defines all surfaces that have an artificial cover as a result of human activities such as construction, extraction or waste disposal.

In the **fourth level** (Figure 8), Natural surfaces is governed by the Natural surface aspects discriminative rule.

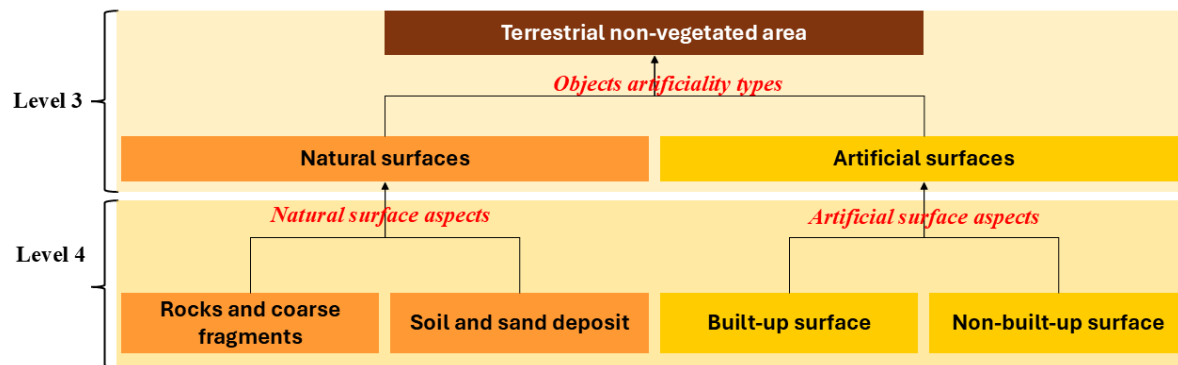
**Natural surface aspects** defines the overall appearance of a surface that is without human modifications or construction. This specific rule defines two major metaclasses:

- **Rocks and coarse fragments** defines natural areas which are characterized by the solid and firm consistency of their surface or by the presence of coarse fragments with these properties. These surfaces are impenetrable with a spade or a hoe. The surface and the coarse materials remain coherent and hard even when moist. At this level, the metaclass refers to areas that are either dominated by a continuous rock surface or covered by coarse rock fragments.
- **Soil and sandy surfaces** defines all natural areas covered by unconsolidated materials resulting from weathering of parent material (including the effects of moisture and temperature) or macro- and microorganisms.

**Artificial surface aspects** defines the surface aspect of areas with an artificial or associated cover. The following main classes are distinguished:

- **Built-up surfaces** defines built-up areas covered by impervious structures adjacent to or connected by roads.
- **Non-built-up surfaces** defines non-built-up areas covered by impervious structures adjacent to or connected by roads.

Figure 8. Third and fourth level for *Terrestrial non-vegetated areas*



Note: The rules governing the subdivision of *Terrestrial non-vegetated areas* are in red.

Source: Author's own elaboration.

## 4.5. Waterbodies and associated areas (rules for differentiating the third and fourth levels of schema metaclasses)

In the **third level**, Water and associated areas is governed by the Waterbody artificiality discriminative rule (Figure 9).

**Waterbody artificiality** defines whether a waterbody or the other associated areas are natural or artificial. The term Artificial is defined as an area where under natural circumstances, no water surface would exist. Therefore, these surfaces are the result of an artefact, such as the construction of a dam, and other artefacts. This specific rule defines two major metaclasses:

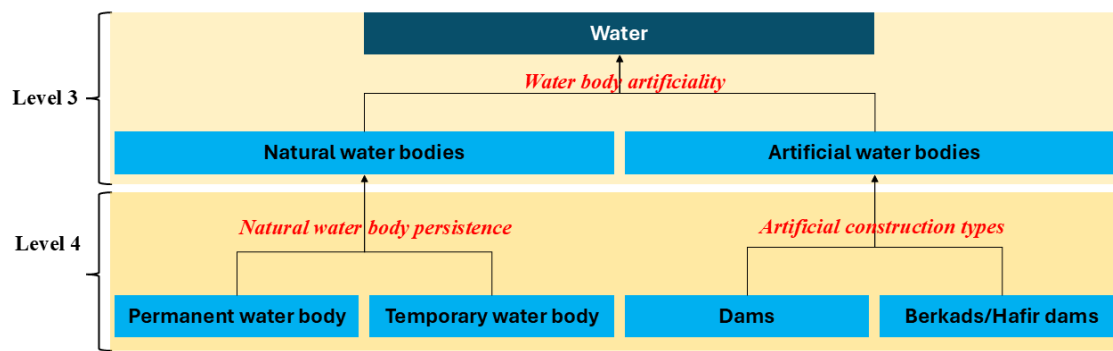
1. **Natural waterbodies** defines all natural lakes and or small natural ponds and all the natural areas where the surface is influenced by the regular presence of water even when it is seasonal, intermittent or sporadic.
2. **Artificial waterbodies** defines all areas where the presence of water result from artificial artefacts.

In the **fourth level** (Figure 9) the Water and associated areas is governed by Natural waterbody persistence and Artificial construction types discriminative rules:

- **Water persistence** defines the persistence extent of the water at or near the surface. This specific rule defines the following major metaclasses:
  - **Permanent natural waterbody** defines an area covered by natural waterbody for the whole year.
  - **Temporary natural waterbody** defines an area where the presence of water is seasonal, sporadically or ephemeral. However, in all those situations the surface aspect of those areas is primarily shaped by the presence of water.

- **Artificial construction types** defines the types of construction of the artificial waterbodies. This specific rule defines the following major metaclasses:
  - **Dams** defines a large, engineered barriers constructed across rivers or streams to impound and control the flow of surface water, creating reservoirs used for water supply, irrigation, energy generation, or flood regulation.
  - **Berkads** defines small, artificial rainwater reservoirs, typically lined with cement or bricks, designed to store seasonal runoff for household and livestock use in arid areas.
  - **Hafir dams** defines shallow, excavated basins with earthen embankments, designed to collect and store seasonal surface runoff for household, livestock and agricultural use in arid areas.

Figure 9. Third and fourth level for *Water and related areas*



Note: The rules governing the subdivision of the waterbody and related areas are in red.

Source: Author's own elaboration.

## 5. Rules governing the separation of the metaclasses at detailed levels

After the fourth level, as described in the previous chapter, there are further specific detailed levels that define the metaclasses with an increasing level of distinctiveness. The rules governing the separation of those metaclasses will be listed and described accordingly. Equally, instead of a simple text definition to characterize the specific metaclasses, a more accurate characterization of each will here be laid out, using the modelling capabilities and rules as defined in the LCML (ISO Standard 19144-2) (FAO and ISO, 2025). Each major metaclass will be therefore represented with an UML-like graphic model in support to its text description. All the metaclasses in these detailed levels are described in Appendix 1.

### 5.1. Natural and semi-natural vegetation (terrestrial)

The discriminative rules are listed and described as follows:

- **Woody cover percentage (1)** is the woody cover percentage defined as the proportion of a particular area of the ground substrate covered by a layer of woody plants, considered at the greatest horizontal perimeter level of each plant. The following cover percentages are considered at this level:
  - **Closed** defines an area where the woody cover ranges from 70 to 100 percent.
  - **Open** defines an area where the woody cover ranges from 10 to 70 percent.
- **Growth form dominance (1)** defines the main domain that characterizes a specific vegetated area. Following the major structural vegetation classifications (as for Natural and semi-natural vegetation) the main criterion is characterized by uppermost canopy layer logic. It defines the following major metaclasses:
  - **Tree-dominated area** defines an area that has the presence of trees with a cover from 10 to 100 percent. For areas where the Trees element cover is less than 10 percent, the areas fall into this class if the cover of the other growth forms (i.e. Shrubs and Herbaceous) is equally sparse (<10 percent).
  - **Shrub-dominated area** defines an area that has the presence of shrubs with a 10 to 100 percent cover and the presence of trees is significantly less than 10 percent. For areas where the Shrubs element cover is less than 10 percent, the area falls into this class if the Trees element cover is not significant (much less than 4 percent), and the Herbaceous element cover is sporadic (present only very few times a year) or less than 10 percent.
- **Tree cover percentage** defines the proportion of a particular area of the ground or substrate covered by a layer of trees, considered at the greatest horizontal perimeter level of each plant. The following cover percentages are considered at this level:
  - **Closed** defines an area where the tree cover ranges from 70 to 100 percent.

- **Open** defines an area where the tree cover ranges from 20 to 70 percent.
- **Very open** defines an area where the tree cover ranges from 10 to 20 percent.
- **Shrub cover percentage** is the proportion of a particular area of the ground or substrate covered by a layer of shrubs, considered at the greatest horizontal perimeter level of each plant. The following cover percentages are considered at this level:
  - **Closed** defines an area where the shrub cover ranges from 70 to 100 percent.
  - **Open** defines an area where the shrub cover ranges from 10 to 70 percent.
- **Growth form layer presence/absence (1)** defines the presence or absence of different growth forms strata. The following aspects are considered:
  - **Savanna** defines an area covered by a consistent main layer of herbaceous vegetation (cover from 10 to 100 percent) and a second layer of scattered woody vegetation (cover substantially less than 10 percent).
  - **Grassland** defines an area covered by a consistent and persistent layer of Herbaceous growth forms (for at least two months a year) and where the woody vegetation component is totally absent.
- **Growth form layer presence/absence (2)** defines the presence or absence of different growth forms strata. One metaclass is considered:
  - **Tree and shrub savanna** defines an area characterized by herbaceous vegetation (20 to 60 percent cover) as the primary layer with scattered trees (2 to 10 percent cover) and shrubs (2 to 10 percent cover) as secondary layers, typically showing seasonal patterns with four to eight months of herbaceous growth.
- **Growth form dominance (2)** defines the main domain that characterizes a specific vegetated area and is mainly characterized by an uppermost canopy layer logic, following the major structural vegetation classifications (as for Natural and semi-natural vegetation). It defines the following major metaclasses:
  - **Tree savanna** defines a vegetation formation dominated by herbaceous cover (20 to 100 percent) with scattered trees (2 to 10 percent cover) as the secondary Woody element, alternating with exposed soil during dry periods.
  - **Shrub savanna** defines a vegetation formation dominated by herbaceous plants (20 to 100 percent cover) with scattered shrubs (2 to 10 percent cover) as the secondary Woody element, alternating with exposed soil during dry periods.
- **Horizontal pattern deposition** defined as the horizontal spatial distribution of vegetation in a certain area. It should not be confused with cover, which defines the spatial arrangement of growth forms such as Trees and Shrubs. A horizontal pattern may be used for a complex land cover class composed of two or more distinct land cover features that will be handled as a unicum independently from scale constraints:

- **Tiger bush woody** defines a distinctive patterned vegetation formation characterized by alternating bands of woody vegetation and bare soil, with specific vegetation strata including shrubs, herbaceous plants and trees arranged in a structured spatial pattern resembling a tiger's stripe pattern.

## 5.2.Natural semi-natural aquatic or regularly flooded vegetation

The discriminative rules governing the subdivision of the natural and semi-natural aquatic or regularly flooded vegetation are listed and described as follows:

- **Water salinity** refers to the concentration of total dissolved solids (TDS), expressed as parts per million in the water. The following major classes are defined:
  - **Woody mangrove** defines coastal woody vegetation adapted to brackish or saline water inundated by regular tidal cycles.
  - **Flooded woody** defines woody vegetation growing in areas regularly inundated by freshwater.
- **Water persistence (1)** refers to the length of time that above ground water remains in a specific area. In this specific case it defines the following types of persistence:
  - **Herbaceous (permanently flooded)** defines above ground water that remains in a specific area for more than nine months a year.
  - **Herbaceous (seasonally flooded)** defines above ground water that remains in a specific area for a period of one to six months.
- **Water persistence (2)** refers to the length of time that above ground water remains in a specific area. In this specific case it defines the following types of persistence:
  - **Permanently flooded woody** defines woody vegetation in areas where above ground water remains for more than nine months a year.
  - **Seasonally flooded woody** defines woody vegetation in areas where above ground water remains for one to six months a year.
- **Growth form dominance** defines the main domain that characterizes a specific vegetated area. Following the major structural vegetation classifications (as for Natural and semi-natural vegetation) the main criterion is characterized by the uppermost canopy layer logic and is defined by the following major metaclasses:
  - **Permanently flooded trees** defines tree-dominated vegetation in areas continuously inundated with freshwater for more than nine months annually.
  - **Permanently flooded shrubs** defines shrub-dominated vegetation in areas continuously inundated with freshwater for more than nine months annually.

- **Growth form dominance** defines the main domain that characterizes a specific vegetated area. Following the major structural vegetation classifications (as for Natural and semi-natural vegetation) the main criterion is characterized by uppermost canopy layer logic. It defines the following major metaclasses:
  - **Seasonally flooded trees** defines tree-dominated vegetation in areas periodically inundated with freshwater for one to six months annually.
  - **Seasonally flooded shrubs** defines shrub-dominated vegetation in areas periodically inundated with freshwater for one to six months annually.

### 5.3. Cultivated and managed vegetation

The discriminative rules governing the subdivision of the cultivated and managed vegetation are listed and described as follows:

- **Crop stratification** defines the simultaneous presence or absence of two or more layers of crops. In this specific case, the following type is considered:
  - **Orchards and other plantations** defines common fruit trees such as orange (*Citrus sinensis*) and apple (*Pirus malus*).

### 5.4. Terrestrial non-vegetated areas

Description of the rules governing the terrestrial non-vegetated are as follows:

- **Rocks and fragments physiognomic aspect** defines the overall appearance of the consolidated material:
  - **Bare rock** defines where the rock surface is continuous except perhaps for a few cracks in the material. Some areas may be covered by shallow layers of soil or there could be isolated pockets of soil or a mixture of both.
- **Natural surface physiognomic aspect** defines the overall appearance of the natural surfaces. It refers to the following aspects:
  - **Loose and shifting sand** defines unconsolidated sandy surfaces with minimal vegetation that are mobile due to wind action, often occurring in arid zones.
  - **Eroded lands** defines areas where soil and surface materials have been significantly degraded by natural processes, particularly water or wind erosion, leading to unstable surfaces with sparse or no vegetation cover.
  - **Bare soil** defines an area covered by stable, unconsolidated material.
  - **Beaches** defines a strip of land surface covered by sand or small stones at the edge of a body of water.
  - **Salt deposits** defines areas characterized by surface accumulation of salt minerals, often resulting from the evaporation of saline water in arid environments.



- **Sandy desert physiognomic aspect** defines refers to the overall aspect of a sandy desert (specifically the presence or absence of dunes):
  - **Sand dunes** defines arid surface covered by loose and shifting sand.
- **Erosion type** defines the process of the removal of solids (such as soil, rock, sediment and other materials) by a number of possible agents such as, wind, water, ice, gravity and living organisms. This materializes in the following natural surface features:
  - **Sheet** defines a type of erosion where there is removal of a uniform layer of soil from the land surface by water.
  - **Gully** defines a type of water erosion that results in the formation of gullies, which are deep channels or trenches in the soil.
  - **Rill** defines a type of water erosion that results in the formation of rills, which are small, closely spaced channels in the soil.
- **Artificial surface physiognomic aspect (1)** defines the overall appearance of the artificial surfaces in built-up surface areas with the following different artificial surface aspects:
  - **Built-up non-linear** defines non-linear built-up areas covered by impervious structures adjacent to or connected by streets (the cover is related to centres of population). Linear elements like (main) roads, railways and communication lines or pipelines occur but are not dominant features. The density of the artificial constructions in relation to the surrounding area can be described separately.
  - **Built-up linear and others** defines linear built-up areas with dominant linear elements like (main) roads, railways and communication lines or pipelines.
- **Artificial surface physiognomic aspect (2)** defines the overall appearance of the artificial surfaces in non-built-up surface areas with the following different artificial surface aspects:
  - **Dump sites** define areas characterized by the absence of the original semi-natural or natural cover or water surface and by the presence of dump site material placed on the surface.
  - **Extraction sites** define areas where the original semi-natural or natural surface has been removed or disturbed due to the extraction of materials such as sand, gravel and minerals.
- **Artificial surface types (and subtypes)** defines the further subdivision of the overall appearance of the artificial surfaces coupled with their main utilization:
  - **Urban areas** defines non-linear built-up areas covered by impervious structures adjacent to or connected by roads.
  - **Camps** defines settlements characterized by temporary or semi-permanent structures made of light materials, typically arranged in clusters with surrounding bare soil areas.
  - **Rural villages** defines built-up areas generally with a low density of impervious structures that are located in rural areas.

- **Infrastructure** defines non-linear built-up structures such as ports and airports.

## 5.5. Water and associated areas

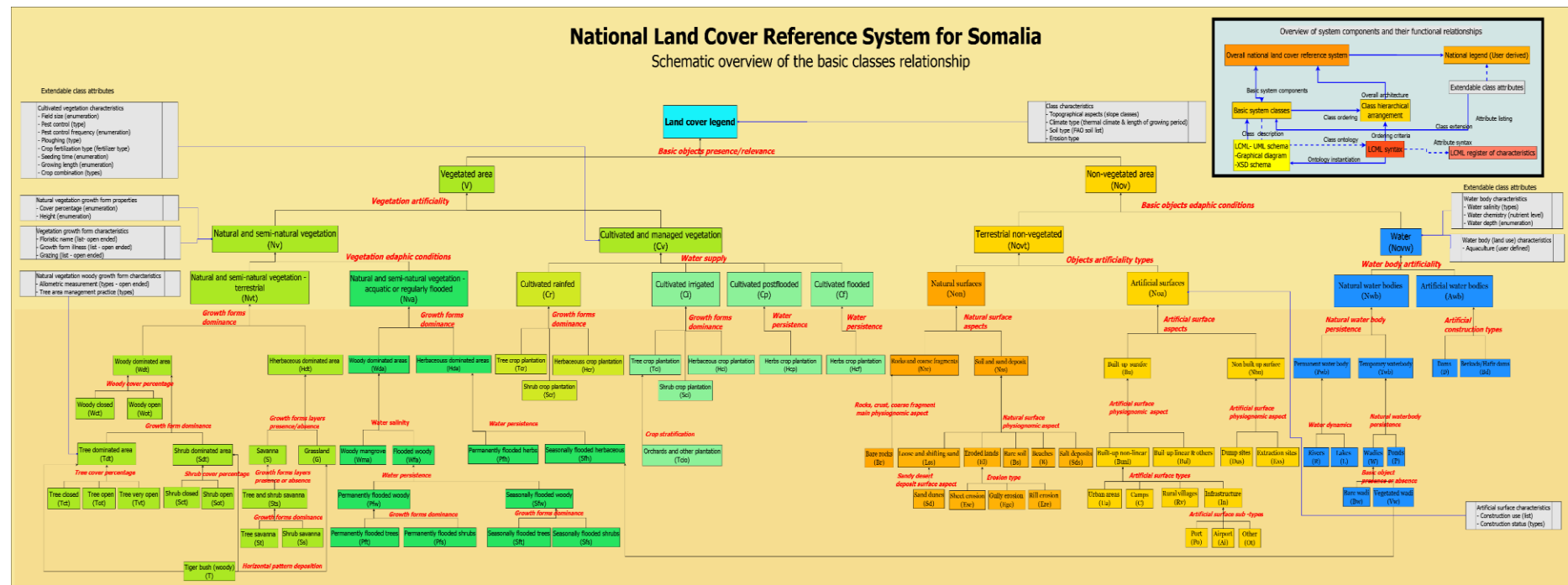
Description of the rules governing waterbodies and associated surfaces are as follows:

- **Water dynamics** defines the presence or absence of a stream in a waterbody with the following aspects:
  - **Rivers** defines natural flowing freshwater bodies with defined channels and directional movement.
  - **Lakes** defines natural or artificial inland waterbodies characterized by standing water dynamics and freshwater salinity.
- **Waterbody persistence** defines the annual duration of above ground water persistence in a specific area, with the following aspects:
  - **Wadis** defines ephemeral watercourses that carry water only during and immediately after rainfall events.
  - **Ponds** defines ephemeral watercourses that carry water only during and immediately after rainfall events.
- **Basic object presence/absence** defines the presence or absence of the vegetation growth form, with the following aspects:
  - **Bare wadis** defines wadis that lack significant vegetation cover and are instead dominated by exposed soil, gravel or sand surfaces with minimal or no plant growth.
  - **Vegetated wadis** defines ephemeral watercourses with noticeable woody vegetation growing in the channel and along the banks.

# 6. National land cover reference system schema for Somalia

This section provides the complete schema for the national LCRS for Somalia (Figure 10). The LCRS is based on six levels, with the levels and their relationships being described in detail in Section 4 and Section 5. The whole schema is divided into major categories of either vegetated (biotic) or non-vegetated (abiotic). The vegetated category is further categorized into natural and cultivated and non-vegetated into terrestrial non-vegetated and water. Concepts and their functional relationship are in the upper right corner of the schema, while the attributes for class expansion are in the grey boxes. The complete class list with descriptions is available in Appendix 1. Detailed steps are available in Appendix 4 for users to use this schema and create their own legend based on their requirements.

Figure 10. Schema of the national land cover reference system (LCRS) for Somalia



Source: Author's own elaboration.

# 7. Using the land cover reference system (LCRS)

## 7.1. A step-by-step guide to its practical utilization

The LCRS is a reference manual for land cover. Its function is to guide the user to better define the list and the thematic content of the thematic classes populating their specific legend. This process may vary from user to user, but it is important to understand that a mapping activity and its derived land cover legends are always local activities. In this context, "local" refers not only to the geographical scope of a future thematic map (regional, national, subnational, or very local) but also to various factors influencing its thematic content. These include the mapping purpose (such as agricultural, rangeland, urban, or forestry), the satellite data used and the methods for extracting land cover information, such as automatic classification or visual interpretation.

A single national legend cannot and will not fully accommodate all specific needs and approaches within a country. In this broader context, the LCRS serves as a flexible reference framework, helping the user determine the required number of classes and customize each class to suit their specific objectives. While offering this flexibility, the system ensures full compatibility and semantic interoperability across all derived legends. The following five steps outline how a hypothetical user can create a tailored legend while adhering to the system's logic.

### Step 1. Understanding the UML-like graphical class in schema

When creating land cover legend or land cover classes using ISO 19144-2/LCML (FAO and ISO, 2025), the user must familiarize themselves with the underlying logic of this standard. However, complex modularization is not required, as the LCRS already provides predefined metaclasses.

The key requirement is understanding LCML's basic logic and, more importantly, learning to interpret the graphical model representing each class (i.e. UML). The thematic content of each class is formalized using an object-based representation, which is visually depicted through a UML-like diagram following ISO 19144-2/LCML rules and its shared dictionary of objects. The LCML shared dictionary includes different object types, each governed by specific rules. A fundamental classification of these elements divides them into three main categories:

1. **Basic elements:** These are the fundamental components of the LCML UML model. A class cannot be created unless at least one of these core elements is included.
2. **Properties of the basic elements:** These are specific attributes that further refine and define the characteristics of basic elements. For example, Trees (basic element) can be more precisely described using properties such as Height, Cover, Leaf type, and Leaf phenology.
3. **Characteristics:** These are qualitative attributes that add specific meaning to a basic element. For example, Trees can have additional characteristics such as Vegetation artificiality (natural or cultivated), distinguishing whether they belong to a natural ecosystem or a managed landscape (such as Orchards or Palm plantations). Characteristics are classified into two types:

Figure 11. Graphical model of the Trees closed class modulization with profile sketch

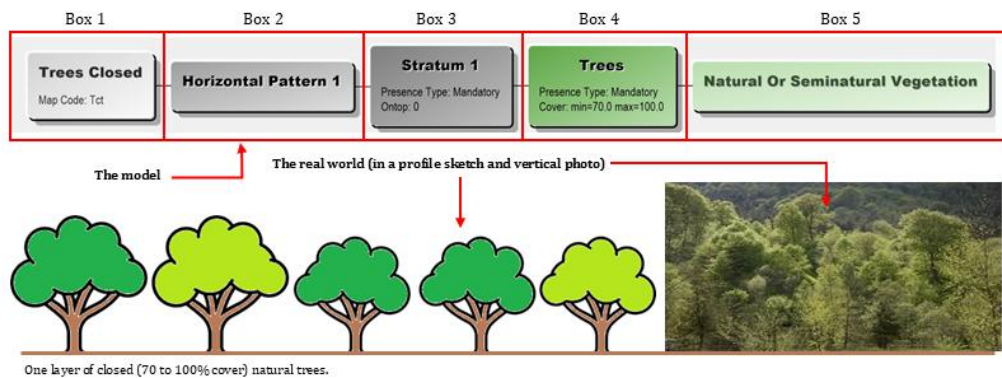


Photo © FAO.

Source: Author's own elaboration

- Basic element characteristics are directly linked to the object itself (such as vegetation artificiality for Trees).
- Class characteristics are attributes that apply to the entire class, such as climate, geographical aspects or soil type.

The graphical model for each class is designed to be self-explanatory, offering a structured and modern alternative to traditional, unstructured text-based class descriptions. Figure 11 is a simplified representation of the Trees closed class, which characterizes dense natural tree cover, typically found in forest areas. By systematically applying LCML logic, this feature is formally described, resulting in the graphical model shown in Figure 11.

The UML diagram in Figure 11 is very systematic and simple to understand and here uses the highlighted (red) five boxes to show how the diagram should be understood:

- Box 1 displays the class name and mapping code. Each class in the LCRS has a unique code, and it is recommended that the user maintains this code in their legend classifications.
- Box 2 represents an LCML element that is a horizontal pattern. The horizontal pattern in LCML logic serves as a specialized rule to describe the spatial distribution of land features in heterogeneous areas. For instance, in urban planning, when a detailed representation of an urban area is required, the horizontal pattern concept can be useful. An urban area comprises both linear features (such as roads or utility pipelines) and non-linear features, which may be built-up (such as buildings used for various purposes) or non-built-up (such as public squares). Additionally, extra urban elements like urban parks may need to be defined. If the class legend requires specifying the proportional distribution of these elements, different horizontal patterns can be applied.
- Box 3 defines strata or layers. In reality, land cover features, particularly vegetation, exhibit both horizontal and vertical characteristics. While horizontal aspects include plant cover,

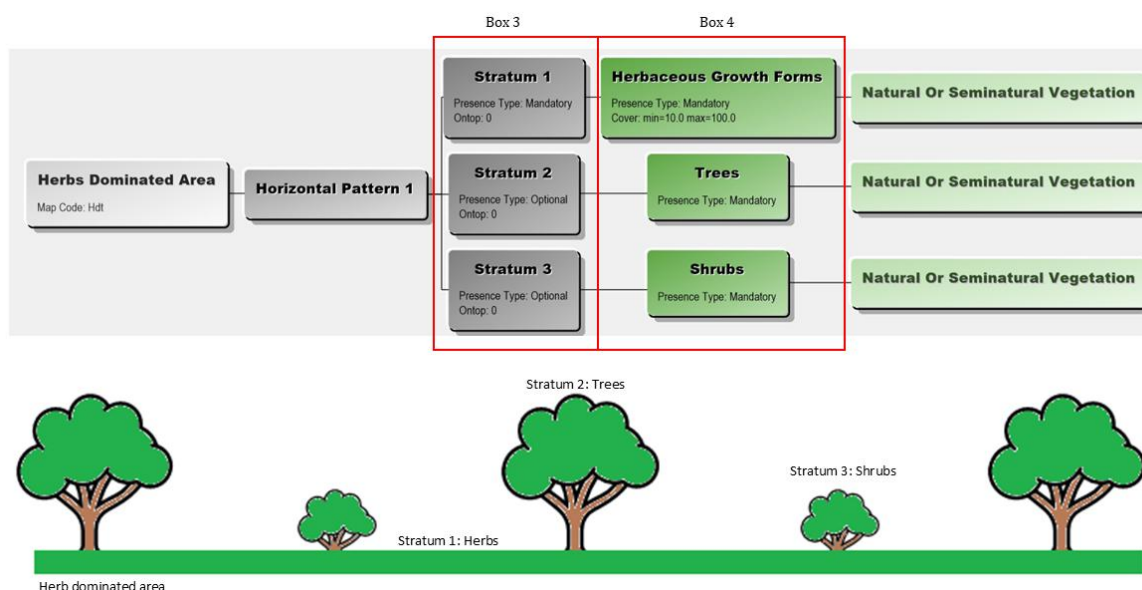
vertical aspects refer to the layering of vegetation. For example, an open tree formation may include an additional layer of small shrubs and another layer of herbaceous plants. The element "strata" is used to model this vertical structure. Here, the specific class Trees closed (Tct) consists of only one stratum (or plant layer).

4. Box 4 is very important as it defines the main basic element that characterizes the class. In this case, the defining element is trees, as well as including additional information:
  - a. **Presence type** defines whether the element is mandatory or optional. Here, the presence is mandatory, meaning that for the class Tct, the defining characteristic is that the Trees element must always be present.
  - b. **Properties** refer to specific attributes associated with the Trees element. Here, the relevant attribute is the cover percentage, indicating that the tree cover ranges from 70 to 100 percent.
5. Box 5 defines the characteristics which are additional qualitative attributes associated with the Trees basic element.

Following are three examples of different scenarios:

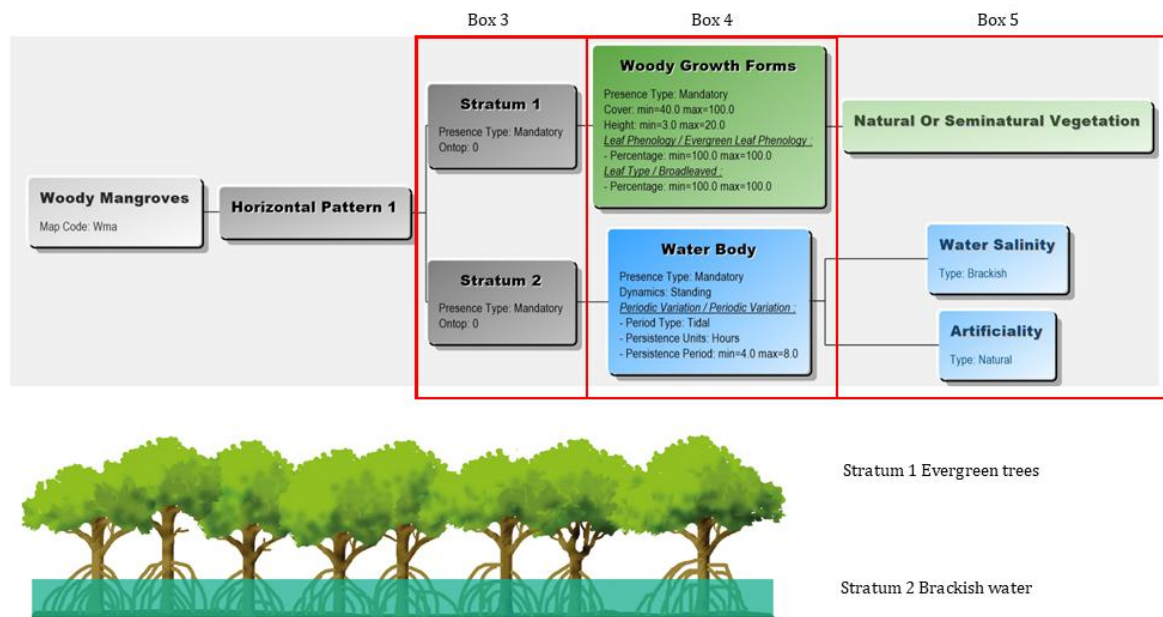
1. **An example with more than one strata** is seen in Figure 12, where three strata are shown in box 3, each one with its basic element and relative attributes in box 4. A Herbaceous-dominated area defines a land feature with a more complex vertical structure of natural vegetation and consists of multiple layers of vegetation growth forms:
  - a. a main layer of open to Closed herbaceous;
  - b. a second layer of Sparse trees; and
  - c. a third layer of Sparse shrubs.

Figure 12. Modelling of a complex vertical layering of natural vegetation



Source: Author's own elaboration

Figure 13. More complex modulization of mangrove area

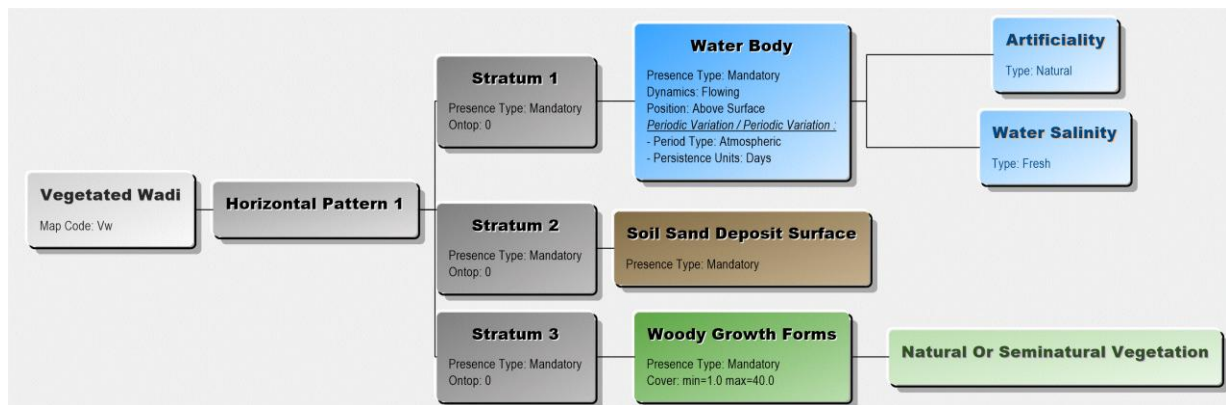


Source: Author's own elaboration

2. **An example with different strata combination** is seen in Figure 13, where two strata are shown in box 3, each with its basic element and related attributes in box 4. This land feature represents mangrove vegetation, characterized by salt-tolerant evergreen trees regularly flooded with brackish water, according to tidal movements with the two strata as follows:
  - a. **The first stratum** is shown with Trees as the primary element, where tree cover ranges between 40 and 100 percent, and all trees are Broadleaved evergreen. Trees are natural rather than cultivated (box 5).
  - b. **The second stratum** is shown with Water as the primary element, where surface water rises and falls every four to eight hours due to tidal movements. Water salinity is Brackish and Artificiality is Natural (box 5).
3. **A more complicated example is shown** in Figure 14 with an open woody vegetation formation on a wadi area. The model is constituted by three different strata:
  - a. **The first stratum** describes Water persistence including attributes such as Water position, Persistence units and Persistence period. Additionally, the Water element includes two extra attributes of Water salinity and Artificiality.
  - b. **The second stratum** represents Soil and sand deposit surface as mandatory, meaning the element is surely present.
  - c. **The third stratum** represents Woody growth forms, characterized by attributes related to Vegetation cover and Artificiality.



Figure 14. Open woody vegetation formation on a wadi area



Source: Author's own elaboration.

## Step 2: Selecting classes to populate the legend

Using the general schema (Figure 10), the user can choose the classes to include in the legend. The schema offers different levels of detail, ranging from broad classifications to more specific ones. The user has the flexibility to navigate the schema and, for example, select a highly detailed vegetation class alongside a more general agricultural class. It is important to note that these are metaclasses, which can later be further customized based on specific needs.

## Step 3: Understanding the selected classes

Once a set of classes has been selected, it is crucial for the user to fully grasp their meaning. This can be achieved by referring to Appendix 1, where a comprehensive description of each class is provided. Additionally, the appendix includes guidance on how the meaning of these metaclasses can be refined or expanded.

## Step 4: Downloading the class graphical model

If the selected classes meet the thematic requirements, the user should proceed to download the graphical class schema for each chosen class. This can be achieved by going to the FAO land cover legend registry (LCLR) where all the graphical schema of each listed classes are documented. These graphs will then constitute the descriptive section of the needed legend documentation (Mushtaq *et al.*, 2022).

To enhance accessibility for users unfamiliar with UML-like representations, each legend graph can be supplemented with additional text descriptions to clarify the thematic significance of the classes. Furthermore, an XSD file for each class will be available for download via LCLR.

## Step 5. Upgrading the class thematic content

if the thematic content of some LCRS metaclasses is not detailed enough and an upgrading is necessary, the user has two options:

1. Install the freeware LCCSv3 software (FAO, 2016), load in the software and the national LCRS legend and upgrade the class selected. When an original metaclass is modified, to ensure harmonization with any other legend derived from the same system, it is important to



maintain the original map code followed by an extra code symbol. For instance, the original map code for Closed trees is “Tct” (metaclass). If this metaclass needs to be upgraded with extra information not present into the LCRS list, an extra code between brackets must be added to the original code “Tct”. Therefore, the code of the upgraded original Closed trees class will be “Tct (aa)”, where “Tct” is the original code and “(aa)” is the extra code. A full description of the different steps to follow are documented in Appendix 3.

2. Contact the person in charge of the national LCRS in the FAO country office, who will support and facilitate the upgrading.

## 8. Conclusion

The Somalia LCRS represents a decisive step toward a more coordinated, data-driven approach to land governance and environmental management. It brings coherence to a historically fragmented landscape of classification systems, empowering institutions to generate, interpret and share land cover data with greater consistency and confidence.

Standardized geospatial information is now a strategic necessity rather than a technical luxury. With this system in place, Somalia is positioned to respond more effectively to climate risks, land degradation and competing resource demands. The ability to tailor land cover legends to local realities while remaining anchored in a nationally and internationally recognized framework ensures both flexibility and integrity in land cover monitoring efforts.

While FAO provided the technical leadership in its formulation, the LCRS is intended for national use, to serve as a shared standard for multiple stakeholders. Ministries, regional institutions, academic bodies, and development partners can now operate from a common reference point, ensuring that land cover data is not only interoperable, but actionable across sectors.

Successful implementation will require continued leadership, institutional commitment and cross-sectoral collaboration. Building capacity across government agencies and investing in systems for quality control, data integration and knowledge exchange will be essential to sustaining its relevance.

This reference system is not simply an end product: it is an enabling platform. It opens the door for more informed policies, targeted interventions and accountable land use planning. As Somalia advances toward its SDGs, the LCRS offers a shared foundation for protecting ecosystems, securing livelihoods and shaping a more resilient and equitable future for all.

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## Appendix 1: Object-based representation of the detailed metaclasses

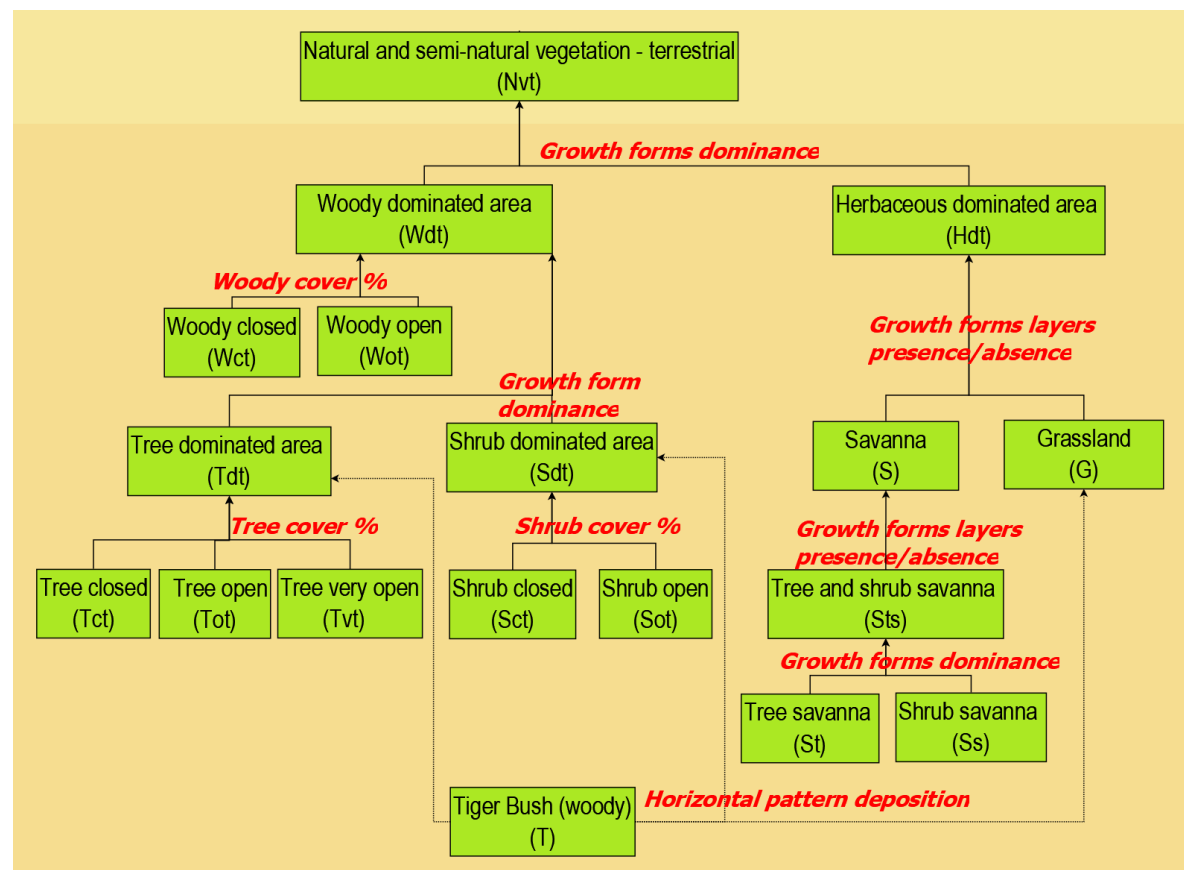
This section of the document focuses on object-based representation of the detailed metaclasses with UML.

### A1.1. Natural and semi-natural vegetation (terrestrial)

Map code/class code: Nvt

Description: This metaclass represents a subdivision of *Natural and semi-natural vegetation* categorized according to edaphic conditions. It includes all terrestrial land features covered by *Natural or semi-natural vegetation*. The structural breakdown is illustrated in Figure A1.1.

Figure A1.1. Structure of the *Natural and semi-natural vegetation (terrestrial)* class



Source: Author's own elaboration.

## Woody dominated area

Map code/class code: Wdt

Description: It corresponds to areas with natural and semi-natural vegetation where the woody growth form is dominant. This class is subdivided into two groups (derived classes at first level): group A and group B. Group A consists of two classes:

1. Woody closed area
2. Woody open area

These are based on different cover ranges from 10 to 100 percent characterizing the dominant Woody basic element. Group B consists of two classes:

1. Tree-dominated areas
2. Shrub-dominated areas

These are based on different growth form dominance characterizing the dominant Trees and Shrubs basic elements.

### *Woody closed*

Map code/class code: Wct

Description: This class is very basic and generic, being constituted by one mandatory stratum that defines the overall class structure. The stratum is constituted by the Trees basic element. The class can have extra strata of Shrubs (4 to 100 percent cover) and Herbaceous (4 to 100 percent cover) depending on the specific local situation (Figure A1.2). However, those eventual extra strata do not modify the main structure of the class. The Trees element has the following attributes:

- Cover ranges from 60 to 100 percent.
- Height ranges from 1.5 to 25 m.
- Vegetation artificiality is defined as natural.

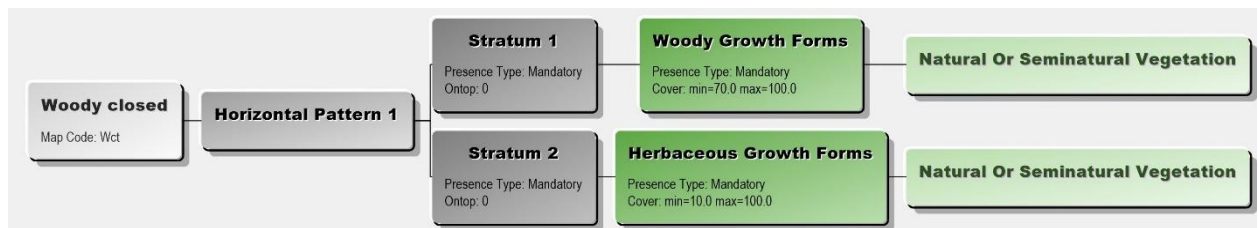
### *Woody open*

Map code/class code: Wot

Description: This class is very basic and generic, being constituted by one mandatory stratum that defines the overall class structure. The stratum is constituted by the Trees basic element. The class can have extra strata of Shrubs (4 to 100 percent cover) and Herbaceous (4 to 100 percent cover) depending on the specific local situation (Figure A1.3). However, those eventual extra strata do not modify the main structure of the class. The Trees element has the following attributes:

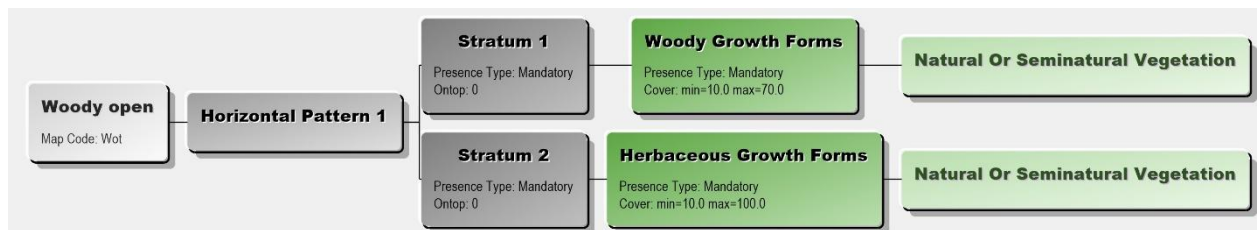
- Cover ranges from 60 to 100 percent.
- Height ranges from 1.5 to 25 m.
- Vegetation artificiality is defined as natural.

Figure A1.2. Unified modelling language (UML) diagram for the *Woody closed* metaclass



Source: Author's own elaboration.

Figure A1.3. Unified modelling language (UML) diagram for the *Woody open* metaclass



Source: Author's own elaboration.

### ***Tree-dominated area***

Map code/class code: Tdt

Description: It corresponds to areas with natural and semi-natural vegetation where the tree growth form is dominant (Figure A1.4). The three main classes are classified as closed, open and very open, according to the different cover ranges that characterize the dominant trees basic element.

### ***Trees closed***

Map code/class code: Tct

Description: This class is very basic, and generic being constituted by one mandatory stratum that defines the overall class structure. The stratum is constituted by the *Trees* basic element. The class can have extra strata of *Shrubs* (10 to 50 percent cover) and *Herbaceous* (10 to 100 percent cover) depending on the specific local situation (Figure A1.5). However, those eventual extra strata do not modify the main structure of the class. The *Trees* element has the following attributes:

- Cover ranges from 70 to 100 percent.
- Vegetation artificiality is defined as natural.

### ***Trees open***

Map code/class code: Tot

Description: This class is very basic and is constituted by an open formation of *Trees* ranging from 20 to 70 percent cover. One mandatory stratum is present and defines the overall class aspect. The stratum is constituted by the *Trees* basic element. The class can have extra strata of *Shrubs* (10 to 60 percent cover) and *Herbaceous* (10 to 100 percent cover), depending on a specific local situation (Figure A1.6). However, those eventual extra strata do not modify the main structure of the class. The *Trees* element has the following attributes:

- Cover ranges from 20 to 70 percent.
- *Vegetation artificiality* is defined as *Natural*.

### ***Trees very open***

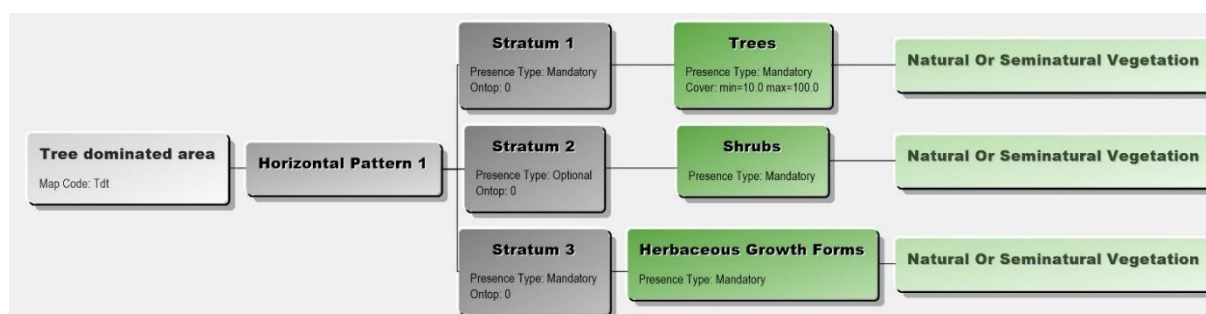
Map code/class code: Tvt

Description: This class is very basic and is characterized by a sparse formation of trees ranging from 20 to 70 percent cover. One mandatory stratum is present and defines the overall class aspect. The stratum is constituted by the *Trees* basic element. The class can have extra strata of *Shrubs* (10 to 60 percent cover) and *Herbaceous* (10 to 100 percent cover) depending on the specific local situation (Figure A1.7). However, those eventual extra strata do not modify the main structure of the class. The *Trees* element has the following attributes:

- Cover ranges from 20 to 70 percent.
- *Vegetation artificiality* is defined as *Natural*.

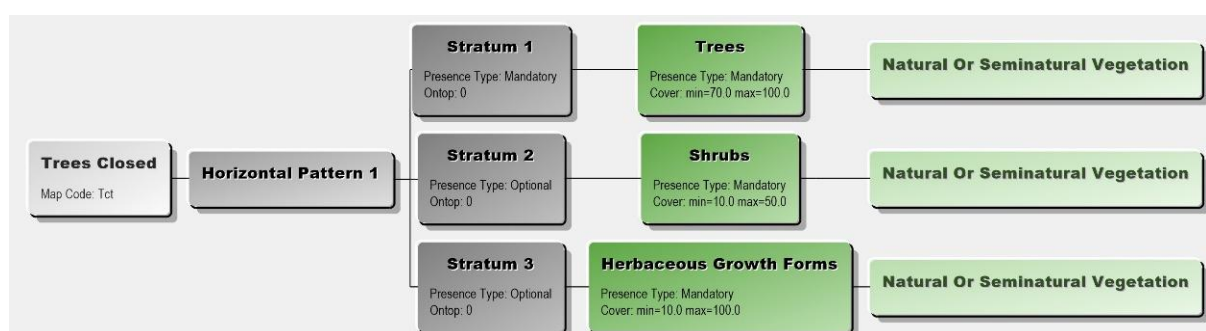


Figure A1.4. Unified modelling language (UML) diagram for the *Tree-dominated area* metaclass



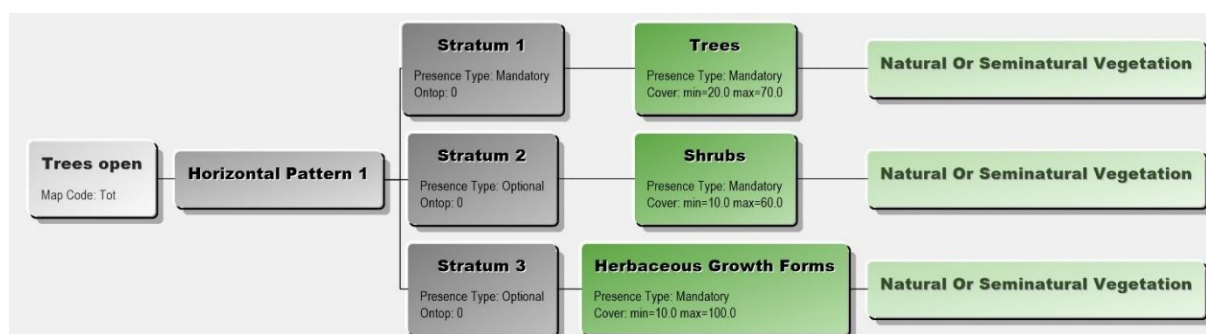
Source: Author's own elaboration.

Figure A1.5. Unified modelling language (UML) diagram for the *Trees closed* metaclass



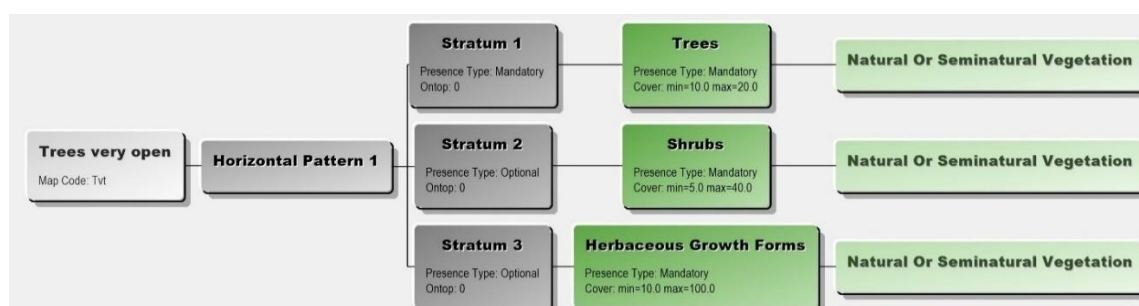
Source: Author's own elaboration.

Figure A1.6. Unified modelling language (UML) diagram for the *Trees open* metaclass



Source: Author's own elaboration.

Figure A1.7. Unified modelling language (UML) diagram for the *Trees very open* metaclass



Source: Author's own elaboration.

### ***Shrub-dominated area***

Map code/class code: Sdt

Description: It corresponds to areas with natural semi-natural vegetation where the growth form shrub is dominant. The Shrubs element ranges from 10 to 100 percent cover (Figure A1.8). The presence of other woody (Shrubs) or non-woody (Herbaceous) growth forms can be present according to the dominance rules.

### ***Shrubs closed***

Map code/ class code: Sct

Description: The class is characterized by a formation of natural shrubs ranging from 70 to 100 percent cover. One mandatory stratum is present and defines the overall class structure and constituted by the Shrubs basic element. The class can have extra strata of Trees (1 to 10 percent cover) and Herbaceous (10 to 100 percent cover) depending on the specific local situation (Figure A1.9). However, those eventual extra strata do not modify the main structure of the class. The Shrubs element has the following attributes:

- Cover ranges from 70 to 100 percent.
- Height ranges from 1 to 5 m.
- Vegetation artificiality is defined as natural.

### ***Shrubs open***

Map code/class code: Sot

Description: The class is characterized by a formation of natural shrubs with a cover ranging from 10 to 70 percent. Two mandatory strata are present and define the overall class structure. The first stratum is constituted by the Shrubs basic element, while the second stratum is constituted by the Herbaceous growth form or Soil or sand deposit surface basic element. The class can have an extra stratum of Trees (cover from 1 to 10 percent) depending on the specific local situation (Figure A1.10). However, those eventual extra strata are only marginal with a negligible cover (especially for the Trees element) and do not modify the main structure of the class. The Shrubs element has the following attributes:

- Cover ranges from 10 to 70 percent.
- Height ranges from 1 to 5 m.
- Vegetation artificiality is defined as natural.

The second stratum contains elements with a temporal sequence relationship between the Herbaceous and Soil or sand deposit surface elements. The Herbaceous element has the following attributes:

- Presence type depends on temporal sequence.
- Sequential period is within the same year with a length of four to eight months.

- Vegetation artificiality is defined as Natural.

The Sand and soil deposit surface element has the following attributes:

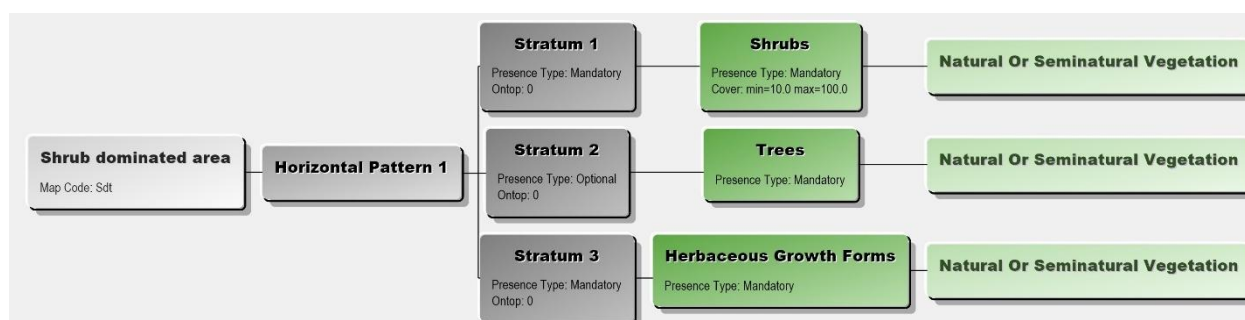
- Presence type depends on temporal sequence.
- Sequential period is within the same year with a length of four to eight months.

## Class expansion for Woody dominated area

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

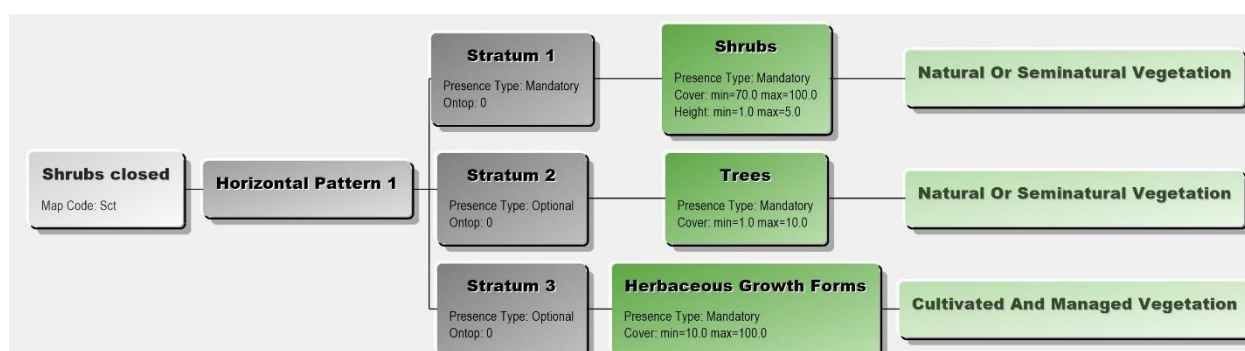
- Physiognomic structural properties, such as:
  - **Height:** The specific height can be used to better characterize the Trees basic element.
  - **Adding different strata of growth forms:** In addition to the main strata characterizing this class, other strata with different growth forms can be added.
  - **Leaf phenology:** The specific leaf phenology for each growth form can be defined.
  - **Leaf type:** The specific leaf type for each growth form can be added.
- Other attributes such as:
  - **Floristic name:** The floristic name of the plant species.
  - **Growth form illness:** The percentage of trees damaged and the illness type.
  - **Burnt status:** The total percentage of trees burnt in the specific area.
  - **Tree area management practices:** The types of forest management undertaken in a specific forest.
  - **Geographic location or altitude:** Any information related to a specific geographic location or altitude ranges.
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.8. Unified modelling language (UML) diagram for the *Shrub-dominated area* metaclass



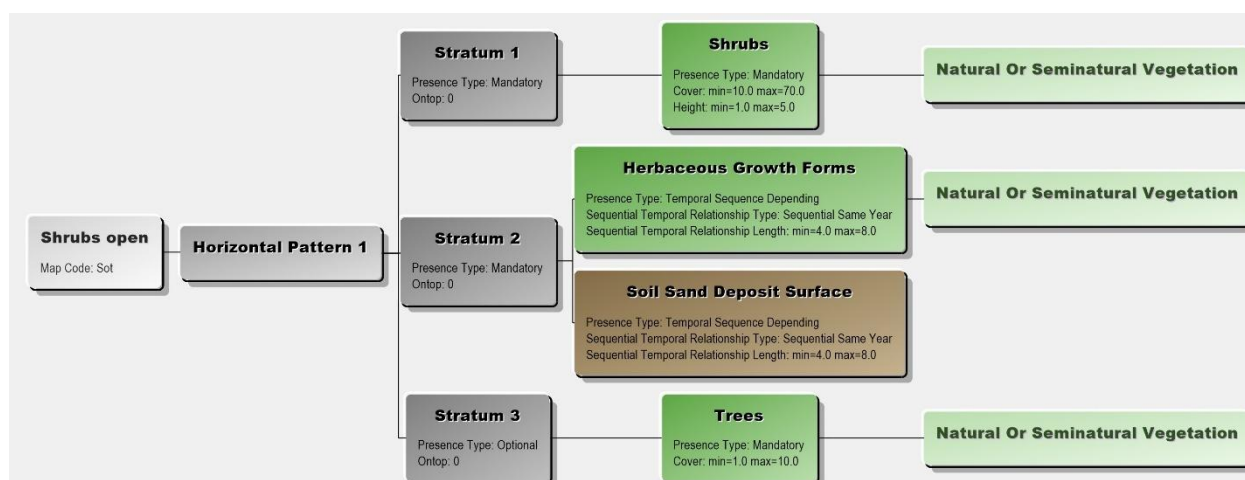
Source: Author's own elaboration.

Figure A1.9. Unified modelling language (UML) diagram for the *Shrubs closed* metaclass



Source: Author's own elaboration.

Figure A1.10. Unified modelling language (UML) diagram for the *Shrubs open* metaclass



Source: Author's own elaboration.

## Herbaceous-dominated area

Map code/class code: Hdt

Description: This level corresponds to an area of natural vegetation with the mandatory presence of *Herbaceous* growth covering 10 to 100 percent of the area.

## Savanna

Map code/class code: S

Description: The class is characterized by a main formation of natural Herbaceous growth forms with scattered woody plants. Two mandatory strata are present, and they define the overall class structure. The first stratum is constituted by the herbaceous basic element in temporal sequence relationship with the Soil or sand deposit surface basic element. The second stratum is constituted by the woody element (Figure A1.11). The term Woody is used when it is not possible or not worth differentiating between the more specific trees and shrubs terms. The Herbaceous stratum has the following attributes:

- Cover ranges from 20 to 100 percent.
- Presence type is a temporal sequence between four and eight months long.
- Vegetation artificiality is defined as natural.
- Height is from 1 to 50 cm.

The Woody stratum has two attributes:

- Cover ranges from 2 to 10 percent.
- Vegetation artificiality is defined as natural.

## Tree and shrub savanna

Map code/class code: Sts

Description: This class defines the generic aspect of tree and shrub savanna. It is constituted by three mandatory strata that define the overall class structure. The first stratum is constituted by one Herbaceous basic element growth form (Figure A1.12) with the following attributes:

- Cover ranges from 20 to 60 percent.
- Presence type is a temporal sequence between four and eight months long.
- Height ranges from 1 to 50 cm.
- Vegetation artificiality is defined as natural.

The second stratum of the Shrubs basic element has the following attributes:

- Cover ranges from 2 to 10 percent.
- Height ranges from 2 to 5 m.

- Vegetation artificiality is defined as natural.

The third stratum of the Trees basic element has the following attributes:

- Cover ranges from 2 to 10 percent.
- Height ranges from 1 to 5 m.
- Vegetation artificiality is defined as natural.

### ***Tree savanna***

Map code/class code: St

Description: This class defines the generic aspect of Tree savanna. It is constituted by two mandatory strata that define the overall class structure. The first stratum is constituted by one Herbaceous basic element in temporal sequential relationship with a Soil or sand deposit surface element (Figure A1.13). The Herbaceous element has the following attributes:

- Cover ranges from 20 to 100 percent.
- Presence type is a temporal sequence between four and eight months long.
- Height ranges from 1 to 50 cm.
- Vegetation artificiality is defined as natural.

The second stratum of the Trees basic element is present with the following attributes:

- Cover ranges from 2 to 10 percent.
- Height ranges from 2 to 10 m.
- Vegetation artificiality is defined as natural.

### ***Shrub savanna***

Map code/class code: Ss

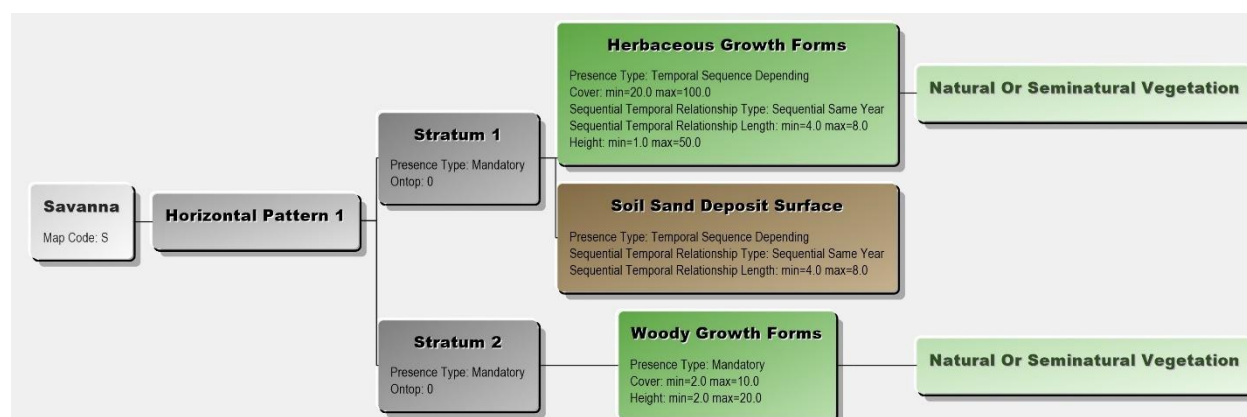
Description: This class defines the generic aspect of *Shrub savanna*. It is constituted by two mandatory strata that define the overall class structure. The first stratum is constituted by one Herbaceous basic element in a temporal sequential relationship with a soil or sand deposit surface element (Figure A1.14). The Herbaceous element has the following attributes:

- Cover ranges from 20 to 100 percent.
- Presence type is a temporal sequence between four and eight months long.
- Height ranges from 1 to 50 cm.
- Vegetation artificiality is defined as natural.

The second stratum of the Shrubs basic element is present with the following attributes:

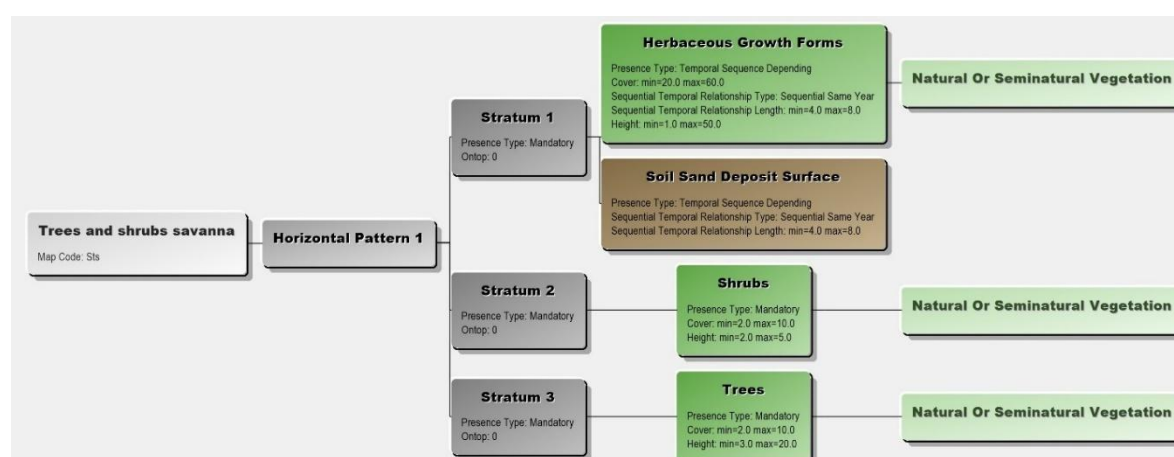
- Cover ranges from 2 to 10 percent.
- Height ranges from 2 to 5 m.
- Vegetation artificiality is defined as natural.

Figure A1.11. Unified modelling language (UML) diagram for the *Savanna* metaclass



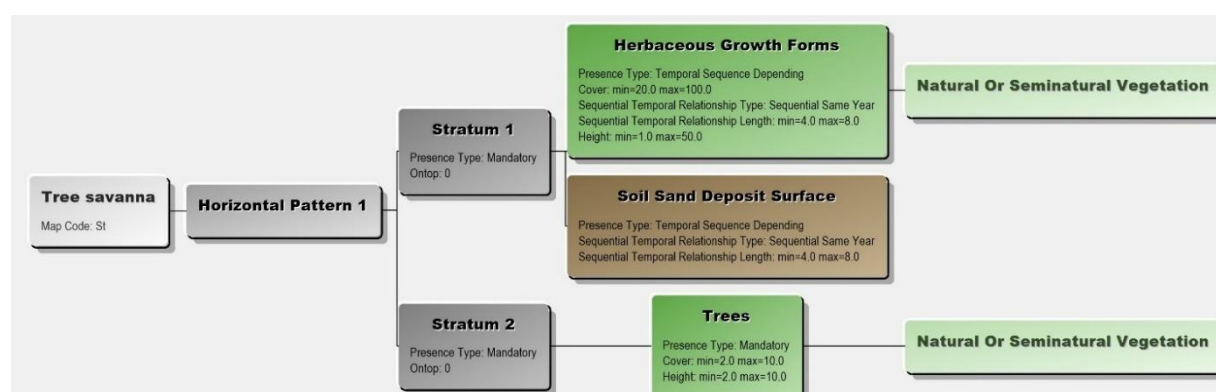
Source: Author's own elaboration.

Figure A1.12. Unified modelling language (UML) diagram for the *Tree and shrub savanna* metaclass



Source: Author's own elaboration.

Figure A1.13. Unified modelling language (UML) diagram for the *Tree savanna* metaclass

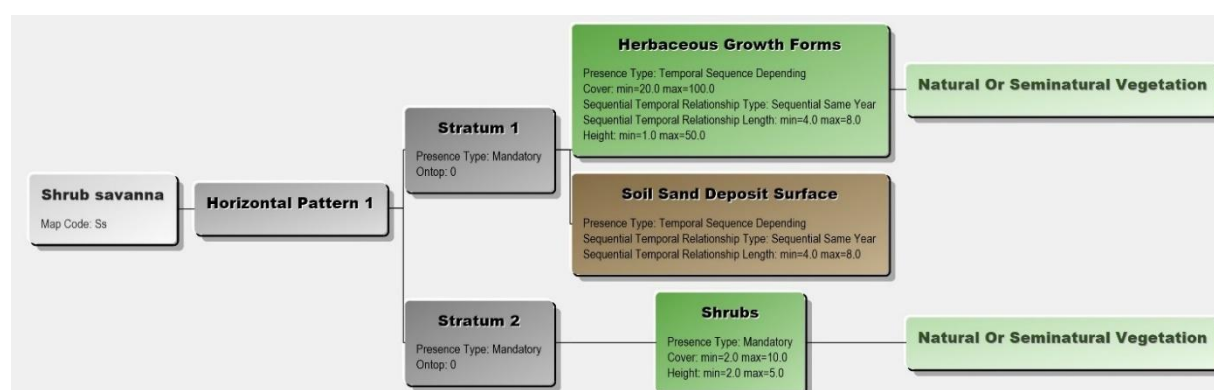


Source: Author's own elaboration.

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Figure A1.14. Unified modelling language (UML) diagram for the *Shrub savanna* metaclass



Source: Author's own elaboration.

## Grassland

Map code/class code: G

Description: This class is characterized by a natural herbaceous cover. The herbaceous cover can range from sparse to open and closed formation. The herbaceous vegetation should persist in the area for at least two months a year. No other types of vegetative Woody growth forms are present. One mandatory stratum is present and defines the overall class structure. The stratum is constituted by the Herbaceous basic element in temporal sequential relationship with a Soil or sand deposit surface element (Figure A1.15). The Herbaceous element has the following attributes:

- Cover ranges from 10 to 100 percent.
- Presence type is a temporal sequence between four and eight months long.
- Vegetation artificiality is defined as natural.

## Tiger bush woody

Map code/class code: T

Description: This class defines the generic aspect of *Tiger bush woody*. It is constituted by two mandatory horizontal patterns and one basic stratum that defines the overall class structure. In the first horizontal pattern (30 to 40 percent cover) the four strata are constituted from three basic elements of Shrubs, Herbaceous and Trees (Figure A1.16) as follows:

For the first stratum with the first Shrubs basic element:

- Cover ranges from 10 to 50 percent.
- Height ranges from 0.8 to 5 m.
- Vegetation artificiality is defined as natural.

For the second stratum with the Herbaceous growth form basic element:

- Cover ranges from 10 to 60 percent.



- Vegetation artificiality is defined as natural.

For the third stratum with the Trees basic element:

- Cover ranges from 5 to 40 percent.
- Vegetation artificiality is defined as natural.

For the fourth stratum with the second Shrubs basic element:

- Cover ranges from 10 to 40 percent.
- Height ranges from 0.3 to 0.8 m.
- Vegetation artificiality is defined as natural.

In the second horizontal pattern (30 to 70 percent cover), one basic mandatory stratum has a presence of bare soil.

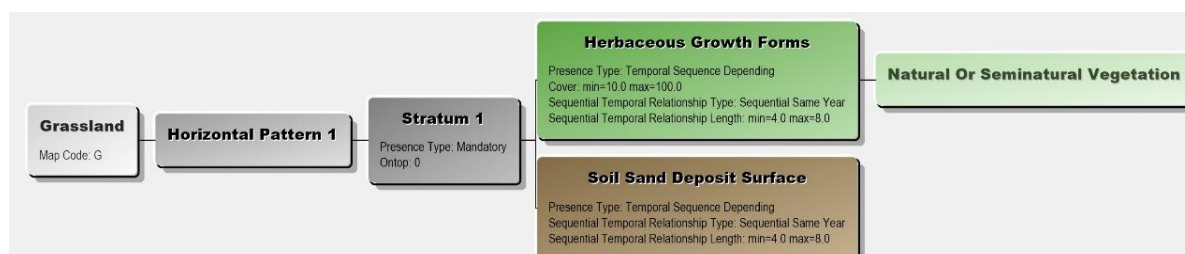
## Class expansion for Herbaceous-dominated area

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

- Physiognomic structural properties:
  - **Height:** A more specific height can be used to better characterize the Herbaceous basic element.
  - **Cover:** A more specific cover range can be defined.
  - **Leaf phenology:** The specific leaf phenology for each growth form can be defined.
  - **Adding different strata of growth forms:** In addition to the main strata characterizing this class, other strata with different growth forms can be added.
- Other attributes:
  - **Floristic name:** The floristic name of the plant species.
  - **Burnt status:** The percentage of herbaceous plants burnt over the total area.
  - **Grazing:** The percentage of the intensity of grazing can be added for the Herbaceous growth form basic element.
  - **Geographic location or altitude:** Any information related to a specific geographic location and or altitude ranges.

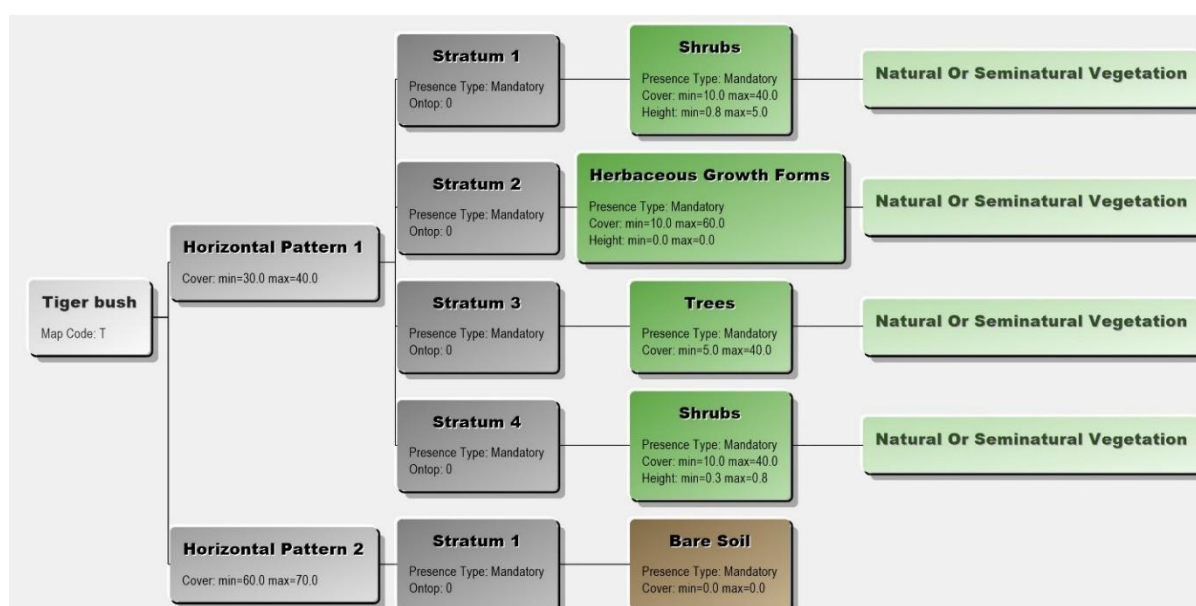
- **Topographical aspects:** any aspects related to the exposition of a slope or its slope gradient.
- **Any other attributes:** other attributes can be derived from the LCML shared dictionary.

Figure A1.15. Unified modelling language (UML) diagram for the *Grassland* metaclass



Source: Author's own elaboration.

Figure A1.16. Unified modelling language (UML) diagram for the *Tiger bush* woody metaclass



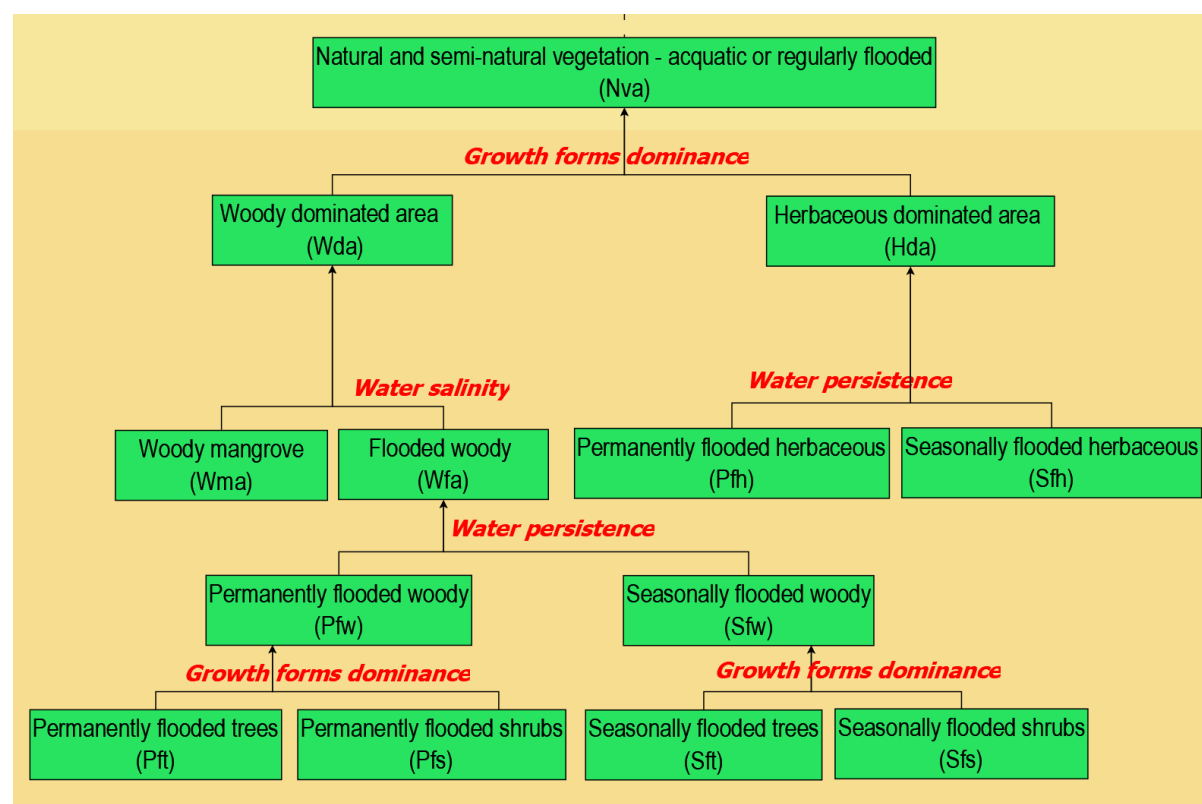
Source: Author's own elaboration.

## A1.2. Natural and semi-natural vegetation (aquatic or regularly flooded)

Map code/class code: Nva

Description: This metaclass represents a subdivision of Natural and semi-natural vegetation categorized according to edaphic conditions. It includes all aquatic land features covered by Natural or semi-natural vegetation. The structural breakdown is illustrated in Figure A1.17.

Figure A1.17. Structure of *Natural and semi-natural vegetation (aquatic or regularly flooded)* class



Source: Author's own elaboration.

### Woody dominated area

Map code/class code: Wda

Description: It corresponds to areas with natural and semi-natural vegetation where the woody growth form is dominant. The woody element ranges from 10 to 100 percent cover (Figure A1.18). The non-vegetated areas such as waterbody is present.

#### Woody mangroves

Map code/class code: Wma

Description: The class is characterized by a formation of natural woody plants along the coastal area regularly flooded by brackish water, according to tidal effects. The cover at this class level is very

generic, varying from sparse to closed. Two mandatory strata are present and define the overall class structure (Figure A1.19). The first stratum is constituted by the Woody basic element with the following attributes:

- Cover ranges from 40 to 100 percent.
- Leaf type is defined as broadleaved.
- Leaf phenology is defined as evergreen.
- Vegetation artificiality is defined as natural.

The second stratum is constituted by the Water basic element with the following attributes:

- Water persistence period is four to eight hours.
- Water period type is defined as Tidal.
- Water salinity is defined as Brackish.
- Water position is defined as Above ground.
- Water artificiality is defined as Natural.
- Water dynamics is defined as standing.

### ***Flooded woody***

Map code/class code: Wfa

Description: The class is the representation of a generic situation where woody vegetation formations are influenced by a seasonal or intermittent flooding of fresh water. The plant cover at this class level is very generic varying from sparse to closed. Two mandatory strata are present and define the overall class structure (Figure A1.20). The first stratum is constituted by the Woody basic element with the following attributes:

- Cover ranges from 10 to 100 percent.
- Vegetation artificiality is defined as natural.

The second stratum is constituted by the water basic element with the following attributes:

- Water salinity is defined as fresh.
- Water position is defined as above ground.
- Water artificiality is defined as natural.

### ***Permanently flooded woody***

Map code/class code: Pfw

Description: This class refers to the woody vegetation that is flooded regularly during the year and which can persist for a long period of time. Two mandatory strata are present and define the overall class structure (Figure A1.21). The first stratum is constituted by the Woody basic element with the following attributes:

- Cover ranges from 10 to 100 percent.
- Vegetation artificiality is defined as Natural.
- Water persistence unit is Months.

The second stratum is constituted by the water basic element with the following attributes:

- Water salinity is defined as Fresh.
- Water position is defined as Above ground.
- Water artificiality is defined as Natural.

### ***Permanently flooded trees***

Map code/class code: Pft

Description: This class refers to the tree vegetation that is flooded regularly during the year and which can persist for a long period of time. The water persistence period is set up in months. Two mandatory strata are present and define the overall class structure (Figure A1.22). The first stratum is constituted by the Trees basic element with the following attributes:

- Cover ranges from 10 to 100 percent.
- Vegetation artificiality is defined as Natural.

The second stratum is constituted by the Water basic element with the following attributes:

- Water salinity is defined as Fresh.
- Water dynamics is defined as Standing.
- Water position is defined as Above ground.
- Water artificiality is defined as Natural.

### ***Permanently flooded shrubs***

Map code/class code: Pfs

Description: This class refers to shrub vegetation that is flooded regularly during the year. The flooding can persist for a long period of time so the water persistence period is measured in months. Two mandatory strata are present and define the overall class structure (Figure A1.23). The first stratum is constituted by the Shrubs basic element with the following attributes:

- Cover ranges from 10 to 100 percent.
- Vegetation artificiality is defined as Natural.

The second stratum is constituted by the Water basic element with the following attributes:

- Water salinity is defined as Fresh.
- Water dynamics is defined as Standing.

- *Water position* is defined as Above ground.
- Water artificiality is defined as Natural.

### ***Seasonally flooded woody***

Map code/class code: Sfw

Description: This class refers to the woody vegetation that is flooded regularly during the year. The flooding is often associated with the rainy season, so the water persistence period is measured in months. Two mandatory strata are present and define the overall class structure (Figure A1.24). The first stratum is constituted by the woody basic element with the following attributes:

- Cover ranges from 10 to 100 percent.
- Vegetation artificiality is defined as natural.

The second stratum is constituted by the Water basic element with the following attributes:

- Water salinity is defined as fresh.
- Water position is defined as above ground.
- Water artificiality is defined as natural.

### ***Seasonally flooded trees***

Map code/class code: Sft

Description: This class refers to tree vegetation that is flooded regularly during the year. The flooding is often associated with the rainy season, so the water persistence period is measured in months. Two mandatory strata are present and define the overall class structure (Figure A1.25). The first stratum is constituted by the trees basic element with the following attributes:

- Cover ranges from 20 to 100 percent.
- Vegetation artificiality is defined as natural vegetation.

The second stratum is constituted by the water basic element with the following attributes:

- Water salinity is fresh.
- Water dynamics is defined as standing.
- *Water position* is defined as above ground.
- Water artificiality is defined as natural.

### ***Seasonally flooded shrubs***

Map code/class code: Sfs

Description: This class refers to shrubby vegetation that is flooded regularly during the year. The flooding is often associated with the rainy season, so the water persistence period is measured in

months. The water persistence period is set up in months. Two mandatory strata are present and define the overall class structure (Figure A1.26). The first stratum is constituted by the Shrubs basic element with the following attributes:

- Cover ranges from 20 to 100 percent.
- Vegetation artificiality is defined as natural.

The second stratum is constituted by the water basic element with the following attributes:

- Water salinity is fresh.
- Water dynamics is defined as standing.
- Water position is defined as above ground.
- Water artificiality is defined as natural.

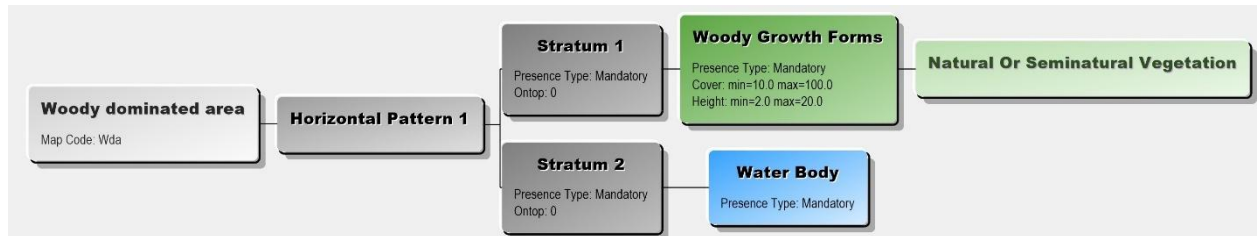
## Class expansion for Woody dominated area (aquatic or regularly flooded)

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

- Physiognomic structural properties:
  - **Height:** A more specific height can be used to better characterize the Trees basic element.
  - **Leaf phenology:** The specific leaf phenology for each growth form can be defined.
  - **Leaf type:** The specific leaf type for each growth form can be added.
  - **Adding different strata of growth forms:** In addition to the main strata characterizing this class, other strata with different growth forms can also be added.
  - **Waterbody:** Water artificiality and Salinity type can also be added.
  - **Water periodic variation:** Period type and Persistence period can also be added.
- Other attributes:
  - **Floristic name:** The floristic name of the plant species.
  - **Growth form illness:** The percentage of trees damaged and the type of disease.
  - **Burnt status:** The percentage of trees burnt over the total area.
  - **Tree area management practices:** The types of forest management undertaken in a specific forest area.

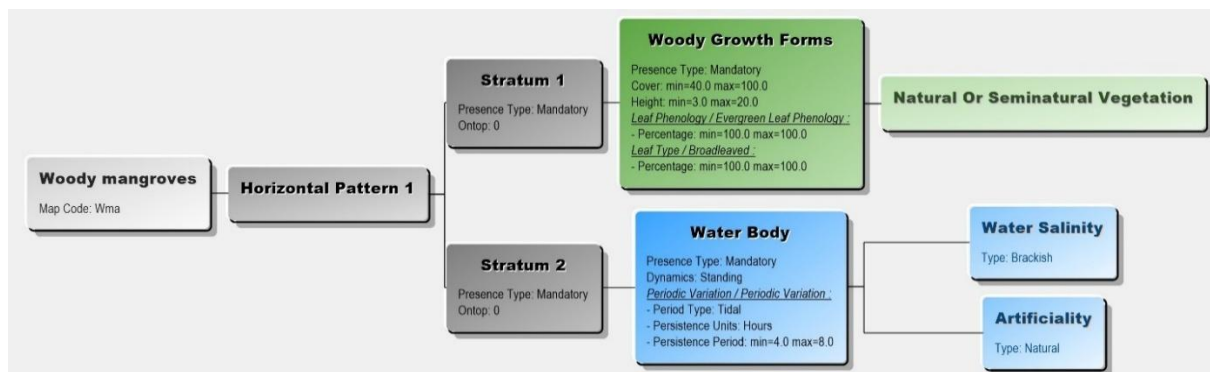
- **Geographic location or altitude:** Any information related to a specific geographic location and or altitude ranges.
- **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.18. Unified modelling language (UML) diagram for the *Woody dominated area* metaclass



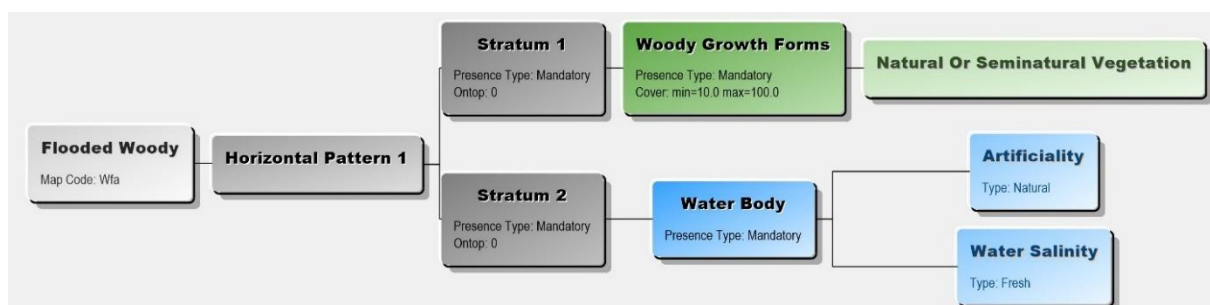
Source: Author's own elaboration.

Figure A1.19. Unified modelling language (UML) diagram for the *Woody mangroves* metaclass



Source: Author's own elaboration.

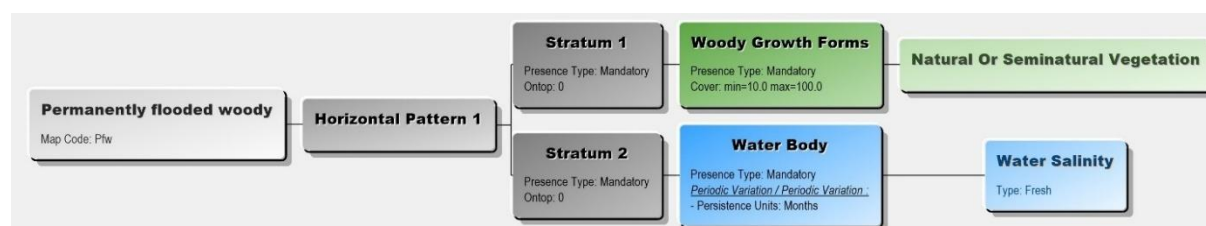
Figure A1.20. Unified modelling language (UML) diagram for the *Flooded woody* metaclass



Source: Author's own elaboration.

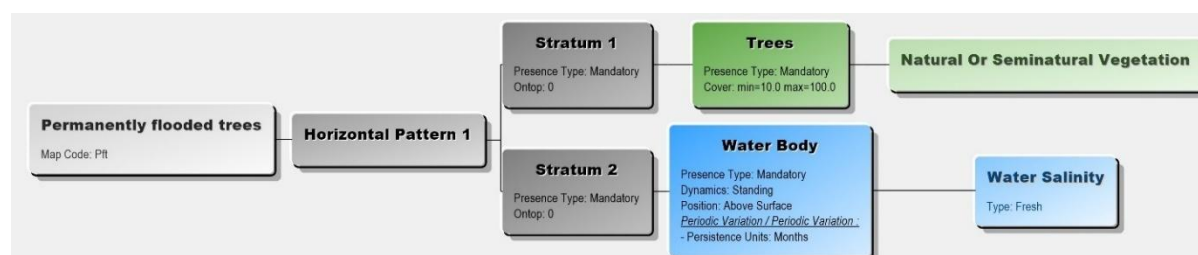


Figure A1.21. Unified modelling language (UML) diagram for the *Flooded woody* metaclass



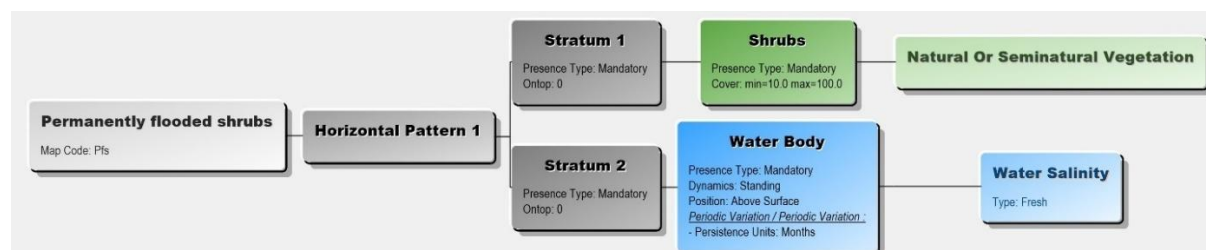
Source: Author's own elaboration.

Figure A1.22. Unified modelling language (UML) diagram for the *Permanently flooded trees* metaclass



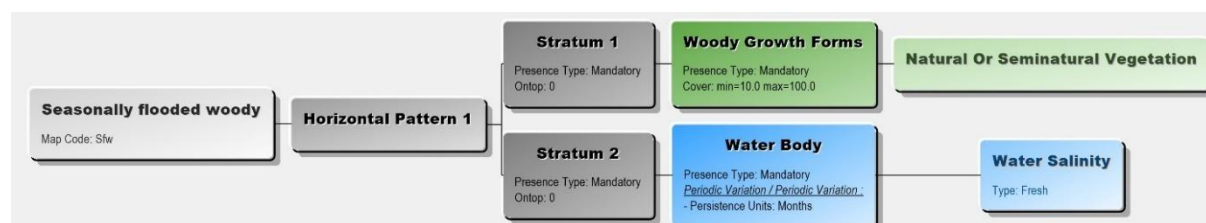
Source: Author's own elaboration.

Figure A1.23. Unified modelling language (UML) diagram for the *Permanently flooded shrubs* metaclass



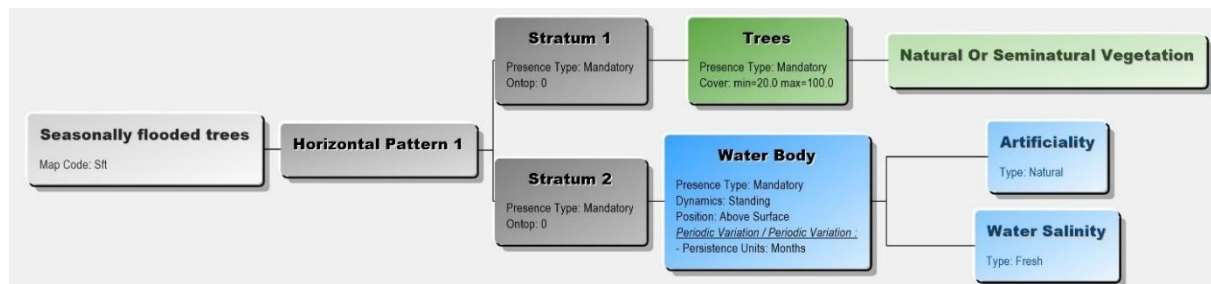
Source: Author's own elaboration.

Figure A1.24. Unified modelling language (UML) diagram for the *Seasonally flooded woody* metaclass



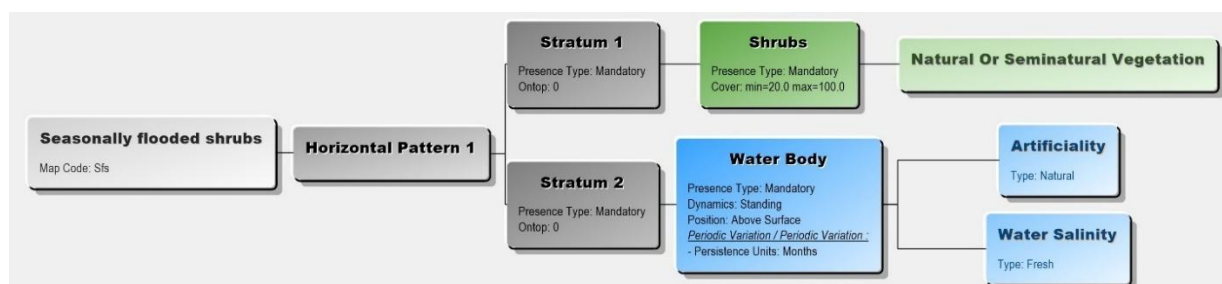
Source: Author's own elaboration.

Figure A1.25. Unified modelling language (UML) diagram for the *Seasonally flooded trees* metaclass



Source: Author's own elaboration.

Figure A1.26. Unified modelling language (UML) diagram for the *Seasonally flooded shrubs* metaclass



Source: Author's own elaboration.

## Herbaceous-dominated areas

Map code/class code: Hda

Description: This level corresponds to permanently or seasonally flooded areas where there is only herbaceous vegetation present (Figure A1.27). The *Woody* growth forms are absent or sporadic.

### *Permanently flooded herbaceous*

Map code/class code: Pfh

Description: This class refers to herbaceous vegetation that is flooded regularly during the year. The flooding can persist over a long period of time, so the water persistence period is measured in months. Two mandatory strata are present and define the overall class structure (Figure A1.28). The first stratum is constituted by the Herbaceous basic element with the following attributes:

- Cover ranges from 60 to 100 percent.
- Vegetation artificiality is defined as natural.

The second stratum is constituted by the water basic element with the following attributes:

- Water salinity is defined as fresh.
- Water position is defined as above ground.
- Water artificiality is defined as natural.

### *Seasonally flooded herbaceous*

Map code/class code: Sfh

Description: This class refers to woody vegetation that is flooded regularly during the year. The flooding is often associated with the rainy season, so the water persistence period is measured in months. Two mandatory strata are present and define the overall class structure (Figure A1.29). The first stratum is constituted by the Herbaceous basic element with the following attributes:

- Cover ranges from 10 to 100 percent.
- Vegetation artificiality is defined as Natural.

The second stratum is constituted by the Water basic element with the following attributes:

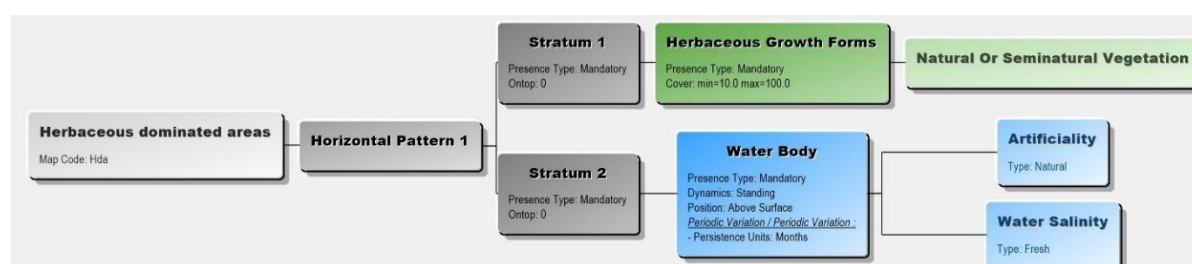
- Water salinity is defined as fresh.
- Water position is defined as above ground.
- Water artificiality is defined as natural.

## Class expansion for Herbaceous-dominated area

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (such as legend scope and terrain knowledge) need different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

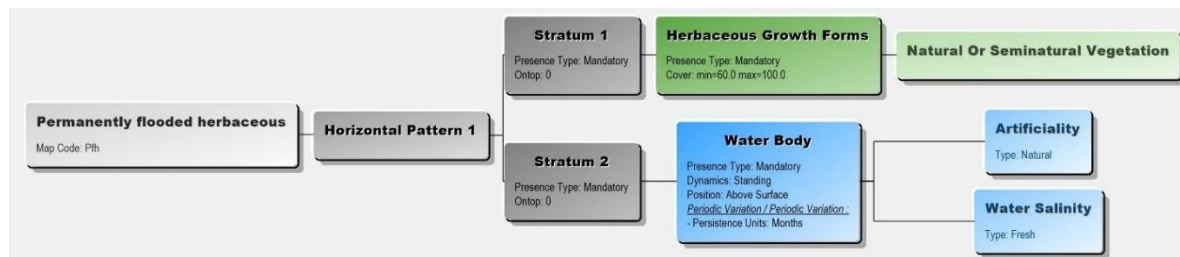
- Physiognomic structural properties:
  - **Height:** A more specific height can be used to better characterize the Herbaceous basic element.
  - **Leaf phenology:** A specific leaf phenology for each growth form can be defined.
  - **Adding different strata of growth forms:** In addition to the main strata characterizing this class, other strata with different growth forms can be added.
  - **Water periodic variation:** The period type and persistence period can be added.
- Other attributes:
  - **Floristic name:** The floristic name of the plant species.
  - **Grazing:** The percentage of the intensity of grazing can be added for Herbaceous.
  - **Geographic location or altitude:** Any information related to a specific geographic location and or altitude ranges.
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.27. Unified modelling language (UML) diagram for the *Herbaceous-dominated areas* metaclass



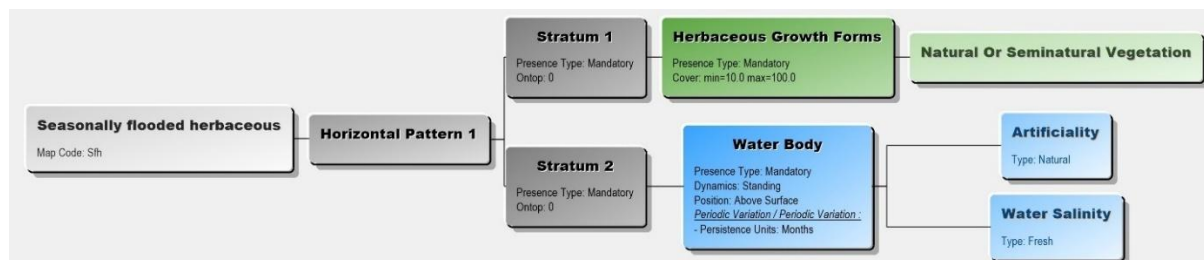
Source: Author's own elaboration.

Figure A1.28. Unified modelling language diagram for *Permanently flooded herbaceous* metaclass



Source: Author's own elaboration.

Figure A1.29. Unified modelling language (UML) diagram for *Seasonally flooded herbaceous* metaclass



Source: Author's own elaboration.

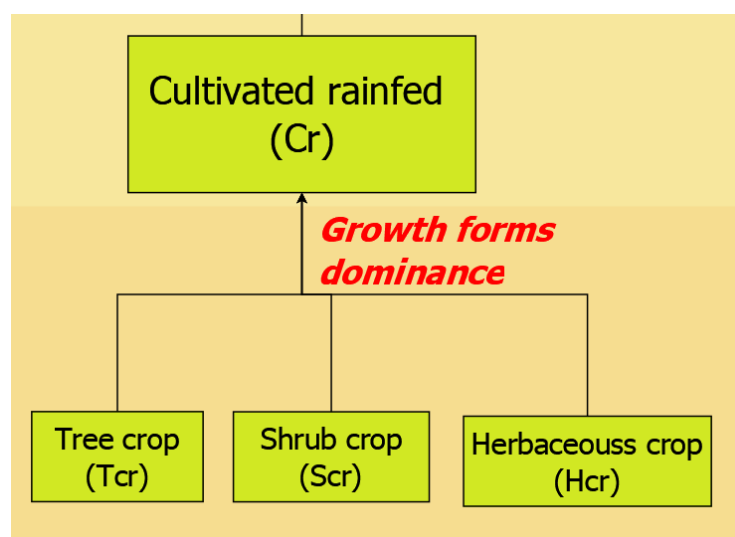
○

## A1.3. Cultivated rainfed

Map code/class code: Cv

Description: This metaclass represents a subdivision of *Cultivated and managed vegetation* categorized according to water supply. It includes rainfed cultivation exists in Somalia. The structural breakdown is illustrated in Figure A1.30.

Figure A1.30. Structure of the *Cultivated rainfed* class



Source: Author's own elaboration.

### Tree crop

Map code/ class code: Tcr

Description: This metaclass is a generic representation of rainfed crops with a tree basic element. The plants are totally reliant on rainfall. One mandatory stratum is present and defines the overall class structure (Figure A1.31) with the following attributes:

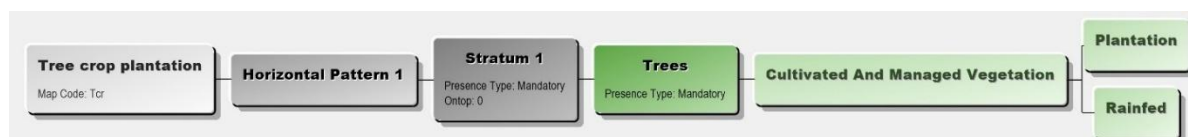
- Vegetation artificiality is defined as cultivated and managed vegetation.
- Water supply is defined as rainfed.

### Class expansion for Tree crop

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

- Physiognomic structural properties:
  - **Height:** A more specific height can be used to better characterize the Trees basic element.
  - **Leaf phenology:** A specific leaf phenology for each growth form can be defined.
  - **Leaf type:** A specific leaf type for each growth form can be added.
  - **Adding different strata of growth forms:** In addition to the main strata characterizing this class, other strata with different growth forms can be added.
- Other attributes:
  - **Floristic name:** The floristic name of the plant species.
  - **Geographic location or altitude:** Any information related to a specific geographic location and or altitude ranges.
  - **Plant spreading geometry:** The geometry of the spreading plant can be added.
  - **Field size:** The field size for Cultivated rainfed can be added.
  - **Irrigation type and percentage:** Any of several means of providing an artificial regular supply of water (in addition to rain) to the crop(s), along with the water percentage can be added.
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.31. Unified modelling language (UML) diagram for the *Tree crop* metaclass



Source: Author's own elaboration.

## Shrub crop

Map code/ class code: Scr

Description: This metaclass is a generic representation of rainfed crops with a shrubs basic element. The plants are totally reliant on rainfall. One mandatory stratum is present and defines the overall class structure (Figure A1.32) with the following attributes:

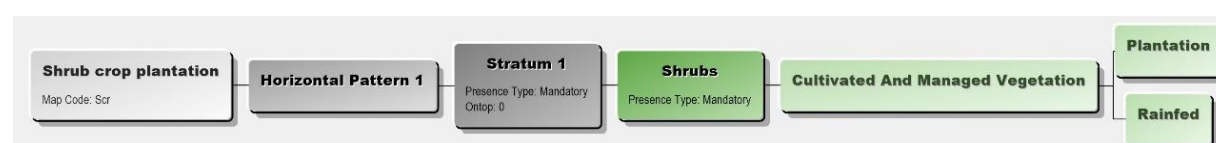
- Vegetation artificiality is defined as cultivated and managed vegetation.
- Water supply is defined as rainfed.

## Class expansion for Shrub crop

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

- Physiognomic structural properties:
  - **Height:** A more specific height can be used to better characterize the Trees basic element.
  - **Leaf phenology:** A specific leaf phenology for each growth form can be defined.
  - **Leaf type:** A specific leaf type for each growth form can be added.
  - **Adding different strata of growth forms:** In addition to the main strata characterizing this class, other strata with different growth forms can be added.
- Other attributes:
  - **Floristic name:** The floristic name of the plant species.
  - **Geographic location or altitude:** Any information related to a specific geographic location and or altitude ranges.
  - **Field size:** The field size for Cultivated rainfed can be added.
  - **Irrigation type and percentage:** Any of several means of providing an artificial regular supply of water (in addition to rain) to the crop(s), along with the water percentage can be added.
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.32. Unified modelling language (UML) diagram for the *Shrub crop* metaclass



Source: Author's own elaboration.



## Herbaceous crop

Map code/class code: Hcr

Description: This metaclass is a generic representation of rainfed crops with the herbaceous basic element, where plants are totally reliant on rainfall. One mandatory stratum is present and defines the overall class structure (Figure A1.33) with the following attributes:

- Vegetation artificiality is defined as cultivated and managed vegetation.
- Water supply is defined as rainfed.
- Crop rotation period is one to three years.

## Class expansion for Herbaceous crop

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

- Physiognomic structural properties:
  - **Height:** A more specific height can be used to better characterize the Herbaceous basic element.
  - **Leaf phenology:** A specific leaf phenology for each growth form can be defined.
  - **Adding different strata of growth forms:** In addition to the main strata characterizing this class, other strata with different growth forms can be added.
- Other attributes:
  - **Floristic name:** The floristic name of the plant species.
  - **Geographic location or altitude:** Any information related to a specific geographic location and or altitude ranges.
  - **Seeding time:** Information about the month that the crop was sown can be added
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.33. Unified modelling language (UML) diagram for the *Herbaceous crop* metaclass



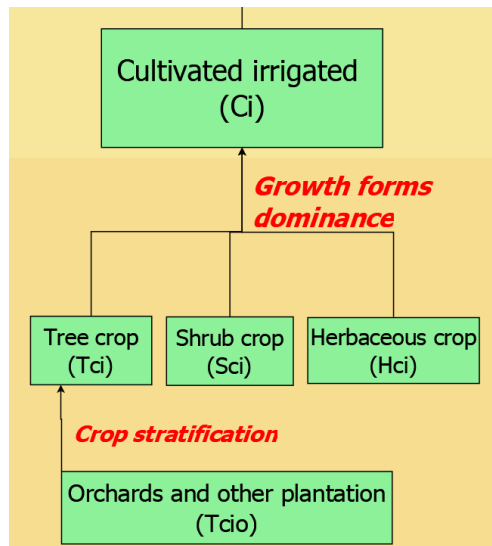
Source: Author's own elaboration.

## A1.4. Cultivated irrigated

Map code/class code: Ci

Description: This metaclass represents a subdivision of *Cultivated and managed vegetation* categorized according to water supply. It includes all irrigated cultivation existing in Somalia. The structural breakdown is illustrated in Figure A1.34.

Figure A1.34. Structure of the *Cultivated irrigated* class



Source: Author's own elaboration.

### Tree crop

Map code/class code: Tci

Description: This metaclass is a generic representation of irrigated crops with a Trees basic element. The plants are fully or mainly relying on irrigation. One mandatory stratum is present and defines the overall class structure (Figure A1.35) with the following attributes:

- Vegetation artificiality is defined as cultivated and managed vegetation.
- Water supply is defined as irrigated.

### Orchards and other plantations

Map code/ class code: Tcio

Description: This metaclass is a generic representation of irrigated tree orchards and other plantations, where the plants are fully or mainly relying on irrigation. One mandatory stratum is present and defines the overall class structure (Figure A1.36) with the following attributes:

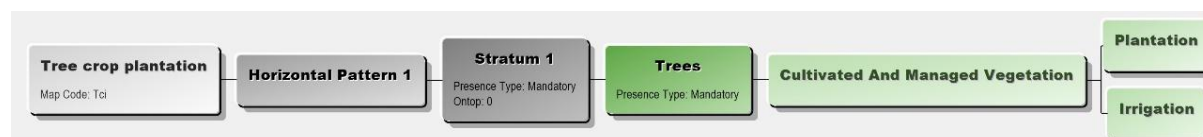
- Vegetation artificiality is defined as Cultivated and managed vegetation.
- Water supply is defined as Irrigated.

## Class expansion for Tree crop

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

- Physiognomic structural properties:
  - **Height:** A more specific height can be used to better characterize the Trees basic element.
  - **Leaf phenology:** A specific leaf phenology for each growth form can be defined.
  - **Leaf type:** A specific leaf type for each growth form can be added.
  - **Adding different strata of growth forms:** In addition to the main strata characterizing this class, other strata with different growth forms can be added.
- Other attributes:
  - **Floristic name:** The floristic name of the plant species.
  - **Geographic location or altitude:** Any information related to a specific geographic location and or altitude ranges.
  - **Plant spreading geometry:** The geometry of the spreading plant can be added.
  - **Field size:** The field size for Cultivated rainfed can be added.
  - **Irrigation type and percentage:** Any of several means of providing an artificial regular supply of water (in addition to rain) to the crop(s), along with the water percentage can be added.
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.35. Unified modelling language (UML) diagram for the *Tree crop* metaclass



Source: Author's own elaboration.

Figure A1.36. Unified modelling language (UML) diagram for the *Orchards and other plantations* metaclass



Source: Author's own elaboration.

## Shrub crop

Map code/class code: Sci

Description: This metaclass is a generic representation of irrigated crops with the Shrubs basic element. The plants are fully or mainly relying on irrigation. One mandatory stratum is present and defines the overall class structure (Figure A1.37) with the following attributes:

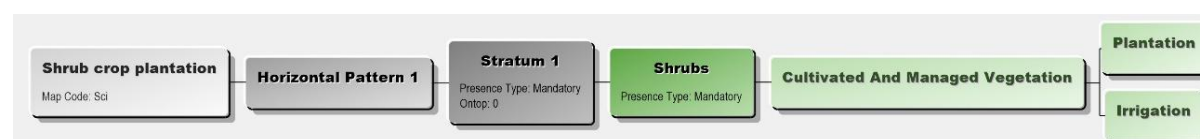
- Vegetation artificiality is defined as Cultivated and managed vegetation.
- Water supply is defined as Irrigated.

## Class expansion for Shrub crop

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

- Physiognomic structural properties:
  - **Height:** A more specific height can be used to better characterize the Trees basic element.
  - **Leaf phenology:** A specific leaf phenology for each growth form can be defined.
  - **Leaf type:** A specific leaf type for each growth form can be added.
  - **Adding different strata of growth forms:** In addition to the main strata characterizing this class, other strata with different growth forms can be added.
- Other attributes:
  - **Floristic name:** The floristic name of the plant species.
  - **Geographic location or altitude:** Any information related to a specific geographic location or altitude ranges.
  - **Field size:** The field size for Cultivated rainfed can be added.
  - **Irrigation type and percentage:** Any of several means of providing an artificial regular supply of water to the crop(s) (in addition to rain) with percentage can be added.
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.37. Unified modelling language (UML) diagram for the *Shrub crop* metaclass



Source: Author's own elaboration.

## Herbaceous crop

Map code/class code: Hci

Description: This metaclass is a generic representation of irrigated crops with a *Herbaceous* basic element. The plants are fully or mainly relying on irrigation. One mandatory stratum is present and defines the overall class structure (Figure A1.38) with the following attributes:

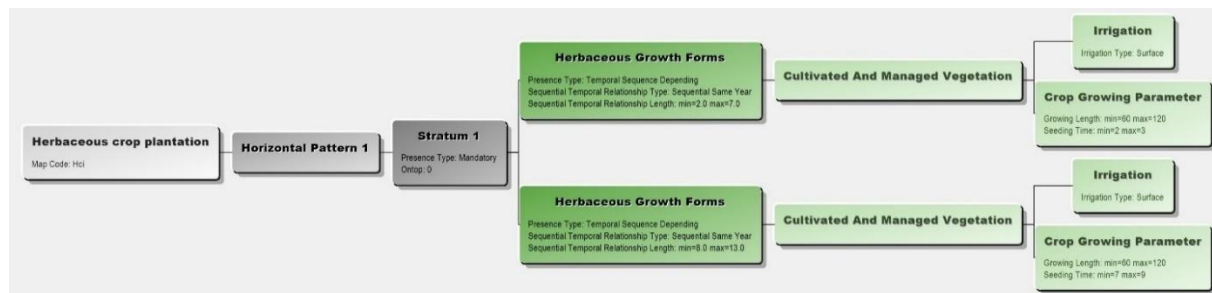
- Vegetation artificiality is defined as Cultivated and managed vegetation.
- Water supply is Irrigated.
- Crop rotation period is from one to three years.
- Presence type is temporal sequence depending.
- Seeding time is from two to three months, or seven to nine months.

## Class expansion for Herbaceous crop

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

- Physiognomic structural properties:
  - **Height:** A more specific height can be used to better characterize the Herbaceous basic element.
  - **Leaf phenology:** A specific leaf phenology for each growth form can be defined.
  - **Adding different strata of growth forms:** In addition to the main strata characterizing this class, other strata with different growth forms can be added.
- Other attributes:
  - **Floristic name:** The floristic name of the plant species.
  - **Geographic location or altitude:** Any information related to a specific geographic location and or altitude ranges.
  - **Seeding time:** Information about the month that crop was sown can be added
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.38. Unified modelling language (UML) diagram for the *Herbaceous crop* metaclass



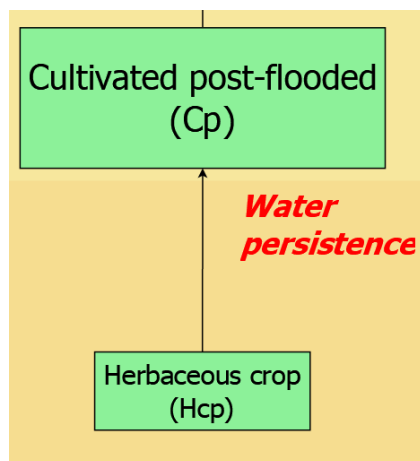
Source: Author's own elaboration.

## A1.5. Cultivated post-flooded

Map code/class code: Cp

Description: The class is characterized by a formation of cultivated plants whose water supply relies on soil moisture retained from previous flooding events, where rainwater has infiltrated the soil and is intentionally used as a reserve for crop establishment. The structural breakdown is illustrated in Figure A1.39.

Figure A1.39. Structure of the *Cultivated post-flooded* class



Source: Author's own elaboration.

## Herbaceous crop

Map code/class code: Hcp

Description: This metaclass is a generic representation of post-flooded crops with a Herbaceous basic element. The plants rely on post-flooding water reserves. Two mandatory strata are present and defines the overall class structure (Figure A1.40). The Herbaceous growth form stratum with the following attributes:

- Vegetation artificiality is defined as Cultivated and managed vegetation.
- Water supply is defined as Post-flooding.

The Waterbody stratum attributes are as follows:

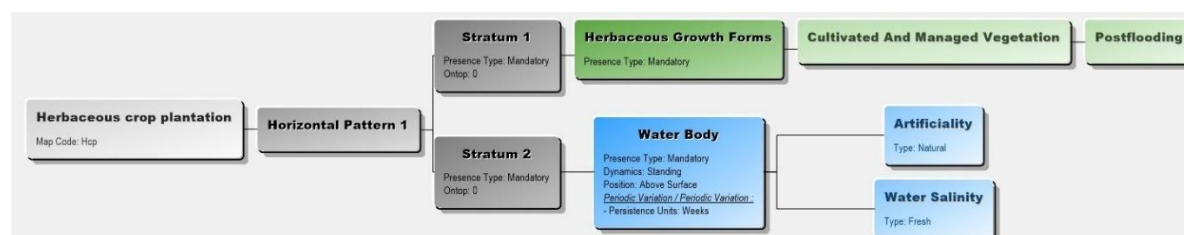
- Water artificiality is defined as Natural.
- Water salinity is defined as Fresh.
- Water dynamics is defined as Standing.

## Class expansion for Herbaceous crop

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

- Physiognomic structural properties:
  - **Height:** A more specific height can be used to better characterize the Herbaceous basic element.
  - **Leaf phenology:** A specific leaf phenology for each growth form can be defined.
  - **Adding different strata of growth forms:** In addition to the main strata characterizing this class, other strata with different growth forms can be added.
- Other attributes:
  - **Floristic name:** The floristic name of the plant species.
  - **Geographic location or altitude:** Any information related to a specific geographic location or altitude ranges.
  - **Seeding time:** Information about the month that crop was sown can be added
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.40. Unified modelling language (UML) diagram for the *Herbaceous crop* metaclass



Source: Author's own elaboration.

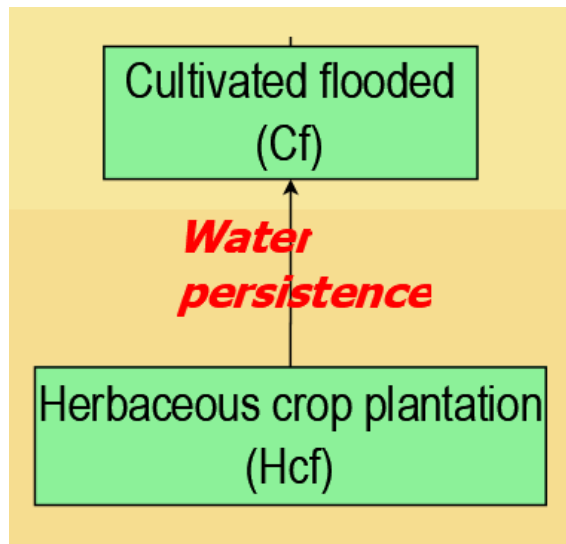


## A1.6. Cultivated flooded

Map code/class code: Cf

Description: The class is characterized by a formation of cultivated plants whose water supply relies on standing water that is intentionally maintained in the field through controlled flooding during part or all of the growing period. The structural breakdown is illustrated in Figure A1.41.

Figure A1.41. Structure of the *Cultivated flooded* class



Source: Author's own elaboration.

## Herbaceous crop

Map code/class code: Hcf

This metaclass is a generic representation of flooded crops with a Herbaceous basic element. The plants rely on flooded water. Two mandatory strata are present and defines the overall class structure (Figure A1.42). The Herbaceous growth form stratum attributes are as follows:

- Vegetation artificiality is defined as Cultivated and managed vegetation.
- Water supply is defined as Post-flooding.

The Waterbody stratum attributes are as follows:

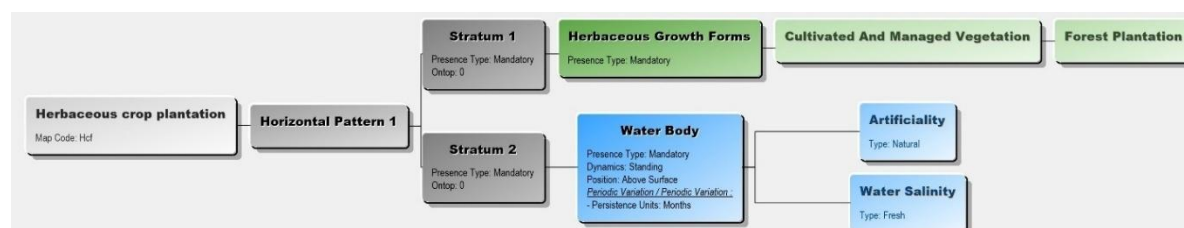
- Water artificiality is defined as Natural.
- Water salinity is defined as Fresh.
- Water dynamics is defined as Standing.

## Class expansion for Herbaceous crop

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

- Physiognomic structural properties:
  - **Height:** A more specific height can be used to better characterize the Herbaceous basic element.
  - **Leaf phenology:** A specific leaf phenology for each growth form can be defined.
  - **Adding different strata of growth forms:** In addition to the main strata characterizing this class, other strata with different growth forms can be added.
- Other attributes:
  - **Floristic name:** The floristic name of the plant species.
  - **Geographic location or altitude:** Any information related to a specific geographic location and or altitude ranges.
  - **Seeding time:** Information about the month that a crop was sown can be added
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.42. Unified modelling language (UML) diagram for *Herbaceous crop* metaclass



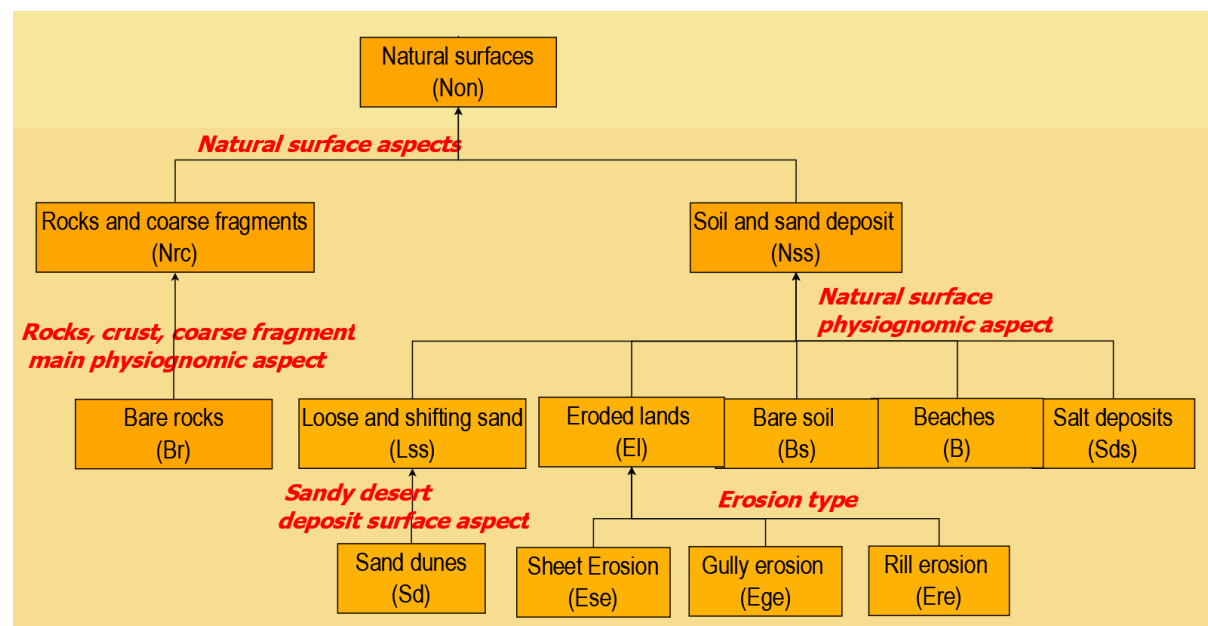
Source: Author's own elaboration.

## A1.7. Natural surfaces

Map code/class code: Non

Description: The metaclass corresponds to areas with natural surfaces where natural vegetation is absent or not significant. The structural breakdown is illustrated in Figure A1.43.

Figure A1.43. Structure of the *Natural surfaces* class



Source: Author's own elaboration.

## Rocks and coarse fragments

Map code/class code: Nrc

Description: This class refers to all-natural areas where the stratum is constituted from consolidated material. The vegetation layer is absent or not significant (>4 percent). The stratum is constituted by a single Rocks and coarse fragments basic element (Figure A1.44).

### *Bare rocks*

Map code/class code: Br

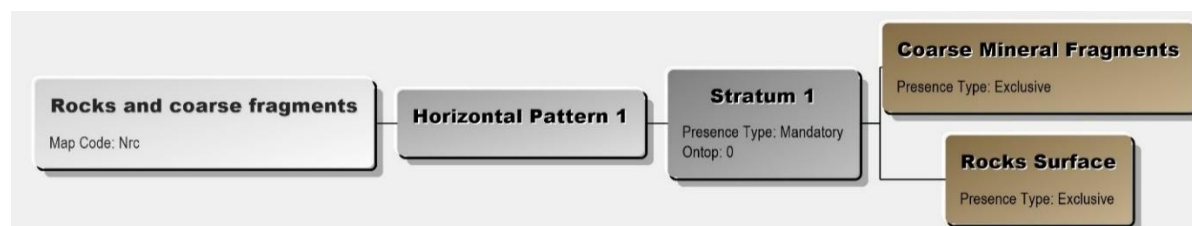
Description: This class refers to all-natural areas where the substratum is constituted mainly by bare rocks. The vegetation layer is absent or not significant (>4 percent). One mandatory stratum is present and defines the overall class structure. The stratum is constituted by a single *Bare rocks* basic element. There is another stratum with an optional presence type and a *Shrubs* basic element with 1 to 5 percent cover (Figure A1.45).

## Class expansion for Rocks and coarse fragments

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

- **Rock type:** The rock type and rock age type for consolidated surface characteristics can be added.
- **Soil type:** The soil type for unconsolidated surface characteristics can be added.
- **Geographic location or altitude:** Any information related to a specific geographic location and or altitude ranges.
- **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.44. Unified modelling language (UML) diagram for the *Rocks and coarse fragments* metaclass



Source: Author's own elaboration.

Figure A1.45. Unified modelling language (UML) diagram for the *Bare rocks* metaclass



Source: Author's own elaboration.

## Soil and sand deposits

Map code/class code: Nss

Description: This class refers to natural areas where the substratum is constituted by unconsolidated material. The vegetation layer is absent or not significant (>4 percent). The stratum is constituted by a single Soil and sand deposits basic element (Figure A1.46).

### *Loose and shifting sand*

Map code/class code: Lss

Description: This class refers to natural desertic areas, characterized by loose and shifting sand. The vegetation layer is absent or not significant (>4 percent). Two mandatory strata are present and define the overall class structure. The first stratum is constituted by a single Loose and shifting sand basic element. The second stratum is constituted by a Vegetation growth form basic element (1 to 5 percent cover) and Vegetation artificiality is defined as Natural or semi-natural (Figure A1.47).

### *Sand dunes*

Map code/class code: Sd

Description: This class refers to all desertic sandy areas where the loose and shifting sand creates sand dunes. At this generalized level, no distinction is made on the form of the dunes itself. The vegetation layer is absent or not significant (>4 percent). One mandatory stratum is present and defines the overall class structure (Figure A1.48). The stratum is constituted by a single Dunes basic element.

### *Eroded lands*

Map code/class code: El

Description: This class refers to natural areas where the substratum is constituted by unconsolidated material (bare soil). The vegetation layer is absent or not significant (>4 percent). The stratum is constituted by a single Bare soil basic element (Figure A1.49). Another Woody growth form stratum may be present (1 to 10 percent cover), with Vegetation artificiality defined as Natural or semi-natural.

### *Sheet erosion*

Map code/class code: Ese

Description: This class is derived from Eroded lands and refers to natural areas where the substratum is constituted by unconsolidated material with the Erosion sheet type. The stratum is constituted by one Bare soil basic element (Figure A1.50).

### *Gully erosion*

Map code/class code: Ege

Description: This class is derived from Eroded lands and refers to natural areas where the substratum is constituted by unconsolidated material with the gully type erosion. The stratum is constituted by a single Bare soil basic element (Figure A1.51).

### ***Rill erosion***

Map code/class code: Ere

Description: This class is derived from eroded lands and refers to natural areas where the substratum is constituted by unconsolidated material with the rill type erosion. The stratum is constituted by one Bare soil basic element (Figure A1.52).

### ***Bare soil***

Map code/class code: Bs

Description: This class refers to all-natural areas where the substratum is constituted from different types of bare soils. The vegetation layer is absent or not significant (>4 percent). One mandatory stratum is present and defines the overall class structure (Figure A1.53). The stratum is constituted by one *Bare soil* basic element.

### ***Beaches***

Map code/class code: B

Description: This class refers to all-natural areas next to the ocean where the substratum is constituted from unconsolidated material: mainly loose and shifting sand. The vegetation layer is absent or not significant (>4 percent). One mandatory stratum is present and defines the overall class structure (Figure A1.54). The stratum is constituted by a Loose and shifting sand or Coarse mineral fragments basic elements with the Exclusive presence type.

### ***Salt deposits***

Map code/class code: Sds

Description: This class is very basic, and generic, being constituted by one mandatory stratum that defines the overall class structure. The stratum is constituted by one Inorganic deposits basic element with a Salt flat type (Figure A1.55).

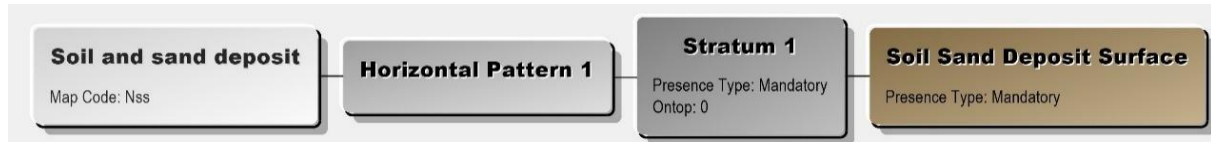
## **Class expansion for Soil and sand deposits**

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

- **Rock type:** A rock type and rock age type for consolidated surface characteristics can be added.
- **Soil type:** A soil type for unconsolidated surface characteristics can be added.

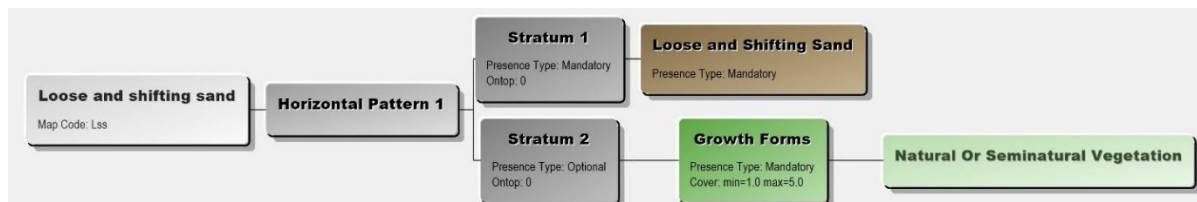
- **Geographic location or altitude:** Any information related to a specific geographic location and or altitude ranges.
- **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.46. Unified modelling language (UML) diagram for the *Soil and sand deposits* metaclass



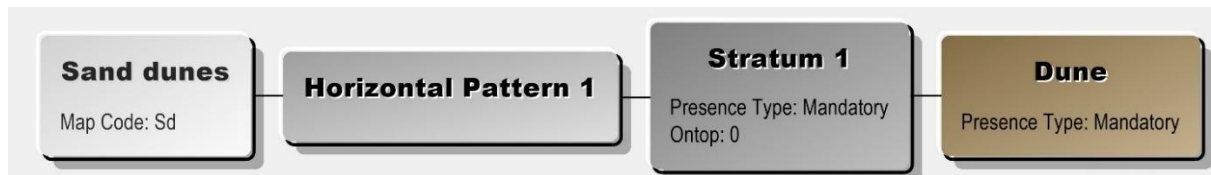
Source: Author's own elaboration.

Figure A1.47. Unified modelling language (UML) diagram for the *Loose and shifting sand* metaclass



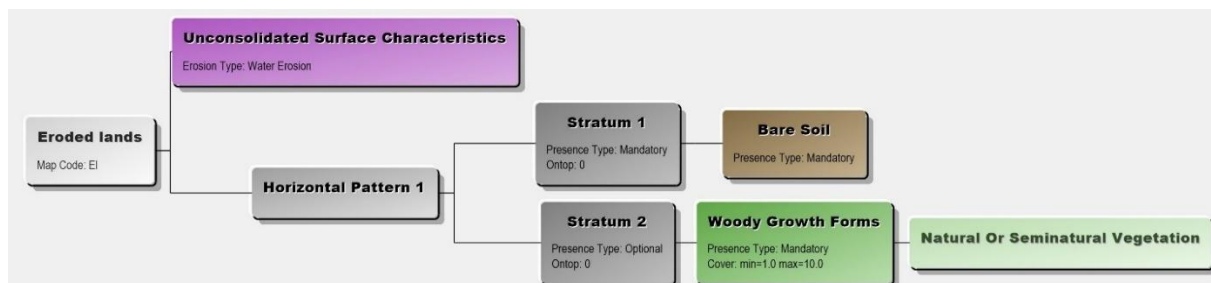
Source: Author's own elaboration.

Figure A1.48. Unified modelling language (UML) diagram for the *Sand dunes* metaclass



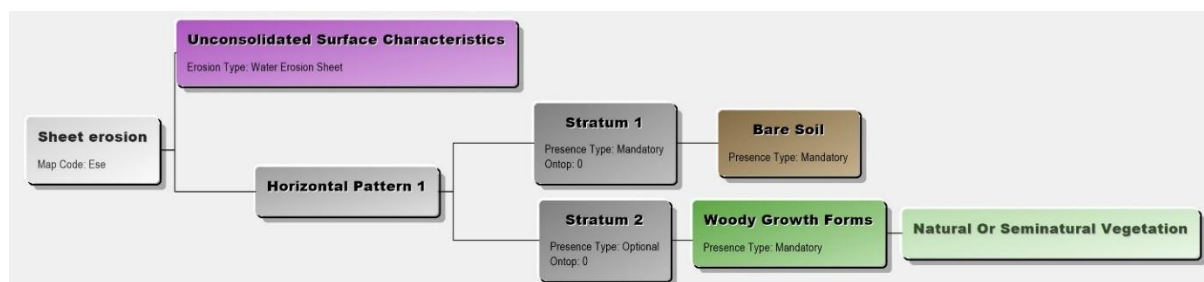
Source: Author's own elaboration.

Figure A1.49. Unified modelling language (UML) diagram for the *Eroded lands* metaclass



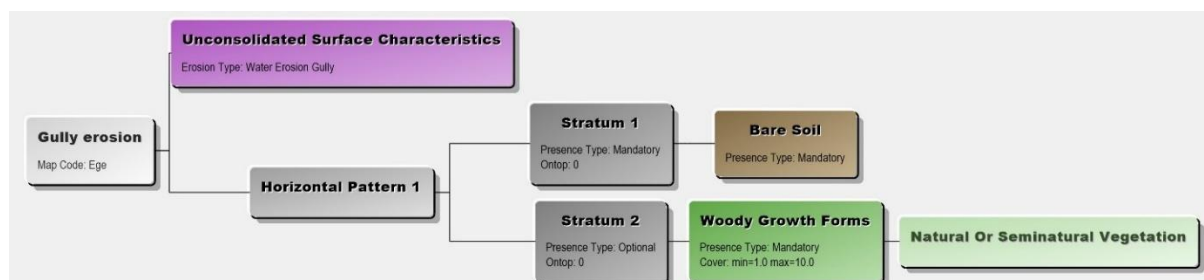
Source: Author's own elaboration.

Figure A1.50. Unified modelling language (UML) diagram for the *Sheet erosion* metaclass



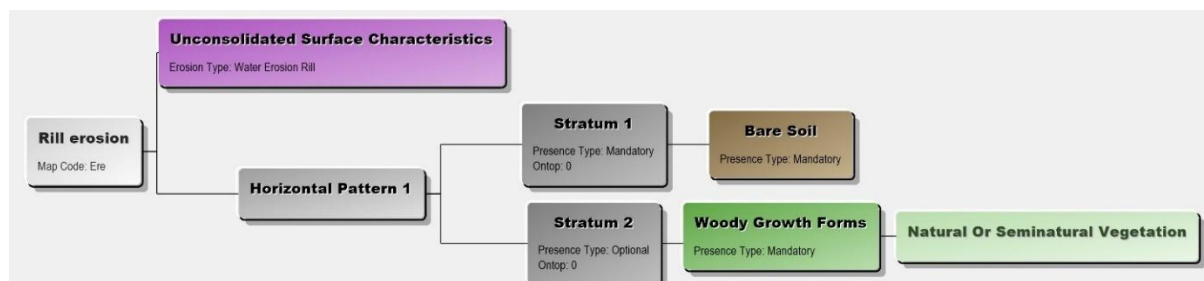
Source: Author's own elaboration.

Figure A1.51. Unified modelling language (UML) diagram for the *Gully erosion* metaclass



Source: Author's own elaboration.

Figure A1.52. Unified modelling language (UML) diagram for the *Rill erosion* metaclass



Source: Author's own elaboration.

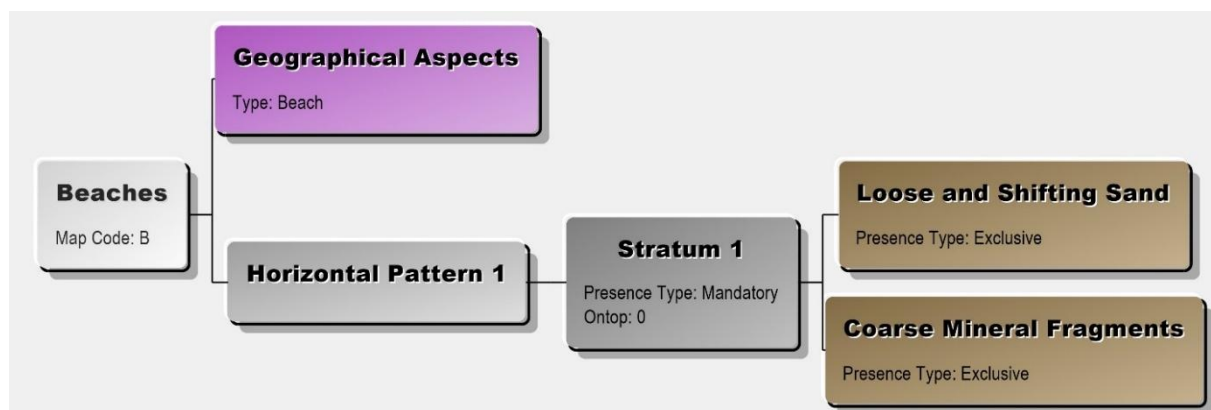
Figure A1.53. Unified modelling language (UML) diagram for the *Bare soil* metaclass



Source: Author's own elaboration.

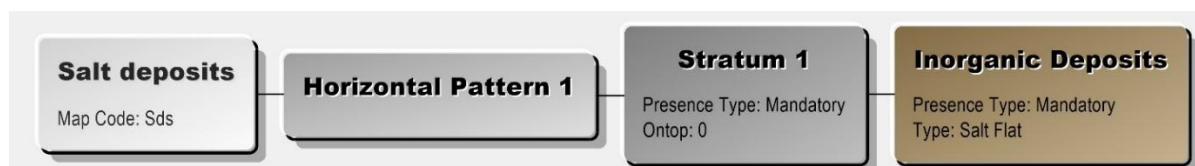


Figure A1.54. Unified modelling language (UML) diagram for the *Beaches* metaclass



Source: Author's own elaboration.

Figure A1.55. Unified modelling language (UML) diagram for the *Salt deposits* metaclass



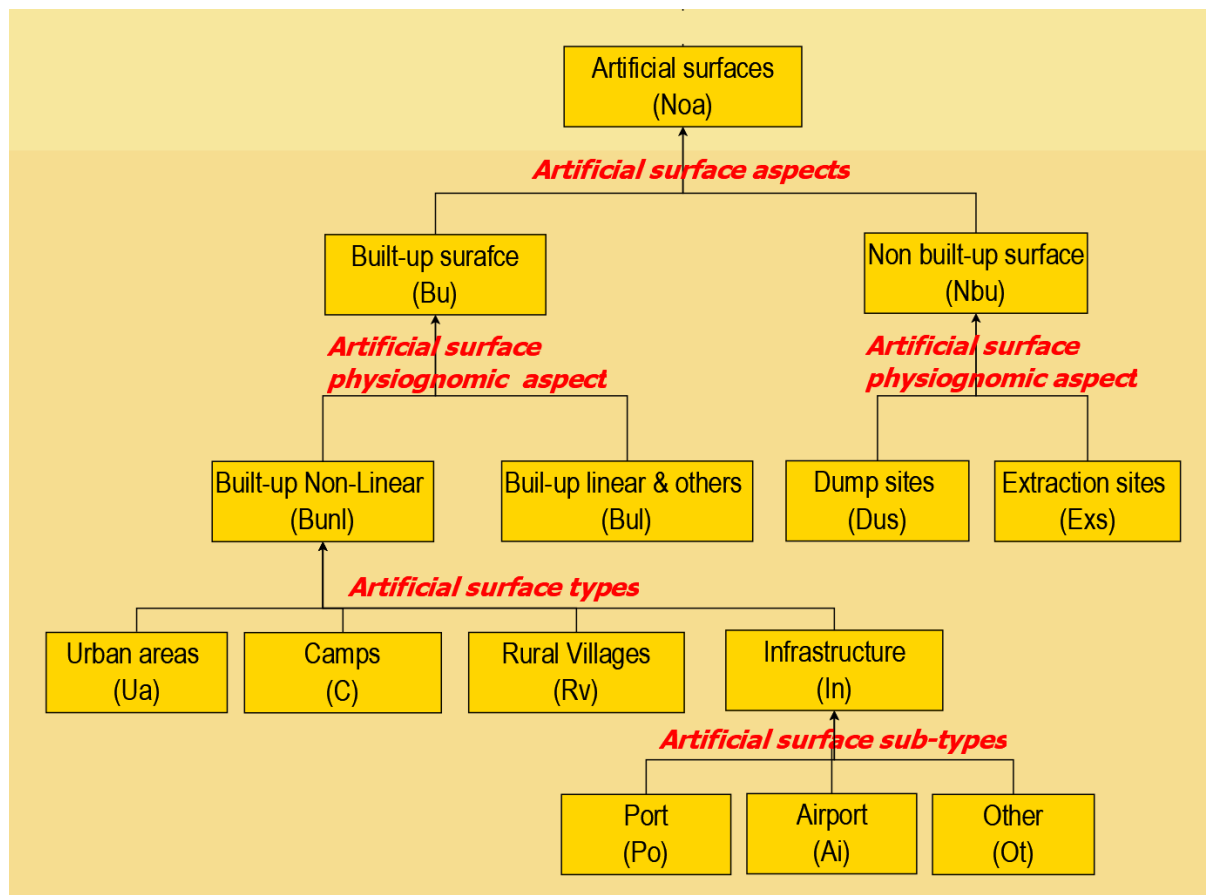
Source: Author's own elaboration.

## A1.8. Artificial surfaces

Map code/class code: Noa

Description: It corresponds to areas that have been artificially created by human activities. The structural breakdown is illustrated in Figure A1.56.

Figure A1.56. Structure of the *Artificial surfaces* class



Source: Author's own elaboration.

## Built-up surface

Map code/class code: Bu

Description: The class corresponds to artificial and impervious surfaces that are paved with hard materials or are built-up areas. One mandatory stratum is present and defines the overall class structure (Figure A1.57). The stratum is constituted by a single Built-up surface basic element.

### ***Built-up non-linear***

Map code/class code: Bunl

Description: This class represents a type of built-up surface paved with hard materials. It includes one mandatory stratum that defines the overall class structure (Figure A1.58). The stratum comprises a single Built-up non-linear basic element.

### ***Urban areas***

Map code/class code: Ua

Description: The class corresponds to artificial and impervious surfaces that are paved with hard materials and are built-up with a relatively consistent cover of buildings. The presence of subsurfaces is recognized as part of the Artificial surface types. Two mandatory strata are present and define the overall class structure (Figure A1.59). The first stratum consists of two mandatory basic elements: Buildings and Roads, and optional basic elements such as Other constructions and Other artificial surfaces. The second stratum includes a mandatory presence type for cultivated and managed Woody growth forms, with a 1 to 5 percent cover.

### ***Camps***

Map code/class code: C

Description: This class is composed of two mandatory horizontal patterns and one stratum that defines the overall structure of the class (Figure A1.60). In the first horizontal pattern (40 to 50 percent cover), the stratum is constituted by a single Building with light construction material basic element. In the second horizontal pattern (30 to 70 percent cover), there is one basic mandatory stratum that has a Bare soil basic element.

### ***Rural villages***

Map code/class code: Rv

Description: It corresponds to buildings or built-up surfaces in rural areas. The built-up surfaces are often mixed or interspersed into crop fields. The presence of subsurfaces is recognized as part of the Artificial surface types. Two mandatory strata are present and define the overall class structure (Figure A1.61). The first stratum consists of two mandatory basic elements: Buildings and Other constructions. The second stratum includes a mandatory presence type of a Bare soil basic element. This class also includes a third stratum with the optional presence type for a Cultivated and managed woody growth form, with a 1 to 5 percent cover.

### ***Infrastructure***

Map code/class code: In

Description: The class corresponds to artificial and impervious surfaces which are paved with hard materials and have built-up constructions used as infrastructure (such as transportation and

communications). The presence of subsurfaces is recognized with types of *Artificial surface* (Figure A1.62).

### **Port**

Map code/class code: Po

Description: The class corresponds to artificial and impervious surfaces which are paved with hard materials designated for marine transportation. The presence of subsurfaces is recognized with types of Artificial surface. One mandatory stratum is present and defines the overall class structure. The stratum is constituted by a single Built-up non-linear basic element with a Port attribute (Figure A1.63).

### **Airport**

Map code/class code: Ai

Description: The class corresponds to artificial and impervious surfaces which are paved with hard materials. One mandatory stratum is present and defines the overall class structure. The stratum is constituted by one Built-up non-linear basic element, with an Airport attribute (Figure A1.64).

### **Others**

Map code/class code: Ot

Description: It corresponds to artificial and impervious surface which are paved with hard materials. One mandatory stratum is present and defines the overall class structure. The stratum is constituted by one Built-up non-linear basic element, with an Others attribute (Figure A1.65).

### ***Built-up linear and others***

Map code/class code: Bul

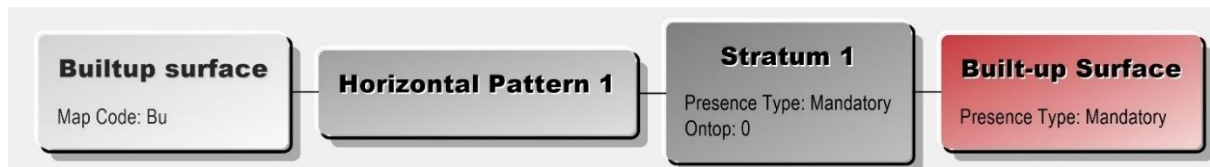
Description: This class represents a type of built-up surface paved with hard materials, categorized as built-up linear. It includes one mandatory stratum that defines the overall class structure. The stratum comprises a single Linear surface basic element (Figure A1.66).

## **Class expansion for Built-up surface**

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

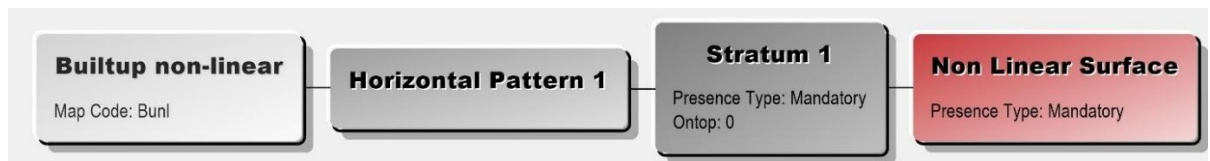
- Physiognomic structural properties are as follows:
  - **Height:** A more specific height can be used to better characterize the Built-up surface basic element.
- Other attributes are as follows:
  - **Road type:** Types of roads can be added.
  - **Construction status:** The status of the construction objects can be added.
  - **Construction use:** The type of construction use can be added.
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.57. Unified modelling language (UML) diagram for the *Built-up surface* metaclass



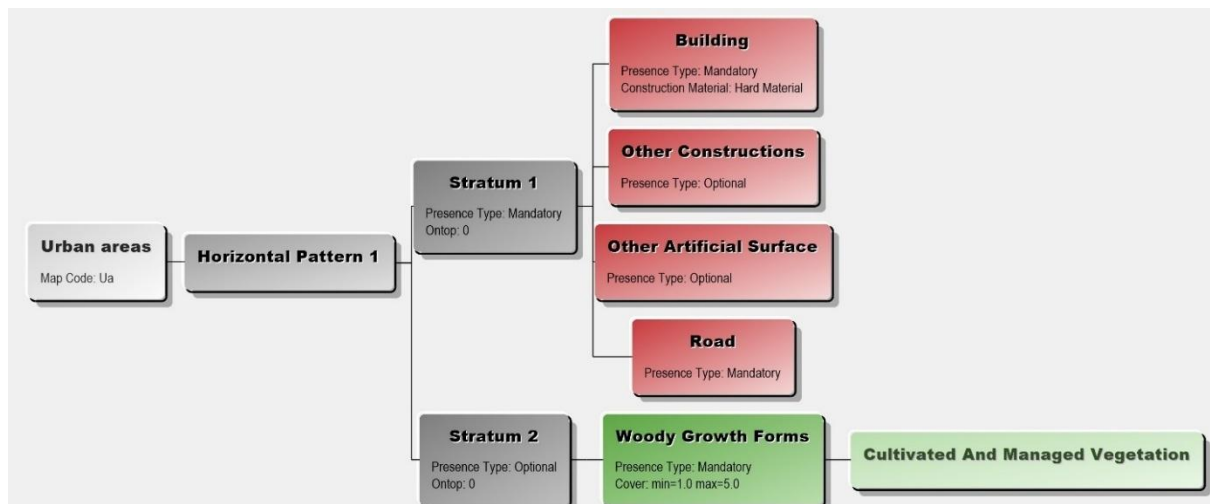
Source: Author's own elaboration.

Figure A1.58. Unified modelling language (UML) diagram for the *Built-up non-linear* metaclass



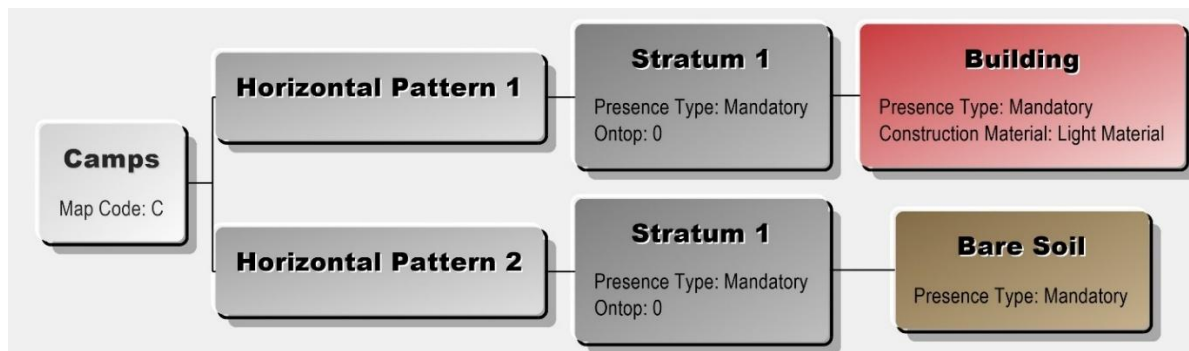
Source: Author's own elaboration.

Figure A1.59. Unified modelling language (UML) diagram for the *Urban areas* metaclass



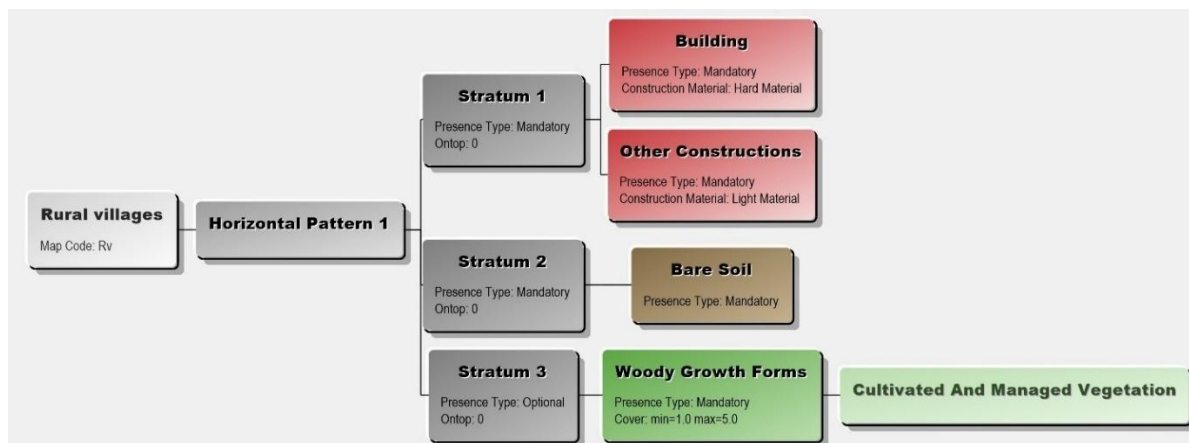
Source: Author's own elaboration.

Figure A1.60. Unified modelling language (UML) diagram for the *Camps* metaclass



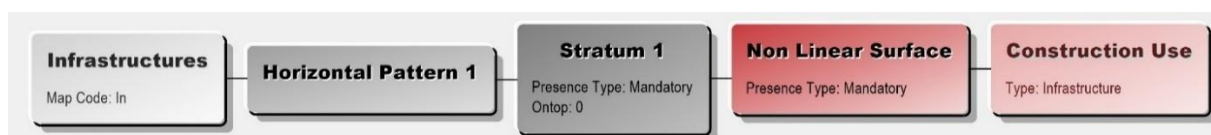
Source: Author's own elaboration.

Figure A1.61. Unified modelling language (UML) diagram for the *Rural villages* metaclass



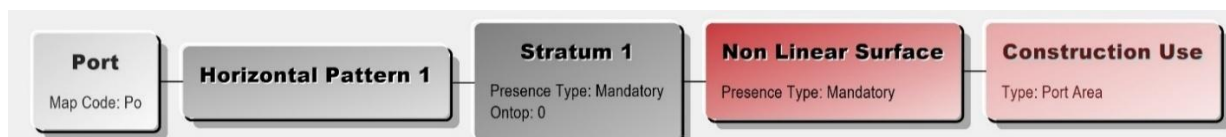
Source: Author's own elaboration.

Figure A1.62. Unified modelling language (UML) diagram for the *Infrastructure* metaclass



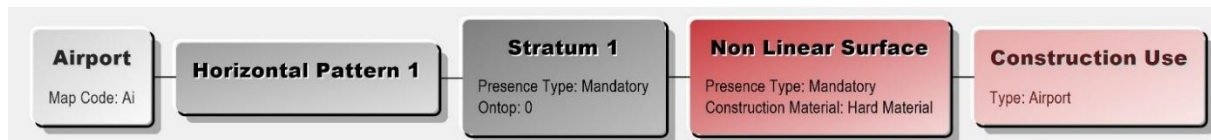
Source: Author's own elaboration.

Figure A1.63. Unified modelling language (UML) diagram for the *Port* metaclass



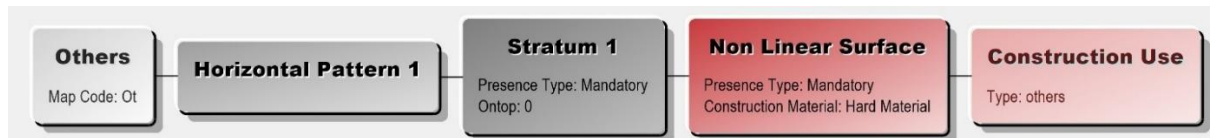
Source: Author's own elaboration.

Figure A1.64. Unified modelling language (UML) diagram for the *Airport* metaclass



Source: Author's own elaboration.

Figure A1.65. Unified modelling language (UML) diagram for the *Others* metaclass



Source: Author's own elaboration.

Figure A1.66. Unified modelling language (UML) diagram for the *Built-up linear and others* metaclass



Source: Author's own elaboration.

## Non-built-up surface

Map code/class code: Nbu

Description: The class corresponds to artificial and impervious surfaces that are paved with hard materials or are non-built-up areas. One mandatory stratum is present and defines the overall class structure. The stratum is constituted by a single Non-built-up surface basic element (Figure A1.67).

### *Dump sites*

Map code/class code: Dus

Description: Dump sites are defined as land used for disposal of waste materials. One mandatory stratum is present and defines the overall class structure. The stratum is constituted by a single Dump sites basic element (Figure A1.68).

### *Extraction sites*

Map code/class code: Exs

Description: Extraction sites are defined by the absence of the original land cover which is removed by human activity or machinery for extraction of sand, stone, mineral, or coal. One mandatory stratum is present and defines the overall class structure. The stratum is constituted by a single Extraction sites basic element (Figure A1.69).

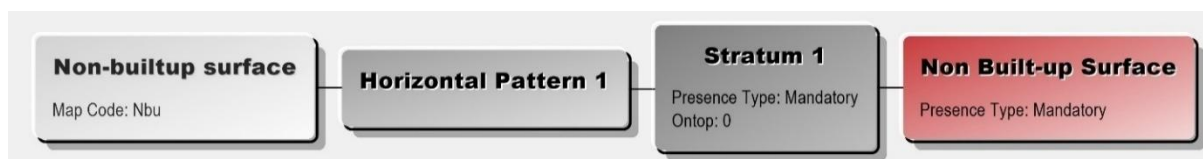
## Class expansion for Non-built-up surface

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

- Physiognomic structural properties are as follows:
  - **Cover:** The cover percentage can be added for the Non-built up surface basic element.
- Other attributes:
  - **Dump types:** Types of dump sites can be added.
  - **Extraction type:** Types of extraction sites can be added.
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

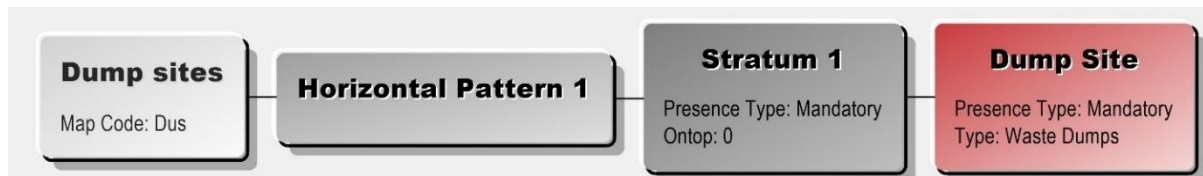


Figure A1.67. Unified modelling language (UML) diagram for the *Non-built-up surface* metaclass



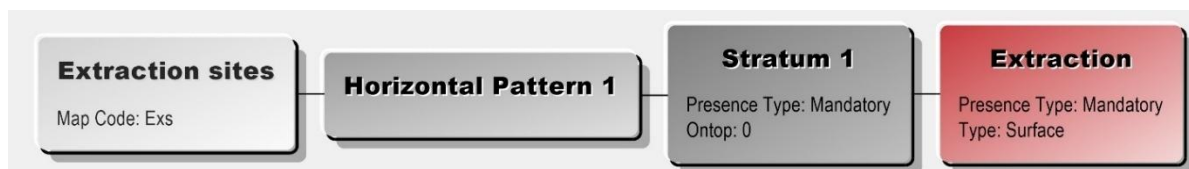
Source: Author's own elaboration.

Figure A1.68. Unified modelling language (UML) diagram for the *Dump sites* metaclass



Source: Author's own elaboration.

Figure A1.69. Unified modelling language (UML) diagram for the *Extraction sites* metaclass



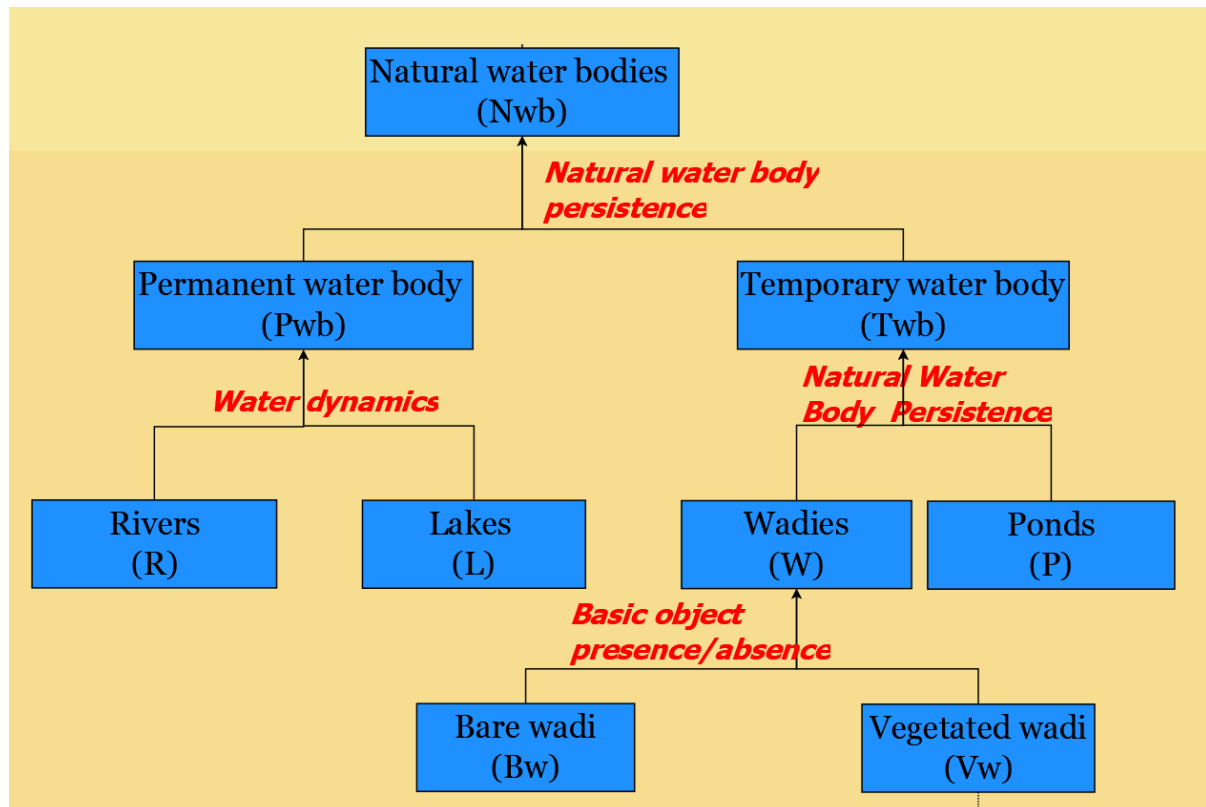
Source: Author's own elaboration.

## A1.9. Natural waterbodies

Map code/class code: Nwb

Description: It corresponds to water features formed by natural processes without significant human modification. The Natural waterbodies metaclass is subdivided into two main subclasses based on natural waterbody persistence. The structural breakdown is illustrated in Figure A1.70.

Figure A1.70. Structure of the *Natural waterbodies* class



Source: Author's own elaboration.

### Permanent waterbodies

Map code/class code: Pwb

Description: The class corresponds to natural waterbodies which are permanent features in an area. They are further categorized based on water dynamics. The stratum is constituted by one Waterbody basic element (Figure A1.71) with the following attributes:

- Periodic variation is defined as Daily.
- Water dynamics is defined as Standing.
- Waterbody artificiality is defined as Natural.
- Water salinity is defined as Fresh.

## Rivers

Map code/class code: R

Description: Rivers are natural watercourses which serves as water drainage channels (Figure A1.72). One mandatory stratum is present and defines the overall class structure. The stratum is constituted by a Waterbody basic element with the following attributes:

- Water dynamics is defined as Flowing.
- Waterbody artificiality is defined as Natural.
- Water salinity is defined as Fresh.

## Lakes

Map code/class code: L

Description: Lakes are any natural standing water reservoirs that are bigger than a pond and have an irregular shape. This class consists of one mandatory stratum and a Waterbody basic element (Figure A1.73) with the following attributes:

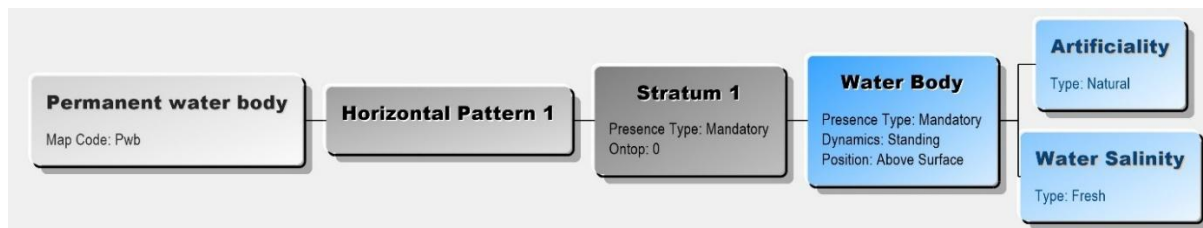
- Periodic variation is defined as Monthly.
- Water dynamics is Standing.
- Water salinity is Fresh.

## Class expansion for Natural waterbodies

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

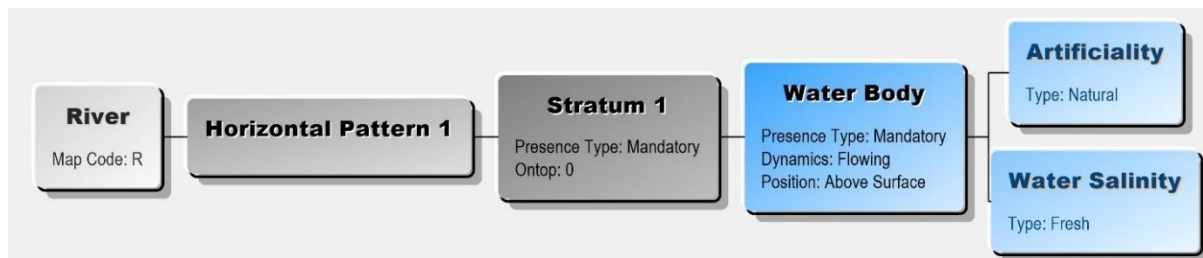
- Other attributes are as follows:
  - **Waterbody dynamics:** The water type (such as Flowing or Standing) can be added.
  - **Waterbody position:** The position of the waterbody can be added.
  - **Waterbody artificiality:** The water artificiality can be added.
  - **Water salinity:** The water salinity type can be added.
  - **Water depth:** The water depth can be added.
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.71. Unified modelling language (UML) diagram for the *Permanent waterbodies* metaclass



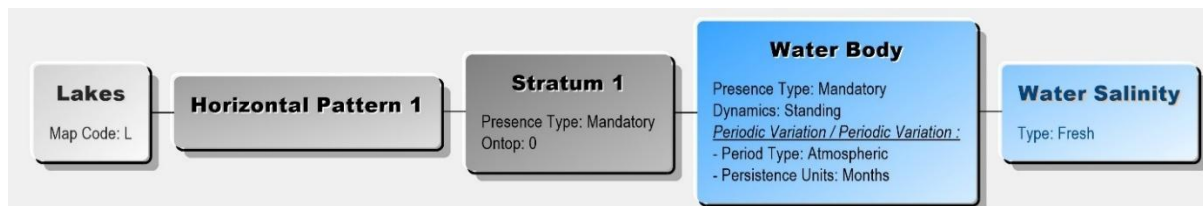
Source: Author's own elaboration.

Figure A1.72. Unified modelling language (UML) diagram for the *Rivers* metaclass



Source: Author's own elaboration.

Figure A1.73. Unified modelling language (UML) diagram for the *Lakes* metaclass



Source: Author's own elaboration.

## Temporary waterbodies

Map code/class code: Twb

Description: The class corresponds to natural waterbodies which are temporary in an area. They are further categorized based on water dynamics. The stratum is constituted by one Waterbody basic element (Figure A1.74) with the following attributes:

- Periodic variation is defined as Daily.
- Water dynamics is defined as Standing.
- Waterbody artificiality is defined as Natural.
- Water salinity is defined as Fresh.

## *Wadis*

Map code/class code: W

Description: This class describe the generic aspect of a wadi. It is constituted from three mandatory strata that define the overall class structure. The strata are constituted by three basic elements of Waterbody, Natural surfaces and Woody growth form (Figure A1.75). The Waterbody basic element has the following attributes:

- Periodic variation is defined as Daily.
- Water dynamics is defined as Flowing.
- Waterbody artificiality is defined as Natural.
- Water salinity is defined as Fresh.

Natural surfaces has the following attributes:

- The basic element is Soil and sand deposits.
- Presence type is a defined as Mandatory.

Woody has the following attributes:

- Cover is defined as 1 to 40 percent.
- Artificiality is defined as Natural.

## ***Bare wadis***

Map code/class code: Bw

Description: The class is derived from wadis and is constituted by two mandatory strata that define the overall class structure. The strata are constituted by Waterbody and Natural surfaces basic elements (Figure A1.76).

The Waterbody basic element has the following attributes:

- Periodic variation is defined as Daily.
- Water dynamics is defined as Flowing.
- Waterbody artificiality is defined as Natural.
- Water salinity is defined as Fresh.

Natural surfaces has the following attributes:

- The basic element is Soil and sand deposits.
- Presence type is a defined as Mandatory.

### ***Vegetated wadis***

Map code/class code: Vw

Description: The class is derived from Wadis and is constituted by two mandatory strata that define the overall class structure. The strata are constituted by one Waterbody basic element and a Woody growth form as a basic element (Figure A1.77). The waterbody basic element has the following attributes:

- Periodic variation is defined as Daily.
- Water dynamics is defined as Flowing.
- Waterbody artificiality is defined as Natural.
- Water salinity is defined as Fresh.

The Woody growth form has the following attributes:

- Cover is 1 to 40 percent.
- Artificiality is defined as Natural.

### ***Ponds***

Map code/class code: P

Description: This class describe the generic aspect of ponds. It is constituted by one mandatory stratum that defines the overall class structure. The stratum is constituted by a single Waterbody basic element (Figure A1.78) with the following attributes:

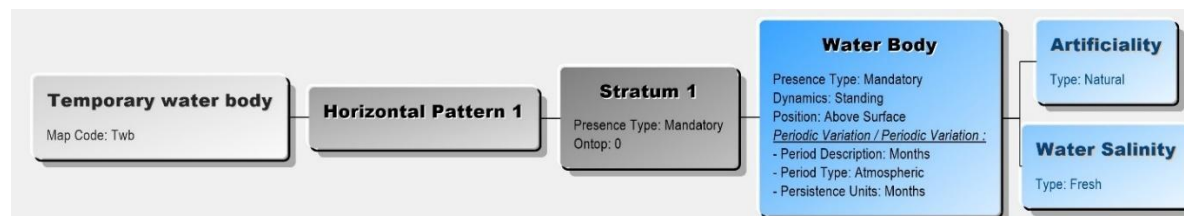
- Periodic variation is defined as Monthly.
- Water dynamics is defined as Standing.
- Waterbody artificiality is defined as Natural.
- Water salinity is defined as Fresh.

## Class expansion for Temporary waterbodies

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

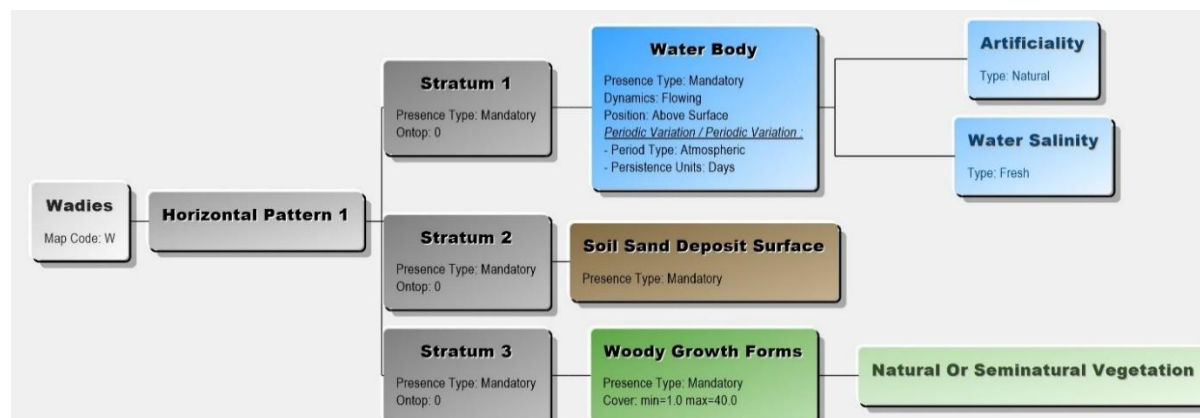
- Other attributes:
  - **Waterbody dynamics:** The water type can be added, between Flowing or Standing.
  - **Waterbody position:** The position of the waterbody can be added.
  - **Waterbody artificiality:** The water artificiality can be added.
  - **Water salinity:** The water salinity type can be added.
  - **Water depth:** The water depth can be added.
  - **Water periodic variation:** The period type and persistence period can be added.
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.74. Unified modelling language (UML) diagram for the *Temporary waterbodies* metaclass



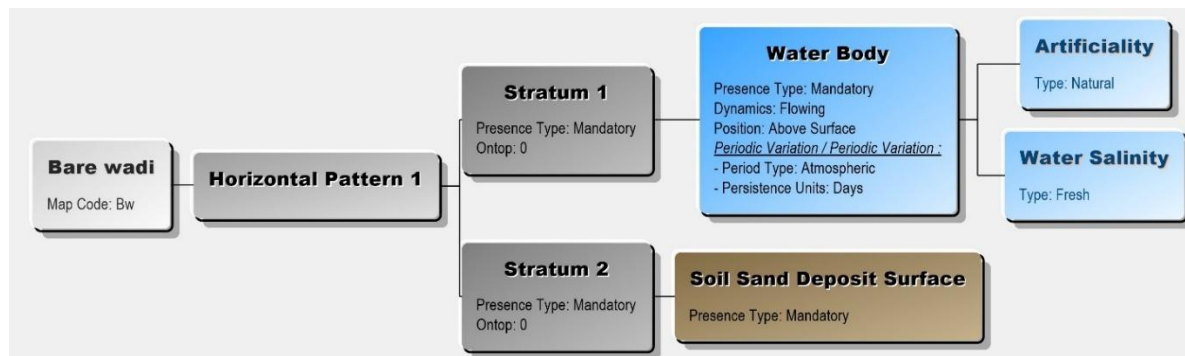
Source: Author's own elaboration.

Figure A1.75. Unified modelling language (UML) diagram for the *Wadis* metaclass



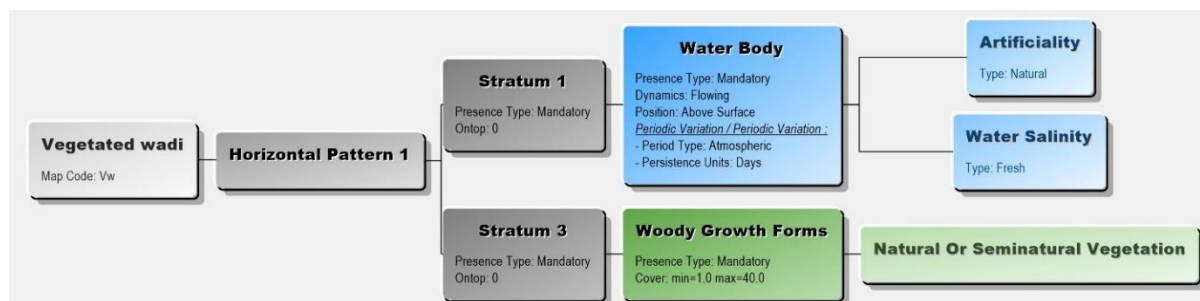
Source: Author's own elaboration.

Figure A1.76. Unified modelling language (UML) diagram for the *Bare wadis* metaclass



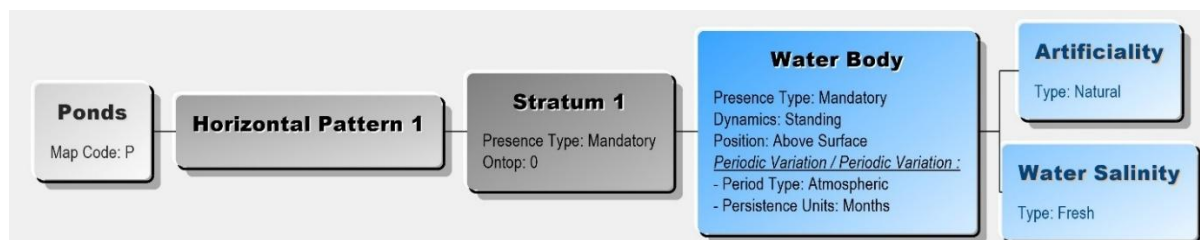
Source: Author's own elaboration.

Figure A1.77. Unified modelling language (UML) diagram for the *Vegetated wadis* metaclass



Source: Author's own elaboration.

Figure A1.78. Unified modelling language (UML) diagram for the *Ponds* metaclass



Source: Author's own elaboration.

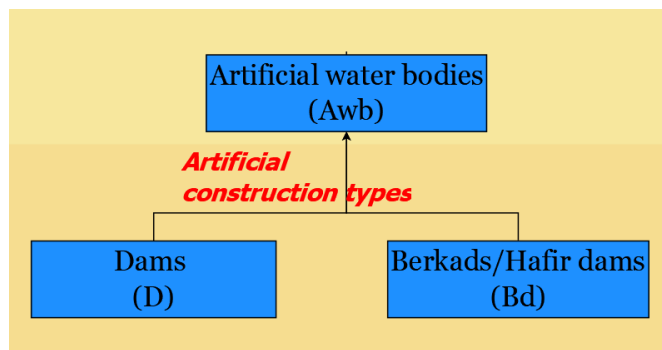


## A1.10. Artificial waterbodies

Map code/class code: Awb

Description: The class corresponds to areas where the persistence of water is due to the presence of an artificial artefact. The structural breakdown is illustrated in Figure A1.79.

Figure A1.79. Structure of the *Artificial waterbodies* class



Source: Author's own elaboration.

### Dams

Map code/class code: D

Description: This class describe the generic aspect of dams. It is constituted by one mandatory strata and defines the overall class structure. The stratum is constituted by a single Waterbody basic element (Figure A1.80) with the following attributes:

- Periodic variation is defined as Monthly.
- Water dynamics is defined as Standing.
- Waterbody artificiality is defined as Artificial.
- Water salinity is defined as Fresh.

### Berkads

Map code/class code: Bd

Description: This class describe the generic aspect of Berkads. It is constituted by one mandatory stratum and defines the overall class structure. The stratum is constituted by a single Waterbody basic element (Figure A1.81) with the following attributes:

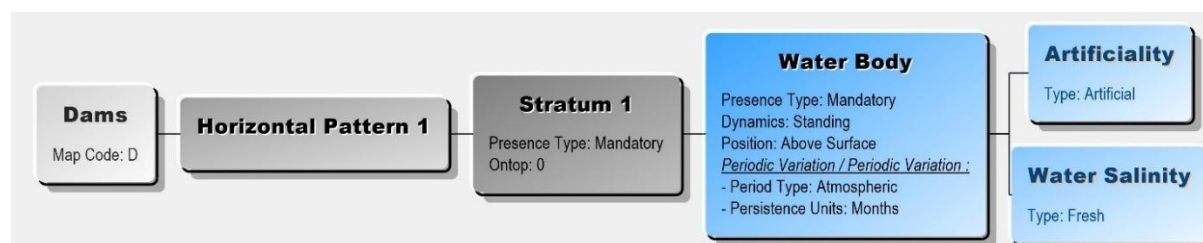
- Periodic variation is defined as Monthly.
- Water dynamics is defined as Standing.
- Waterbody artificiality is defined as Artificial.
- Water salinity is defined as Fresh.

## Class expansion for Artificial waterbodies

The national reference schema gives the user a wide range of possibilities with which to expand and enrich the thematic details of each class. This can be done using the shared dictionary of attributes of the ISO 19144-2 standard (FAO and ISO, 2025). Following this process, each further additional enrichment of legend categories will ensure a full harmonization process of different legends that for different reasons (legend scope and terrain knowledge) have different thematic attributes to define their classes. In this specific context, the suggested class thematic enrichment attributes are as follows:

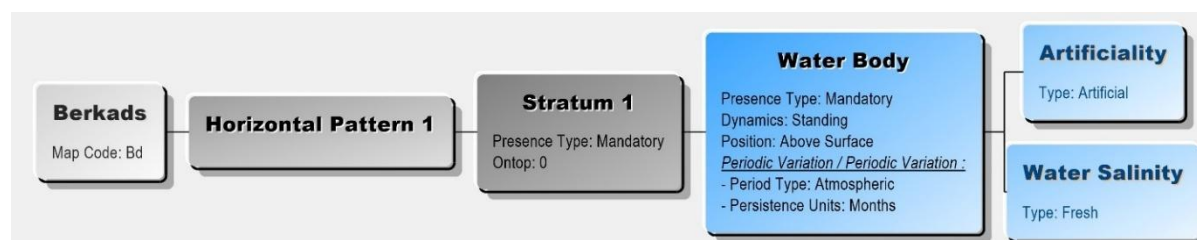
- Other attributes are as follows:
  - **Waterbody dynamics:** Flowing or standing water (for example) can be added as water type.
  - **Waterbody position:** The position of the waterbody can be added.
  - **Waterbody artificiality:** Water artificiality can be added.
  - **Water salinity:** Water salinity type can be added.
  - **Water depth:** Water depth can be added.
  - **Water periodic variation:** The period type and persistence period can be added.
  - **Any other attributes:** Other attributes can be derived from the LCML shared dictionary.

Figure A1.80. Unified modelling language (UML) diagram for the *Dams* metaclass



Source: Author's own elaboration.

Figure A1.81. Unified modelling language (UML) diagram for the *Berkads* metaclass



Source: Author's own elaboration.

## Appendix 2: Field photographs

Figure A2.1, Figure A2.2 and Figure A2.3 show field photographs for the *Natural and semi-natural vegetation* features in the reference system.

Figure A2.1. Woody mangroves



© FAO

Figure A2.2. Herbaceous-dominated area



© FAO/Mohamed Zubeyr Abdulkadir



Figure A2.3. Tiger bushes



© FAO



Figure A2.4, Figure A2.5 and Figure A2.6 show field photographs for the *Cultivated and managed vegetation* features in the reference system.

Figure A2.4. Tree crop irrigated (date palm)



© IMC/Jamal Abdullahi

Figure A2.5. Orchards and other plantations



© FAO/Isak Amin



Figure A2.6. Herbaceous crop (maize)



© FAO/Mahmud Utaama



Figure A2.7, Figure A2.8 and Figure A2.9 show field photographs for *Terrestrial non-vegetated* features in the reference system.

Figure A2.7. Gully erosion



© FAO/ Sulieman Abdullahi

Figure A2.8. Built-up



© FAO/Mahamud Utaama



Figure A2.9. Rural villages



© FAO/Abdullahi Kilas



Figure A2.10, A2.11 and A2.12 show field photographs for *Water* features in the reference system.

Figure A2.10. River



© FAO/Abdullahi Kilas

Figure A2.11. Artificial waterbody



© FAO/ Ismail Taxta

Figure A2.12. Dam



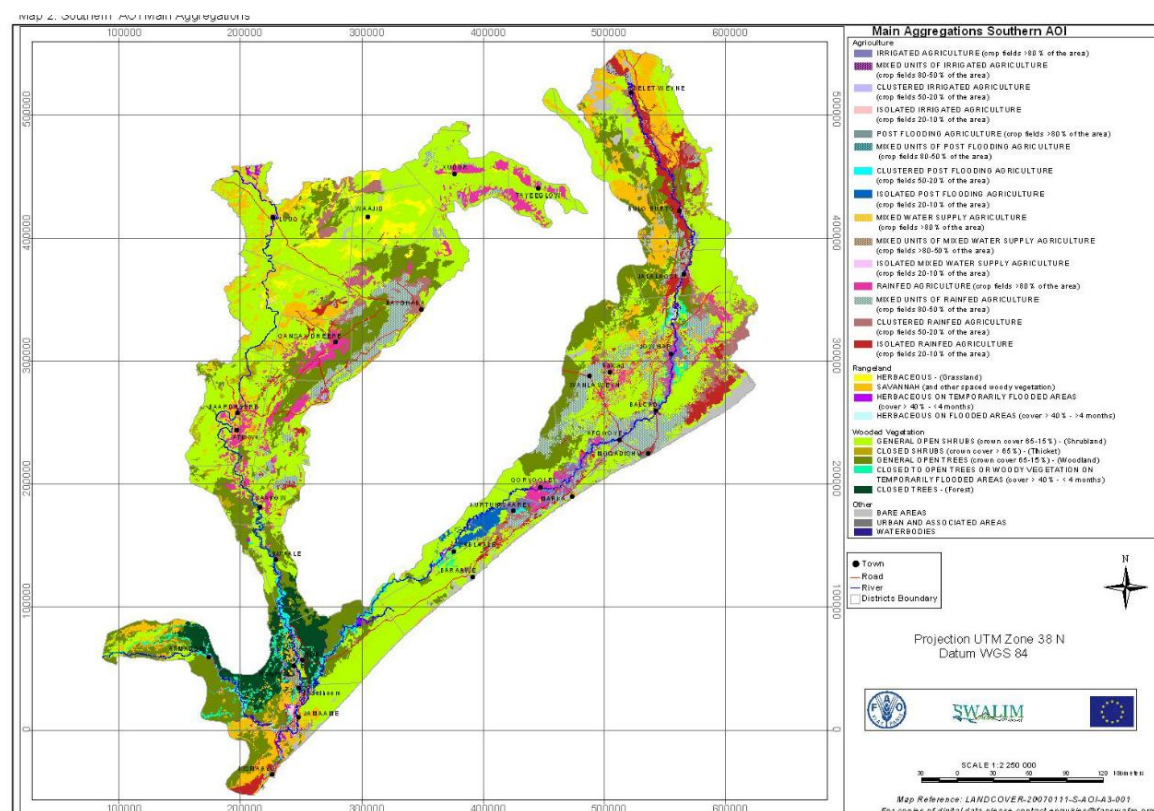
© FAO/Hiis Mohamed



# Appendix 3: Existing land cover maps of Somalia

Figure A3.1, Figure A3.2 and Figure A3.3 show the existing land cover maps that have been developed for Somalia.

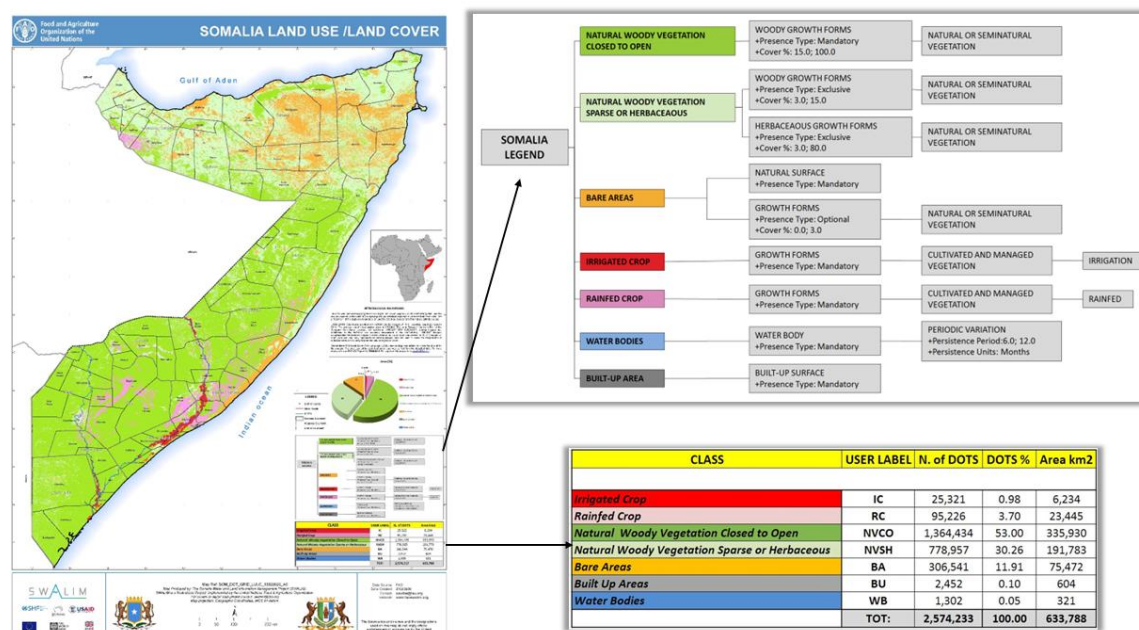
Figure A3.1. Land cover map of riverine agriculture cover for southern Somalia



Note: Refer to the disclaimer on page ii for the names and boundaries used in this map.

Source: Monaci, L., Downie, M. & Oduori, S. 2007. L-03 Land Cover of Selected Study Areas in Somaliland and Southern Somalia. Project Report No L-03. Nairobi, FAO SWALIM. [https://faoswalim.org/resources/site\\_files/L-03%20Land%20Cover%20of%20Selected%20Areas%20in%20Somaliland%20and%20Southern%20%20Somalia.pdf](https://faoswalim.org/resources/site_files/L-03%20Land%20Cover%20of%20Selected%20Areas%20in%20Somaliland%20and%20Southern%20%20Somalia.pdf)

Figure A3.2. Land cover map of Somalia derived from a statistical dot-grid analysis (produced in 2014)

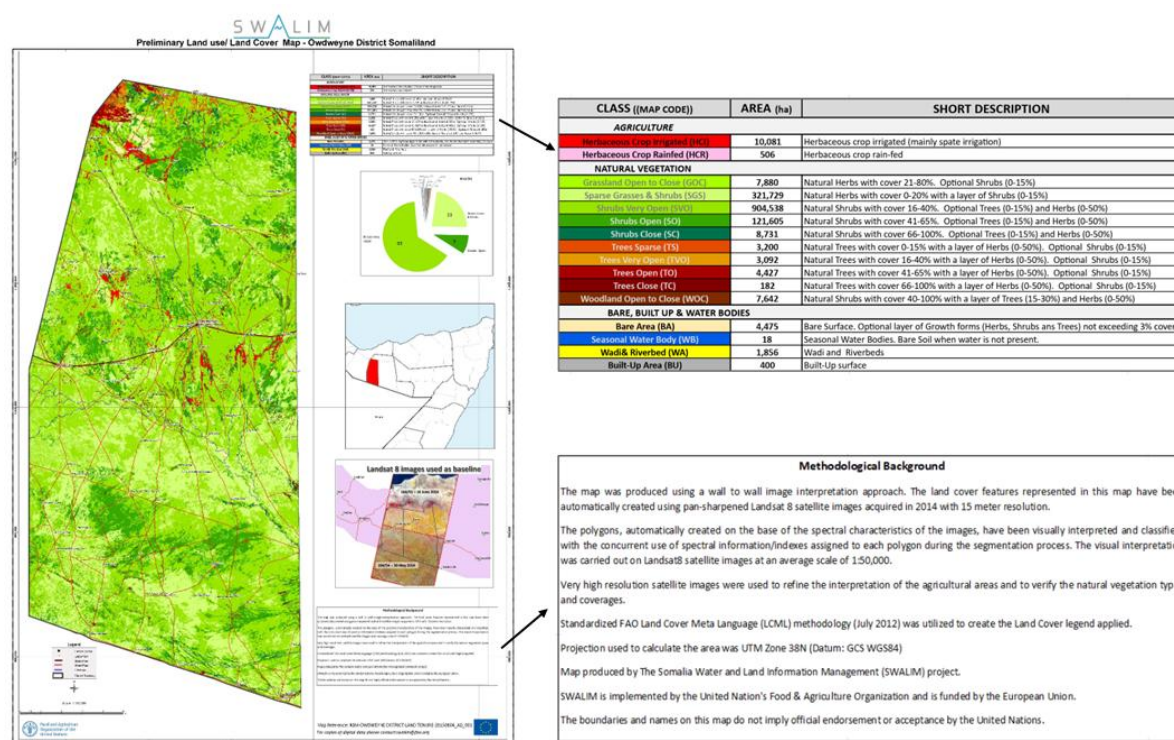


Note: Refer to the disclaimer on page ii for the names and boundaries used in this map.

Sources: **FAO SWALIM (Somalia Water and Land Information Management)**. 2014. *Somalia Land cover map*. Nairobi.

[https://faoswalim.org/resources/maps/land\\_cover.pdf](https://faoswalim.org/resources/maps/land_cover.pdf)

Figure A3.3. Land cover map of Owdweyne district, Somalia (produced in 2015)



Note: Refer to the disclaimer on page ii for the names and boundaries used in this map.

Source: FAO SWALIM (Somalia Water and Land Information Management). 2015. *Somalia Land cover map*. Nairobi.  
<https://faoswalim.org/>

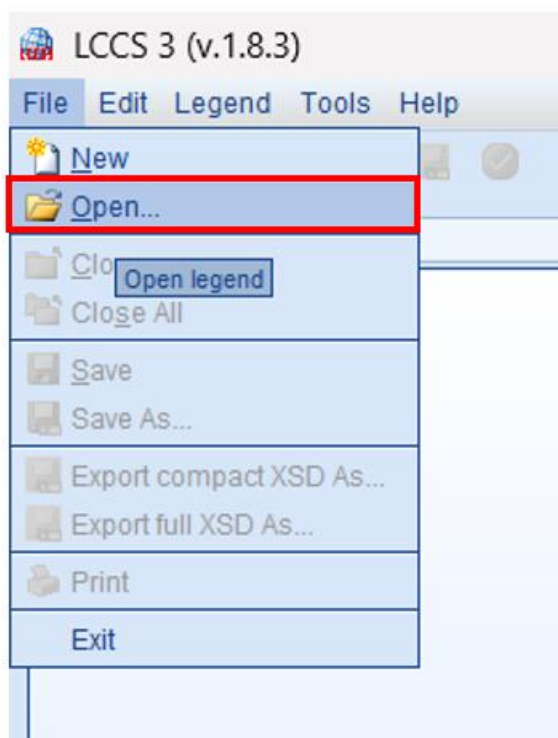
# Appendix 4: Using the Land Cover Classification System version 3 when deriving a legend from the land cover reference system

**Step 1:** Download and install the LCCSv3 software, using a PC Windows computer, as mentioned in the user manual (FAO, 2016).

**Step 2:** Download the .lccs file of the LCRS for Somalia from the FAO land cover legend registry (Mushtaq *et al.*, 2022).

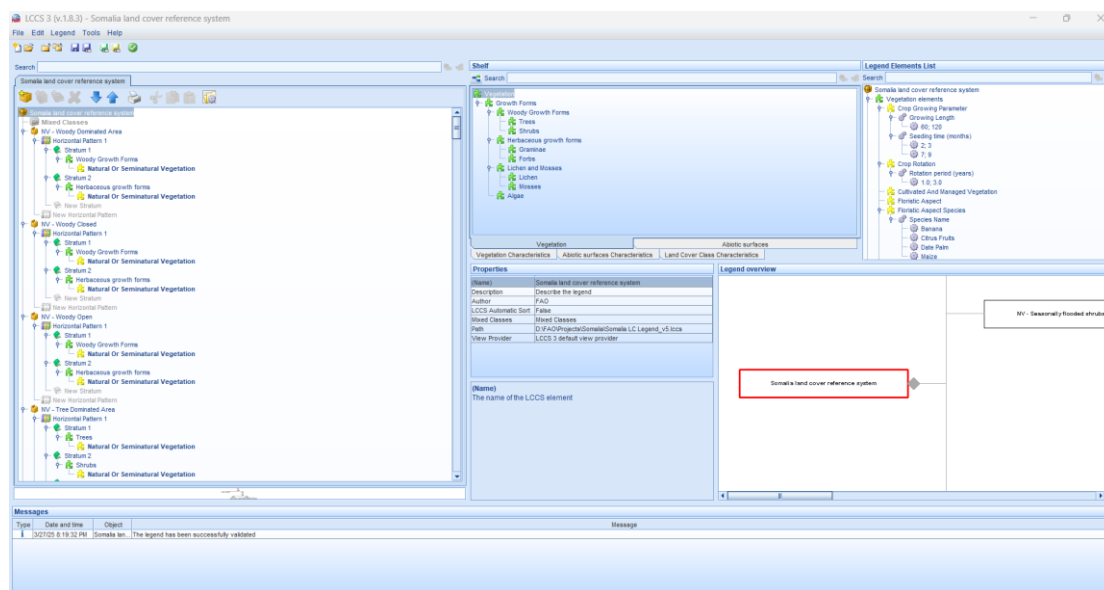
**Step 3:** Load the file in the software and create or modify the national or local legend.

Open the LCCS 3 software and click on “file -> open (LCCS file: .lccs)”, as shown here:



Source: FAO. 2016. Land Cover Classification System User Manual. Rome. <https://openknowledge.fao.org/items/ea8e72e646104c7a85718fcbdd24e346>

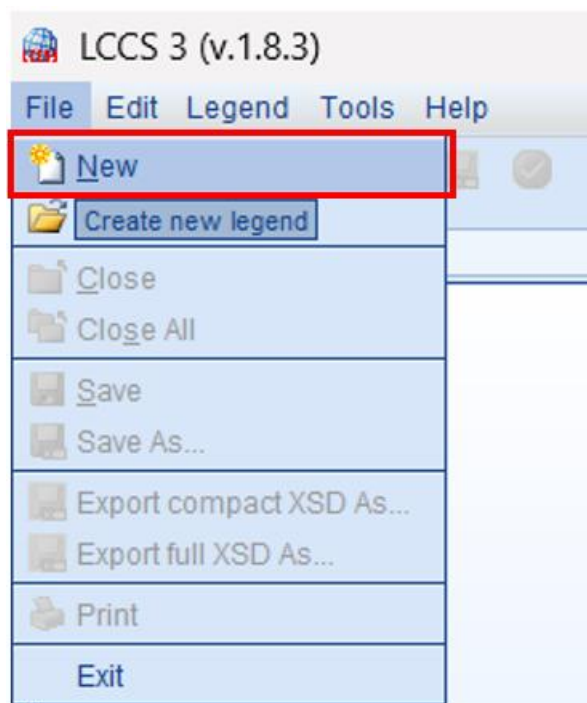
The LCRS file for Somalia will open, as seen here:



Source: FAO. 2016. Land Cover Classification System User Manual. Rome. <https://openknowledge.fao.org/items/ea8e72e646104c7a85718fcbdd24e346>

**Step 4:** Create a new file “Legend for Somalia” for legend creation using the reference system.

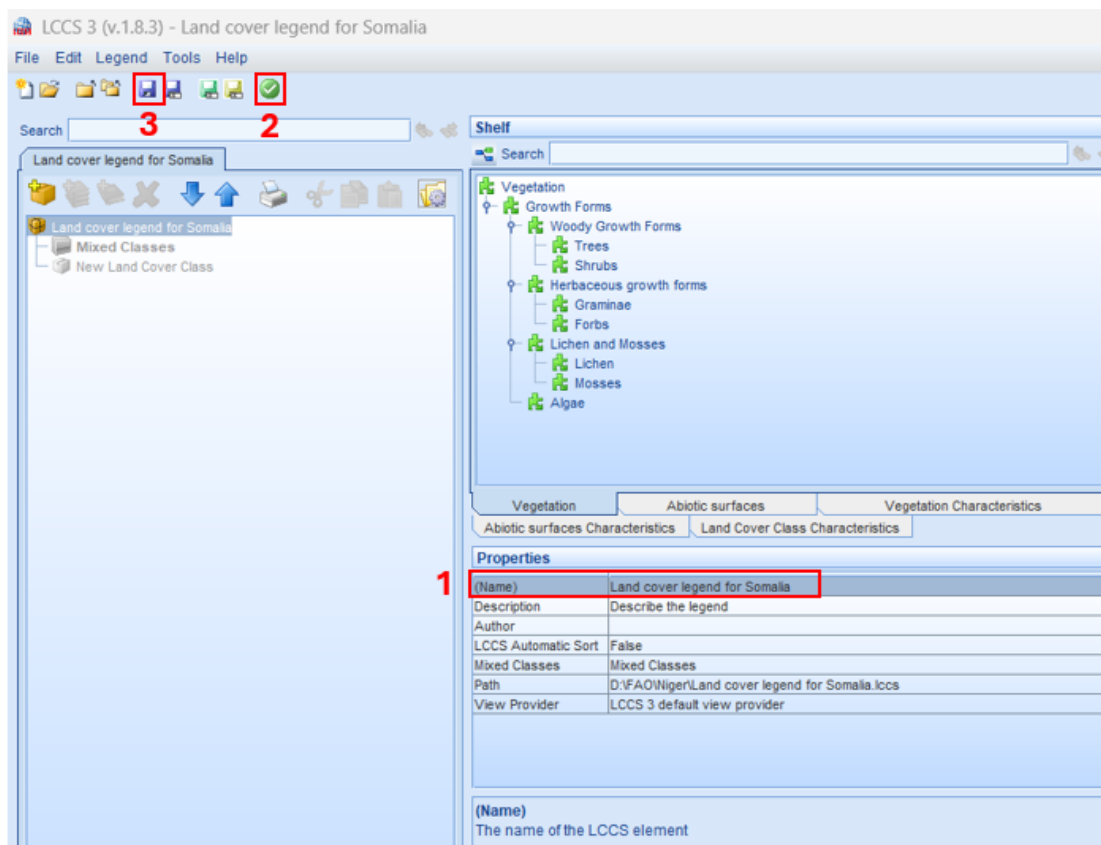
Click on the file “-> New”, as seen here:



Source: FAO. 2016. Land Cover Classification System User Manual. Rome. <https://openknowledge.fao.org/items/ea8e72e646104c7a85718fcbdd24e346>

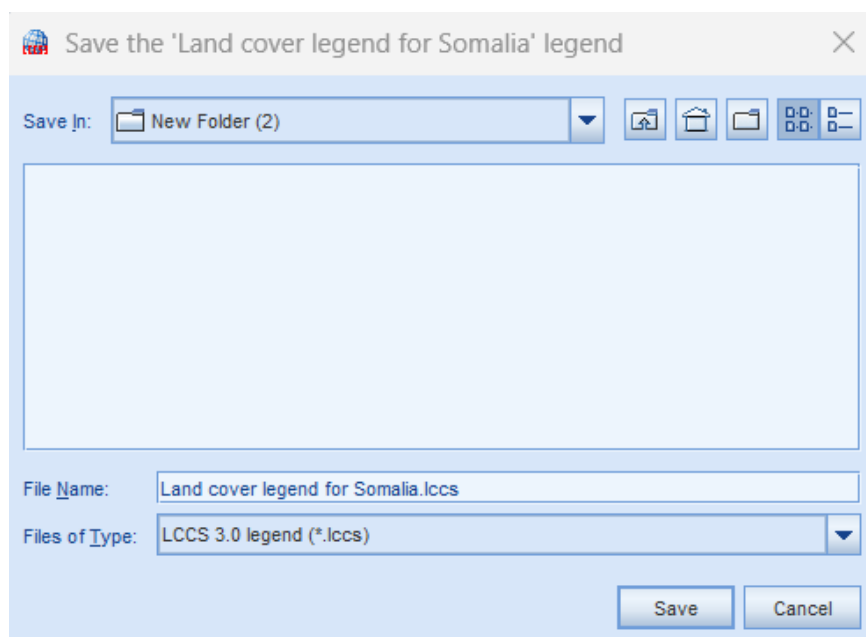


Edit the name of the legend (1), validate (2) and save it in LCCS format (3), as seen here:



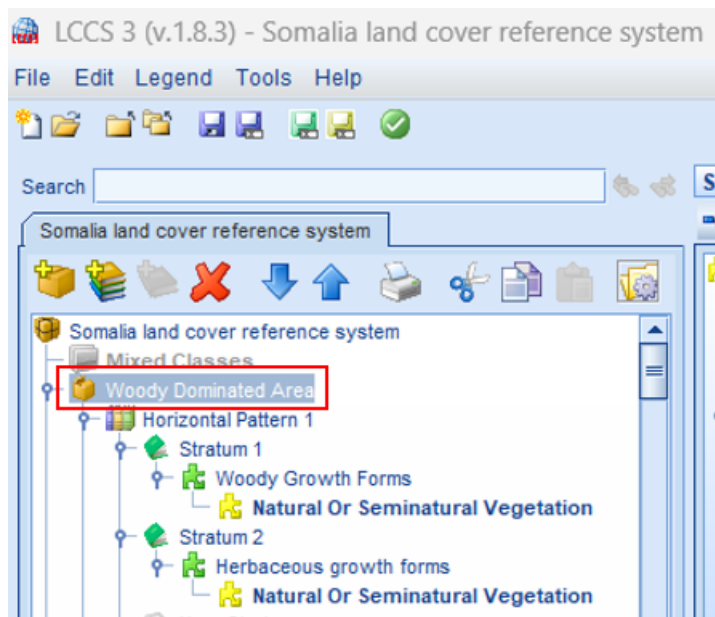
Source: FAO. 2016. Land Cover Classification System User Manual. Rome. <https://openknowledge.fao.org/items/ea8e72e646104c7a85718fcbdd24e346>

When saving a legend, a new window will open. Give the folder path and click on “Save”, as seen here:



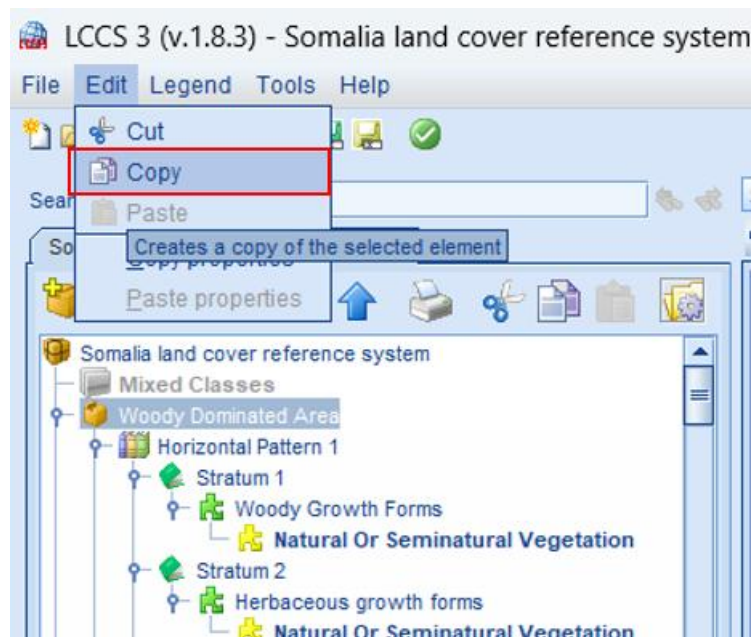
Source: FAO. 2016. Land Cover Classification System User Manual. Rome. <https://openknowledge.fao.org/items/ea8e72e646104c7a85718fcbdd24e346>

**Step 5:** Click on the national LCRS file tab and select the required class, as seen here:



Source: FAO. 2016. Land Cover Classification System User Manual. Rome. <https://openknowledge.fao.org/items/ea8e72e646104c7a85718fcbdd24e346>

Click “Edit” and copy the selected class, as seen here:



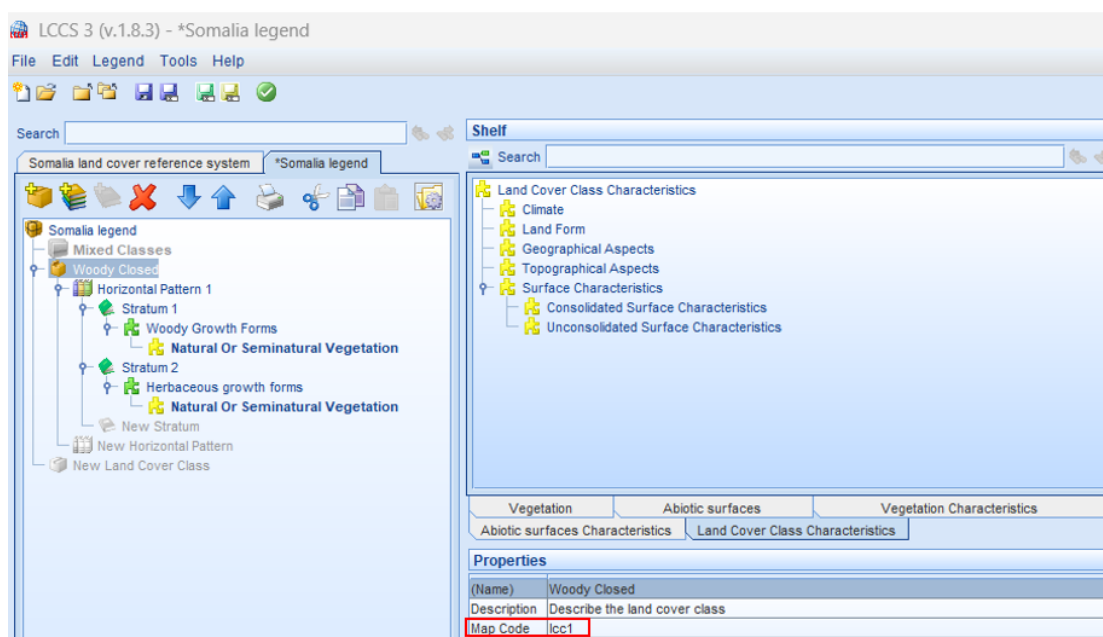
Source: FAO. 2016. Land Cover Classification System User Manual. Rome. <https://openknowledge.fao.org/items/ea8e72e646104c7a85718fcbdd24e346>

**Step 6:** Paste this class in a new file, “Legend for Somalia” (take care about the sequence of the classes while copying and pasting it in the new legend), as seen here:



Source: FAO. 2016. Land Cover Classification System User Manual. Rome. <https://openknowledge.fao.org/items/ea8e72e6-4610-4c7a-8571-8fcbdd24e346>

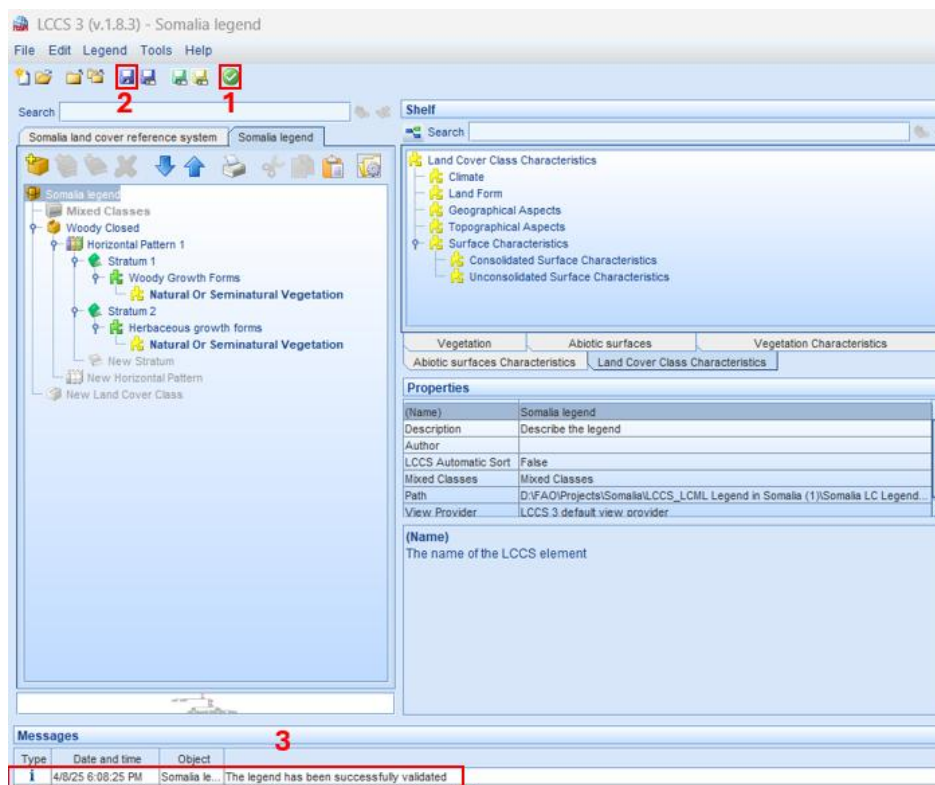
**Step 7:** After pasting the class in a new legend file, edit the map code if the class attributes need changing, or additional detail is needed in a specific class, as seen here:



Source: FAO. 2016. Land Cover Classification System User Manual. Rome. <https://openknowledge.fao.org/items/ea8e72e6-4610-4c7a-8571-8fcbdd24e346>

Note: Users can add further details, features or attributes to a metaclass.

**Step 8:** Each time the user provides detail to the metaclass, click on the validate button (1). It will give a notification in the message section (3). In case of validation, click on the save button (2). The class will be updated and saved in a given path, as seen here:



Source: FAO. 2016. Land Cover Classification System User Manual. Rome. <https://openknowledge.fao.org/items/ea8e72e646104c7a85718fcbdd24e346>

# LAND COVER REFERENCE SYSTEM FOR **SOMALIA**

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